Flexible Computing Loads, Electricity Grid Resilience, and Carbon Emission Reduction:

A Tale of Bitcoin Mining and AI Training

(Extremely preliminary working draft)

Abstract

We develop an analytical framework for *flexible computing loads* (FCLs)—intensive electricity demands that can scale elastically in real time (Bitcoin mining is an archetypal FCL; AI training an imperfect alternative). In a model of an integrated grid with endogenous renewable and fossil capacity, we show that if the amortized capital cost of FCL hardware falls below a threshold—a condition currently met by Bitcoin mining—introducing FCLs with demand response may generate a *triple win*: higher utility profit, lower blackout probability, and lower carbon emissions. This occurs as FCLs (1) incentivize renewable investment by monetizing otherwise uncompensated renewable surplus and (2) reduce fossil usage by substituting for peak-hour fossil backups. We find these benefits also prevail in deregulated competitive markets and are amplified under rate-of-return regulation if FCL revenues are passed to consumers as lower retail prices. Legacy capacity, however, may reverse the carbon-reduction effect in the short run. Our analysis lays a foundation for evaluating the carbon-reduction potential of FCLs and provides guidance for designing demand response programs to harness them.

Keywords: AI, Bitcoin, blockchain, carbon emissions, demand response, electricity pricing policy and regulation, flexible computing loads, mining, renewable energy, sustainability

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1 Introduction

Bitcoin mining, and more broadly proof-of-work blockchain operations, have long been criticized for their high electricity consumption. Media headlines frequently compare Bitcoin mining's energy use to that of a mid-sized country (Forbes, 2022), and some studies have made the bold claim that Bitcoin mining alone could contribute to a 2°C rise in global temperatures (Mora, Rollins, Taladay, Kantar, Chock, Shimada and Franklin, 2018). Some scholars have even pointed to their policy analyses linking Bitcoin mining to carbon emissions as a primary justification for China's ban of Bitcoin mining in September 2021.[1] Given rising Bitcoin prices, recent political shifts, and increasing adoption by governments and corporations—including recent initiatives for strategic Bitcoin reserves (U.S. Senator Lummis, 2024; The Washington Post, 2024) and the emergence of Bitcoin treasury companies—the energy demand from Bitcoin mining is likely to grow further, making a rigorous assessment of its environmental impact more urgent than ever.

Meanwhile, the explosive growth of artificial intelligence has created an even larger class of electricity-intensive computing operations. Training a frontier AI model today can require tens of thousands of specialized processors running continuously for weeks, with energy demands rivaling those of a mid-sized data center. Like Bitcoin mining, such batch training workloads are elastically scalable: the GPU cluster can to a large extent be scaled up or down in real time without disrupting later operations, so the jobs can in principle be scheduled around grid conditions. This shared property—the ability to absorb or release electricity demand in real time without disrupting later operations—defines what we call *flexible computing loads* (FCLs). Bitcoin mining is the archetypal FCL; AI training is a close but imperfect cousin. Developing a rigorous analytical framework for FCLs as a class—rather than treating Bitcoin mining and AI in isolation—is one of the contributions of this paper.

[1] Private communication. On China's ban, see https://www.reuters.com/world/china/china-central-bank-vows-crackdown-cryptocurrency-trading-2021-09-24/.

While it is indisputable that Bitcoin mining (and FCL in general) consumes vast amounts of electricity, we argue that a closer analysis of its economic structure reveals a strikingly different possibility. With appropriately designed demand response programs, the high electricity demand from Bitcoin mining may actually *lower* net carbon emissions. To see why, note two characteristics of the electricity system that our model makes precise. First, renewable energy sources (e.g., wind or solar) feature high fixed costs but near-zero variable costs and imperfect correlation between supply and demand. This supply-demand mismatch forces grids to rely on fossil "peaker" plants for balancing, imposing carbon costs and undermining the commercial case for renewable investment. Second, Bitcoin mining is *memoryless*: every joule of hashing contributes equally to expected rewards regardless of when it occurs, so miners are perfectly indifferent between running now and running later. This temporal flexibility—unique among large electricity consumers—makes Bitcoin mining an ideal shock absorber for intermittent renewable supply. Mining can be dialed up during renewable surplus (earning revenue on power that would otherwise be curtailed) and dialed down during scarcity (freeing fossil capacity for consumers), simultaneously raising the profitability of renewable investment and reducing the need for fossil backup.

It is important to stress that this carbon-reduction effect is not automatic. It requires that the demand response program be appropriately designed, and it depends on key parameters—most importantly the amortized hardware cost of mining equipment relative to the expected per-megawatt revenue from flexible dispatch. Our paper characterizes these conditions precisely. When they are satisfied, mining generates a *triple win*: lower carbon emissions, lower blackout probability, and higher utility profit. When they fail, the reverse holds. The even-handed characterization of both cases is, we argue, the central contribution of a formal analysis that informal advocacy cannot provide.[2]

[2]Some practitioners and policy advocates have argued informally that Bitcoin mining can act as a flexible grid resource (Ibañez and Freier, 2023; Bitcoin Policy Institute, 2024), and sporadic empirical anecdotes support this channel (Forbes, 2024). Our paper provides the analytical foundation for evaluating when and

This paper develops the first rigorous analytical framework for studying these questions. Unlike conventional electricity uses—heating, lighting, industrial processes—where curtailment imposes an immediate service disruption, FCLs can be elastically scaled up or down in real time without disrupting later operations.

Our baseline model features a regulated electricity system in which a profit-maximizing utility jointly decides how much renewable capacity, fossil backup capacity, and FCL capacity to install. The key trade-off is between the high fixed cost of renewable capacity (which generates electricity at zero marginal cost but intermittently) and the high variable cost of fossil backup (which generates reliably but emits carbon). We show that integrating FCLs via a demand response program solves this trade-off through two reinforcing channels. First, FCLs monetize renewable surplus that would otherwise be curtailed, raising the revenue from renewable investment and thus incentivizing more capacity. Second, FCLs substitute for fossil backup during hours when renewables just fall short of consumer demand, reducing the installed fossil capacity at the optimum. Together, these channels can lower total carbon emissions while increasing utility profit.

We characterize the exact conditions under which the profit and reliability gains occur and show that they coincide; the carbon reduction obtains under additional distributional conditions that we also characterize precisely. The mining preference condition takes the intuitive form that the amortized capital cost of FCL hardware must be below the expected per-megawatt value of flexible dispatch, which equals the per-MWh profit margin times the fraction of hours the grid has spare fossil capacity.

Beyond the baseline model, we analyze four extensions that are critical for external validity and policy relevance. First, we consider *timing variations*: many grids have legacy fossil or renewable capacity that is not freely adjustable. We show that the carbon improvement channel critically relies on endogenous fossil capacity choice. In the short run,

why the mechanism operates, and when it does not.

when both fossil and renewable capacities are fixed (Case C), the demand response program *unambiguously increases* carbon: FCLs dispatch on spare fossil power, raise the plant's utilization rate, and no capacity substitution offsets this effect. Critics who observe high fossil utilization rates at mining facilities sited near gas plants are empirically correct—for those legacy-capacity environments. Our result is that this *short-run* carbon cost is the flip side of the *long-run* triple win: the same FCL revenue that increases fossil utilization in the short run finances the renewable capacity expansion that reduces both fossil utilization and carbon in the long run. A renewable-only dispatch constraint (capping FCL dispatch at the realized renewable surplus $(-\tilde{x})^+$) eliminates even the short-run carbon cost, at the expense of narrowing the FCL operating window and reducing the program's profitability. Second, we embed the model in the *Averch–Johnson rate-of-return regulation* framework, where the retail electricity price is determined endogenously. Under ROR regulation, mining revenue creates a "double win": in addition to the standard capacity subsidy channel, it is passed through to consumers as a lower retail electricity price. Third, we analyze *deregulated competitive electricity markets*, showing that the competitive equilibrium replicates the regulated outcome with the market's scarcity price playing the role of the regulated penalty, and that FCLs alleviate the missing money problem facing *renewable* generators by converting zero-price surplus hours into fossil-marginal hours at price c, inducing additional renewable investment $K_R^* > K_R^\circ$. Fourth, we examine the *scope of the grid*: whether the model describes a price-taking regional grid or one with market power in the global hash market, and how coordination failures arise when many grids simultaneously adopt demand response programs.

Finally, we quantify the model's predictions using current market data. Prevailing Bitcoin prices, reward per block (the "coinbase" reward), and falling renewable capital costs imply that the viable range for FCL adoption is currently roughly 2.5 times wider than under 2022 conditions. For AI training, the viable hardware cost range depends critically on the

curtailable fraction θ and training revenue R^{AI}: under plausible frontier-model assumptions ($\theta = 0.5$, $R^{\text{AI}} \approx$ \$200–400/MWh), the threshold is comparable to or wider than Bitcoin's current threshold. These quantitative assessments suggest that the carbon-reduction potential of FCLs is not just a theoretical possibility but a practical reality under current market conditions.

Key contributions. This paper makes three contributions to the intersection of energy economics, blockchain, and operations management.

First, we provide the first analytical characterization of FCLs as a grid resource, unifying Bitcoin mining and AI training within a common framework that identifies when and why flexible computing loads can reduce carbon emissions. Previous work on demand response (Aïd, Possamaï and Touzi, 2022; Fattahi, Dasu and Ahmadi, 2023) does not consider the specific properties of computing loads (memoryless, globally competitive, transparent revenue), and previous work on Bitcoin mining (Halaburda and Yermack, 2023; Cong, He and Li, 2021) does not develop a full welfare analysis within an electricity system model.

Second, we provide tractable comparative-static results across a wide range of market structures—regulated utilities, rate-of-return regulation, and competitive markets—enabling a unified policy analysis of FCL programs across different electricity system designs.

Third, we identify and formalize the key differences between Bitcoin mining and AI training as FCL technologies, including the inference latency constraint, revenue opacity, and power surge behavior, providing a theoretic foundation for the emerging debate about whether data centers can serve as flexible grid resources (Knittel, Senga and Wang, 2025).

The rest of the paper is organized as follows. Section 2 surveys related literature. Section 3 presents the model. Section 4 analyzes the baseline simultaneous optimization. Section 5 studies timing variations with pre-installed capacity. Section 6.1 analyzes rate-of-return regulation. Section 6.2 analyzes deregulated competitive markets. Section 7 examines

grid scope, mining revenue endogeneity, and the multi-grid coordination externality. Section 8 compares Bitcoin mining and AI training as FCL technologies. Section 9 presents the quantitative assessment and supporting evidence from ERCOT grid data. Section 10 concludes. Appendices contain proofs, the formal Nash equilibrium derivation for multi-grid FCL programs, the analysis of the $R \leq c$ parameter region, and alternative regulatory objective functions.

2 Related Literature

Our work contributes to the intersection of four distinct streams of literature: research on Bitcoin mining and blockchain, demand response programs, renewable energy investments and electricity markets, and the economics of artificial intelligence and data centers.

Research on Bitcoin mining. On the theory side, Biais, Bisiere, Bouvard and Casamatta (2019) model Bitcoin mining as a coordination game and identify the possibility of multiple equilibria. Cong, He and Li (2021) studies the dynamics of mining pool formation and shows that the rise of mining pools may lead to significant electricity consumption. Prat and Walter (2021) propose a model that uses the Bitcoin–USD exchange rate to predict the computing power of the network. Arnosti and Weinberg (2022) show that Bitcoin mining has a tendency toward significant centralization given even small initial differences in variable costs. Bertucci, Bertucci, Lasry and Lions (2024) models Bitcoin mining as a mean-field game.

On the empirical side, the most directly related paper is Halaburda and Yermack (2023), who study the operations and financial valuations of 13 cryptocurrency mining companies listed on NASDAQ and find that miners using Texas wind power are offline more than other miners in a more erratic pattern, yet are more profitable due to extremely low prices

during renewable surplus periods. Their findings directly support the mechanism we model. Using Bitcoin prices as an instrument, Benetton, Compiani and Morse (2023) find that cryptomining entry raises electricity prices for nearby households and small businesses in upstate New York. Unlike our paper, the setting that leads to their finding does not feature demand response programs that curtail demand during peak hours, a key economic force behind our results.

Several industry reports also provide supporting evidence for our model. Ibañez and Freier (2023) contain numerous verbal arguments on why Bitcoin mining may be environmentally friendly that echo key aspects of this paper. Bitcoin Policy Institute (2024) document data patterns consistent with Bitcoin mining's flexible electricity demand. There is also extensive anecdotal evidence of grids using Bitcoin mining to absorb surplus electricity, including documented cases in Texas, Pakistan, and Alberta.

The flexibility of Bitcoin mining at the hardware level is grounded in ASIC design. Miners have been reported to switch operations "at a minute's notice" (Bloomberg, 2021; CNBC, 2022; Forbes, 2021), and technical product manuals (BITMAIN Support, 2022) confirm that rapid on/off cycling is a design feature. Yaish, Tochner and Zohar (2022) show that miners can repeatedly switch machines to manipulate difficulty in ways that increase profits, confirming that rapid cycling is economically feasible. Yaish and Zohar (2023) further document two firms—Layer1 Technologies (2022) and Blockstream (2022)—that designed ASICs *explicitly* as grid shock absorbers, underscoring that hardware-level flexibility is an intentional design feature rather than an artifact. Before Bitcoin mining was banned in China, miners routinely migrated between hydro-rich regions as the wet and dry seasons shifted renewable availability (Bitcoin.comNews, 2019; decrypt, 2020; Cointelegraph, 2020)—an early large-scale demonstration of FCL demand flexibility in practice. Kawaguchi, Noda and Komiyama (2022) provide econometric evidence that miners adjust their electricity consumption in response to profitability shocks, confirming elastic demand response in the field.

Research on electricity markets and renewable investment. Electricity supply has historically been treated as a natural monopoly subject to rate-of-return (ROR) regulation, under which utilities recover prudently incurred costs plus a regulated return on invested capital (Train, 1991; Jamison, 2008). The landmark pathology of this regime is the Averch–Johnson overcapitalization effect (Averch and Johnson, 1962): because allowed profits scale with the rate base, regulated firms over-invest in capital relative to the social optimum. This observation directly informs our Section 6.1. The theoretical critique of cost-of-service regulation motivated the development of incentive regulation mechanisms; the landmark contribution is Laffont and Tirole (1986), which formalizes the regulator's problem as a principal-agent relationship and shows that optimal contracts must trade off productive efficiency against information rents captured by the firm. Abito (2020) estimates structurally that moving to optimal incentive regulation could reduce costs by 20–30 percent for some utilities.

Beginning in the 1990s, dissatisfaction with ROR regulation motivated electricity sector restructuring: separating generation from transmission, introducing bid-based wholesale markets, and creating independent system operators (Joskow, 1997, 2003). Evidence on restructuring's efficiency effects is largely positive (Fabrizio, Rose and Wolfram, 2007; Davis and Wolfram, 2012; Cicala, 2022): Cicala (2015) shows that ROR regulation inflates fuel procurement costs by roughly 10 percent relative to competitive procurement, and Abito, Han, Houde and Van Benthem (2024) identifies the specific channels—removal of cost pass-through provisions, repositioning in input markets, and unwinding of legacy contracts—through which divestiture raised plant efficiency. Market power and its impact on electricity prices remain a concern, however (Wolak, 2003; Hortaçsu and Puller, 2008; Bushnell, Mansur and Saravia, 2008; MacKay and Mercadal, 2024).

Electric Power Research Institute (2024) project significant near-term load growth from data centers and identify grid reliability risks. Complementing the operational analysis,

Van Nieuwerburgh (2026) characterizes the financial architecture of this expansion, noting that the massive scale of AI infrastructure is driving a shift toward third-party ownership and private credit financing. Gete and Mercader (2026) document the emerging datacenter mortgage market and analyze originate-to-distribute incentives that arise when lenders securitize data center loans, raising questions about underwriting standards in an asset class whose long-run viability depends on sustained electricity availability. Satchwell, Mims Frick, Cappers, Sergici, Hledik, Kavlak and Oskar (2025) survey large-load tariffs—real-time pricing, demand charges, and interruptible service agreements—through which utilities manage financially and operationally challenging loads; the NARUC rate design primer (Abal, Hedman, Butterworth and Kneeland, 2021) provides the foundational cost-allocation framework underpinning these instruments and is directly relevant for FCL program design.

Among academic work on renewable investment, Aflaki and Netessine (2017) show that carbon pricing may sometimes discourage renewable adoption due to intermittency challenges. Kök, Shang and Yücel (2018) show that flat pricing can encourage solar adoption. Kök, Shang and Yücel (2020) find that renewable and flexible energy sources complement each other, whereas inflexible sources act as substitutes. Peng, Wu and Souza (2024) study the interplay between renewable generation and energy storage. Andonov and Rauh (2024) document the increasing entry of private equity in U.S. electricity markets, especially renewable investment. Bai and Wu (2023) find that private equity buyouts of fossil fuel power plants reduce output-scaled CO_2 and NO_x emissions by improving production efficiency. Darmouni, Lehrer and Sastry (2025) point out that the high fixed cost of renewables justifies ex-ante power purchase agreements. These papers provide context for understanding FCLs as a complementary mechanism for enabling renewable investment.

Research on demand response programs. A stream of research has explored demand response programs in electricity markets. Peura and Bunn (2015) examine how peak electric-

ity producers set prices under uncertain demand. Aïd, Possamaï and Touzi (2022) analyze optimal demand response contracts. Fattahi, Dasu and Ahmadi (2023) propose a scenario-based approach to optimize direct load control contracts. Lohmann and Rebennack (2017) develop a long-term power generation expansion model that factors in short-term demand response. Agrawal and Yücel (2022) investigate the role of baseline adjustments in demand-response programs. Together, these studies highlight the importance of demand response in balancing supply and demand, improving grid stability, and reducing costs. However, none examines how such programs can leverage the specific properties of computing loads to facilitate renewable investment—the central contribution of this paper.

Research on carbon pricing and externalities. Our analysis of carbon taxes and FCL programs as complementary instruments (Section 6.2) connects to the canonical literature on environmental externalities. The Pigouvian case for carbon taxation dates to Pigou (1920); its application to electricity markets was formalized in Baumol and Oates (1988) and extended to intermittent renewables by Fischer and Newell (2008), who show that renewable portfolio standards and carbon taxes interact non-trivially when generation is stochastic. Acemoglu, Aghion, Bursztyn and Hemous (2012) study directed technical change under environmental constraints, showing that carbon taxes can tilt innovation toward clean technologies—a mechanism that parallels the FCL investment channel we model. Ambec and Crampes (2012) analyze the complementarity between renewable energy and flexible backup generation, providing theoretical foundations closely related to our capacity substitution result. The empirical evidence on carbon pricing is reviewed in Goulder and Schein (2013): while carbon taxes are first-best in theory, second-best instruments including renewable subsidies (Investment Tax Credits, Production Tax Credits) and demand response programs are often more politically feasible. Our paper provides a foundation for evaluating FCL demand response as one such second-best instrument, characterizing precisely when it

substitutes for—and when it complements—explicit carbon pricing.

Research on AI, data centers, and electricity. The electricity demands of artificial intelligence have received growing attention. Electric Power Research Institute (2024) project that U.S. data centers will require 90–100 GW of additional generation capacity by 2030, placing AI on a scale comparable to current Bitcoin mining. International Energy Agency (2024) document that global data centers consumed approximately 415 TWh in 2024, and project this to exceed 800 TWh by 2030 under a high-growth scenario. Feher, Garcia-Appendini and Mihet (2025) provide evidence that U.S. data center development has been substantially subsidized by public resources, raising questions about efficient capacity allocation and the distributional implications of concentrating energy-intensive compute in specific localities. Gargano and Giacoletti (2026) document the scope of U.S. state tax incentives for data center siting, finding that interstate competition has generated large public subsidies for an industry whose energy demands are increasingly material to grid planning—a policy context directly relevant to the demand response programs we analyze.

The flexibility of AI training loads has attracted emerging interest. Modern distributed training frameworks support checkpointing—pausing mid-run without losing progress—which is directly analogous to Bitcoin miners powering down ASICs and restarting (Jiang, Lin et al., 2024). Hyperscalers (Google, Microsoft, Amazon) have begun using temporal flexibility to reduce energy costs and participate in demand response programs (Acun et al., 2023; Radovanović, Koningstein et al., 2023). The feasibility of demand-response participation by data centers is documented in Texas, where ERCOT's large flexible load program includes multiple data center operators alongside Bitcoin miners (Knittel, Senga and Wang, 2025).

The key difference between AI training and Bitcoin mining lies in *AI inference*: serving live queries is latency-constrained in ways that hashing is not, reducing the curtailable fraction of AI electricity demand as inference becomes a larger share of total AI compute

(International Energy Agency, 2024; Sachs, 2024), although some recent attempts in peak-hour pricing for AI inferences may change the situation. Our Section 8 formally models this distinction and characterizes the conditions under which each technology is the superior flexible load.

3 Model

This section presents the overall model setup, which captures a regulated electricity system with stochastic renewable generation, fossil backup, and an optional demand response program that integrates flexible computing loads.

Our baseline model has one strategic decision maker: a regulated utility. The utility chooses three capacity levels—renewable capacity K_R, fossil backup capacity K_F, and flexible computing load capacity K_M (subscripts R, F, M denote *renewable*, *fossil*, and *mining*, respectively)—to maximize expected profit. The utility's profit is then obtained from revenues from consumer electricity sales and computing revenue, minus variable fossil costs, blackout penalties (which the regulator requires it to internalize), and capital costs for all three capacity types. In the baseline model the utility sets K_M directly (in practice, K_M may be implemented through a demand response program that contracts with third-party miners; we model it as a direct capacity choice for simplicity); we relax this assumption in Appendix E. The regulator sets the retail price p exogenously and imposes a blackout penalty β. In the timing variations (Section 5), p remains exogenous; we endogenize it under rate-of-return regulation (Section 6.1) and in deregulated spot markets (Section 6.2).

3.1 Key model ingredients

The model features a market for electricity within a representative period. The key state variable is *residual demand*

$$\tilde{x} \equiv \tilde{D} - K_R \tilde{S}, \tag{1}$$

where $\tilde{D}$ is stochastic (non-FCL) consumer demand drawn from distribution F_D with support on $[0, \bar{D}]$, K_R is installed renewable capacity, and $\tilde{S} \in [0, 1]$ is the stochastic renewable capacity factor (that is, the utilization ratio of installed capacity). By construction, $\tilde{x} \in [-K_R, \bar{D}]$. When $\tilde{x} > 0$, electricity supply from renewables fall short of consumer demand and fossil backup is needed. When $\tilde{x} < 0$, there is a renewable surplus. Let $F(x; K_R)$ denote the CDF of $\tilde{x}$ given renewable capacity K_R, assuming that $F(x; K_R) := 0$ for $x < -K_R$ and $F(x; K_R) := 1$ for $x > \bar{D}$, and with $f(x; K_R)$ denoting the corresponding density on $[-K_R, \bar{D}]$. More renewable capacity shifts the distribution leftward: $\frac{\partial F}{\partial K_R} > 0$ for any given x, meaning a higher K_R reduces the probability of high residual demand.

The supply side consists of: (1) renewable capacity K_R (MW) with zero variable cost and capital cost C_R per MW; (2) fossil backup capacity K_F (MW) with variable cost c per MWh and amortized capital cost C_F per MW (the same subscript convention applies to capital costs: C_R renewable, C_F fossil, C_M mining);[3] and (3) an optional FCL program of capacity K_M (MW) generating revenue R per MWh consumed,[4] with capital cost C_M

[3] We model the fossil source as a flexible, dispatchable plant (e.g., a natural gas peaker). Inflexible conventional sources such as nuclear cannot be rapidly ramped up or down, so they act as *substitutes* for renewables rather than complements (Kök, Shang and Yücel, 2020): their fixed output reduces the need for renewable balancing but does not enable the demand-flexibility mechanism we study. Including nuclear would not alter the key renewable-versus-fossil trade-off that drives our results.

[4] For simplicity, the baseline model assumes the utility directly operates the FCL program and captures R per MWh (e.g., through blockchain rewards or contracted computing fees). Note that R is independent of the retail price p: it reflects the market value of computing output and may exceed or fall short of p depending on market conditions. An alternative, arguably more common, arrangement is that the utility *sells electricity* to third-party mining operators at the regulated retail tariff p per MWh and administers curtailment without owning hardware. In that case, the relevant FCL margin for the utility is $p - c$ rather than $R - c$; since $p > c$, all qualitative results of the baseline carry over with this substitution. See Appendix E for a full development.

per MW (the amortized ASIC hardware cost). All capital costs are expressed as amortized flow costs per MW per hour of the representative period, so they are directly comparable to the per-MWh operating margins below. A blackout occurs when residual demand exceeds total dispatchable capacity and imposes social penalty β per MWh of unserved demand. Consumers pay retail price p per MWh.

Assumption 1 (Parameter regularity)**.** *We maintain throughout the paper that $R > c + C_M$ (computing is profitable enough to justify acquiring hardware and operating it with existing fossil capacity) and $R < c + C_F + C_M$ (computing is not so profitable as to justify acquiring hardware and building new fossil backup capacity). We also assume that the penalty of blackout is adequately high so that $\beta + p > c + C_F$ (it is worth building fossil backup to avoid blackout) and $\beta + p > R$ (it is not worth curtailing normal users, i.e., enduring blackout to support computing).*

All the assumptions are innocuous: If instead we have $R \leq c + C_M$, then we should not observe any profitable miners at all, which is at odds with the reality; If instead we have $R \geq c + C_F + C_M$, then it makes sense for the utility to build infinite amount of fossil capacity to serve infinite amount of FCL, which is also economically unrealistic and outside the scope of this paper. Appendix D will separately analyze the case $R \leq c$.

3.2 Utility company's expected profit

Depending on our specific model variation, the profit-maximizing utility may be able to control all or some of the following variables: renewable capacity K_R, fossil backup capacity K_F, and FCL capacity K_M (during *ex ante* at capacity planning) and the amount of FCL

dispatch $d(\cdot)$ (during *ex post* at real-time operation) to maximize expected profit.

$$\mathbb{E}\Big[p\,\tilde{D} + R \cdot d(\tilde{x}) - c \cdot \Big[\min\{K_F, \max\{\tilde{x} + d(\tilde{x}),\, 0\}\}\Big] \\ - (\beta + p) \cdot (\tilde{x} - K_F)^+ - C_R K_R - C_F K_F - C_M K_M\Big]. \quad (2)$$

In words, the utility earns revenue from consumer electricity sales $p\tilde{D}$, mining revenue $R{\cdot}d(\tilde{x})$, incurs variable fossil costs, absorbs blackout penalties (which the regulator requires it to internalize) and associated lost revenues, and pays capital costs for all three types of capacity.

3.3 Demand response program for flexible computing loads

The utility operates a Demand Response Program that dispatches registered FCL operators. FCL demand under the Demand Response Program is given by

$$d(x) = \min\Big\{\max\{K_F - x,\, 0\},\, K_M\Big\}, \quad (3)$$

where x is the realized value of $\tilde{x}$. This formulation captures the key properties of FCL demand response: (i) when $x \le K_F - K_M$ (equivalently, when net load plus full FCL does not exceed fossil capacity: $x + K_M \le K_F$), FCLs run at full capacity K_M; (ii) when residual demand $x \in (K_F - K_M, K_F)$, FCLs absorb the exact capacity gap $K_F - x$, running on fossil power (the utility earns revenue R and incurs variable cost c per MWh); (iii) when $x > K_F$, there is a shortage of dispatchable capacity (blackout occurs), so all fossil capacity is used to serve consumers and the entire FCL capacity is curtailed.

Figure 1 illustrates the dispatch rule (3) as a function of residual demand x.

The dispatch rule (3) is not an arbitrary assumption; it is the unique solution to the utility's within-period dispatch optimization, as the following lemma establishes.

Lemma 1 (Optimality of the dispatch rule)**.** *Given installed capacities* (K_R, K_F, K_M) *and*

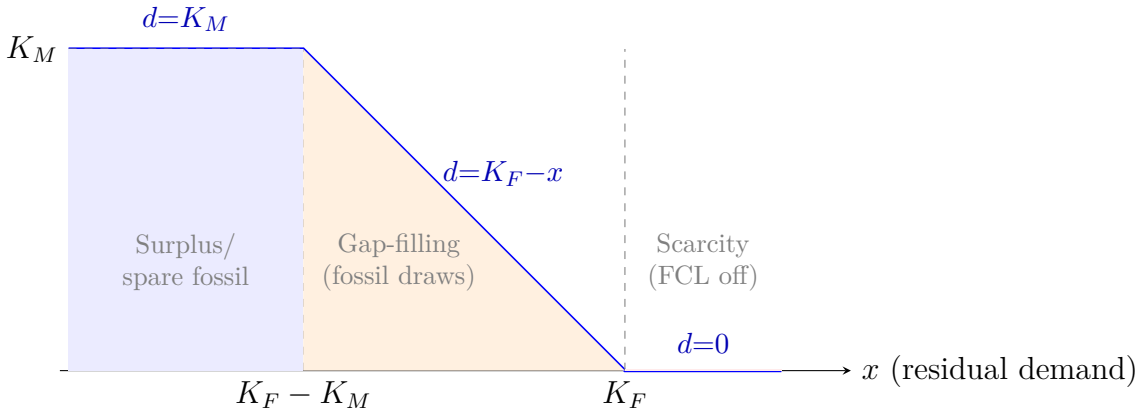


Figure 1: FCL dispatch rule $d(x) = \min\{\max\{K_F - x,\, 0\},\, K_M\}$ as a function of residual demand x. *Blue region*: renewable surplus or deep spare-fossil zone—FCL runs at full capacity K_M. *Orange region*: gap-filling zone ($x \in (K_F - K_M, K_F)$)—FCL absorbs the exact capacity gap $K_F - x$, running on fossil power. *Scarcity region*: $x > K_F$—FCL shuts down, all dispatchable capacity serves consumers; at the social optimum K_F^* is chosen so that this region has zero residual-demand probability, and it therefore appears as an empty (degenerate) zone in the figure. Lemma 1 establishes that this piecewise-linear rule is the unique profit-maximizing dispatch policy.

any realization x of residual demand $\tilde{x}$, the dispatch rule (3) is the unique solution to the per-period dispatch subproblem

$$d^*(x) \;=\; \underset{d\,\in\,[0,\,K_M]}{\arg\max}\; R\cdot d \;-\; c\cdot\Big[\min\{K_F,\,(x+d)^+\} - \min\{K_F,\,x^+\}\Big], \tag{4}$$

subject to consumers having priority over the fossil plant.

Proof. The objective in (4) is FCL revenue minus the incremental fossil cost induced by dispatching d units of computing load. We verify uniqueness in two cases.

Scarcity ($x > K_F$). All fossil capacity is already serving consumers, who still face a blackout of $x - K_F$ MWh. No spare fossil capacity is available for FCL, so the only feasible choice is $d = 0$. Moreover, Assumption 1 ($\beta + p > R$) implies that the social cost of an additional MWh of unserved consumer demand exceeds the mining revenue; hence diverting fossil from consumers to FCL is never profitable even if it were physically feasible. Thus

$d^*(x) = 0 = \min\{\max\{K_F - x, 0\}, K_M\}$.

Non-scarcity ($x \leq K_F$). In this region $K_F - x \geq 0$, so total electricity available for FCL is the renewable surplus $(-x)^+$ (free) plus fossil capacity not needed for consumers, up to $K_F - \max\{x, 0\}$. The marginal benefit of dispatching one additional unit of d is:

- $R > 0$ for each unit drawn from renewable surplus (when $d < -x$ and $x < 0$), since variable cost is zero;
- $R - c > 0$ for each unit drawn from fossil (once all surplus is absorbed), since Assumption 1 gives $R > c$.

In both sub-regions the objective is *strictly increasing* in d. The optimal d is therefore constrained only by the binding feasibility limit: the tighter of the FCL capacity cap $d \leq K_M$ and the total available electricity constraint $d \leq K_F - x$ (fossil capacity net of consumer demand, noting that for $x < 0$ consumers draw nothing from fossil, so up to $K_F + (-x)$ MW is available, which equals $K_F - x$). Hence $d^*(x) = \min\{K_F - x,\ K_M\}$. Since $K_F - x > 0$ for all $x \leq K_F$, this equals $\min\{\max\{K_F - x, 0\}, K_M\}$.

Combining the two cases yields $d^*(x) = \min\{\max\{K_F - x, 0\}, K_M\}$, which is precisely (3). □

Remark 1 (Revenue transparency and self-enforcement)**.** Lemma 1 relies on two conditions from Assumption 1: $R > c$ (FCL is always worth dispatching on spare fossil capacity) and $\beta + p > R$ (FCL always yields to consumers during scarcity). Crucially, verifying $R > c$ in real time requires that R be publicly observable. Bitcoin mining satisfies this transparency requirement by construction: R equals the block reward value divided by global network energy per block, both of which are published continuously by the Bitcoin protocol. The dispatch rule is therefore *self-enforcing*—both the utility and the miner can independently verify that the margin is positive, eliminating the need for costly monitoring or revelation mechanisms. AI training lacks a transparent real-time analogue of R, complicating the design of an incentive-compatible dispatch contract (see Section 8).

Throughout the analysis we treat the per-MWh mining revenue R as exogenous, which is appropriate for any price-taking regional grid. The full discussion of grid scope—including the endogenous-R case, Tullock mining competition, and multi-grid coordination externalities—is deferred to Section 7.

3.4 Simplifying the profit expression.

Plugging in the Demand Response Program dispatch rule (3) into (2) and evaluating expectations using the residual-demand CDF $F(x; K_R)$, the profit function reduces to:

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F(x; K_R)\,\mathrm{d}x - c\int_{-K_M}^{K_F-K_M}\Big(1 - F(x; K_R)\Big)\,\mathrm{d}x \\ - (\beta + p)\int_{K_F}^{\bar{D}}\Big(1 - F(x; K_R)\Big)\,\mathrm{d}x - C_F K_F - C_R K_R - C_M K_M, \quad (5)$$

where $F(x; K_R)$ is the *extended CDF*, defined to equal 0 for $x < -K_R$ and 1 for $x > \bar{D}$. This convention makes (5) valid in *both* regimes: in the large-FCL regime ($K_M > K_R$, assumed throughout this section) the lower limit $-K_M$ falls outside the support $[-K_R, \bar{D}]$, so the extended CDF equals zero there and an integration-by-parts step converts (5) into the algebraically simpler form (10); in the small-FCL regime ($K_M \le K_R$, Appendix G) $-K_M$ lies within the support, the extended CDF equals the ordinary CDF on the integration domain, and (5) applies directly without further simplification. The three integral terms represent expected net mining revenue, expected fossil variable cost, and expected blackout cost, respectively.

Without FCL program (benchmark). Setting $K_M = 0$ gives the no-FCL benchmark:

$$\Pi_0(K_R, K_F) = p\,\mathbb{E}[\tilde{D}] - c\int_0^{K_F}\Big(1 - F(x; K_R)\Big)\,\mathrm{d}x - (\beta + p)\int_{K_F}^{\bar{D}}\Big(1 - F(x; K_R)\Big)\,\mathrm{d}x - C_F K_F - C_R K_R. \quad (6)$$

Differentiating (6) with respect to K_F and setting equal to zero gives the FOC:

$$(\beta + p - c)\big[1 - F(K_F^\circ; K_R^\circ)\big] = C_F \quad \Longrightarrow \quad F(K_F^\circ; K_R^\circ) = \frac{\beta + p - c - C_F}{\beta + p - c}. \tag{7}$$

Blackout probability at the no-FCL optimum is

$$\mathbb{P}(\tilde{x} > K_F^\circ) = 1 - F(K_F^\circ; K_R^\circ) = \frac{C_F}{\beta + p - c} \in (0, 1). \tag{8}$$

Differentiating (6) with respect to K_R and setting equal to zero gives the FOC:

$$c\int_0^{K_F^\circ} \frac{\partial F(x; K_R^\circ)}{\partial K_R^\circ}\,\mathrm{d}x + (\beta + p)\int_{K_F^\circ}^{\bar{D}} \frac{\partial F(x; K_R^\circ)}{\partial K_R^\circ}\,\mathrm{d}x - C_R = 0. \tag{9}$$

Since $\partial F/\partial K_R > 0$, both integrals are positive: the left-hand side is the marginal benefit of renewable capacity, comprising the reduction in expected fossil cost (first term) and the reduction in expected blackout cost (second term), equated to the marginal capital cost C_R.

Remark 2 (Expected fossil output with or without FCL)**.** Under the optimal FCL dispatch rule, the expected fossil output is

$$\mathbb{E}\big[\min\{K_F,\, (\tilde{x} + K_M)^+\}\big] = \int_{-K_M}^{K_F - K_M} (1 - F(x; K_R))\, dx,$$

which is strictly greater than the no-FCL expected output

$$\mathbb{E}[\min\{K_F, \tilde{x}^+\}] = \int_0^{K_F} (1 - F(x; K_R))\, dx.$$

To see the identity, apply the formula $\mathbb{E}[\min\{A, B\}] = \int_0^A \Pr(B > t)\, dt$ with $A = K_F$ and $B = (\tilde{x} + K_M)^+$: this gives $\int_0^{K_F} \Pr(\tilde{x} > t - K_M)\, dt$, and the substitution $x = t - K_M$ yields the stated expression. The mining program thus raises fossil utilization and earns net

computing revenue $R - c$ per MWh of computing load.

4 Baseline Analysis: Simultaneous Capacity Choice

This section analyzes the full joint optimization in which the utility simultaneously chooses renewable capacity K_R, fossil backup capacity K_F, and FCL capacity K_M. This is the natural benchmark when all three investments can be made simultaneously (e.g., when designing a new electricity system from scratch or executing a comprehensive grid upgrade).

Throughout this section, we work under the *large-FCL regime* ($K_M^* > K_R^*$), in which optimal FCL capacity exceeds renewable capacity; the complementary small-FCL regime (Appendix G) yields a wider adoption threshold and a lower blackout probability, so the results below are conservative in those respects. In this regime, the profit function (5) becomes

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F(x; K_R)\,\mathrm{d}x - c\Big(K_F - \int_{-K_R}^{K_F-K_M} F(x; K_R)\,\mathrm{d}x\Big) \\ - (\beta + p)\int_{K_F}^{\bar{D}} \Big(1 - F(x; K_R)\Big)\,\mathrm{d}x - C_F K_F - C_R K_R - C_M K_M. \quad (10)$$

The small-FCL regime ($K_M^* \leq K_R^*$) is analyzed in Appendix G. We also maintain the interior regime ($R < c + C_F + C_M$, Assumption 1); the full-coverage regime is discussed in Section 4.2 below. We first characterize the first-order conditions and optimal allocations (Section 4.1), then compare blackout risk (Section 4.2), carbon emissions (Section 4.3), and the conditions under which all three outcomes improve simultaneously (Section 4.4).

4.1 Optimal allocations: K_R^*, K_F^*, K_M^*

Formally, the profit-maximizing utility's problem is to simultaneously choose renewable capacity K_R, fossil backup capacity K_F, and FCL capacity K_M (and subsequently adopt the

optimal FCL dispatch rule $d(\cdot)$ as in (3)) to maximize expected profit (10).

It will be convenient to define a cutoff value for ASIC hardware cost: $\bar{C}_M \equiv \frac{(R-c)(\beta+p-c-C_F)}{(\beta+p-c)}$, which relates to the net mining revenue $R-c$ and the probability of no blackout when there is no mining (recall (7)). As we will show below, it is a necessary condition for the *large-FCL regime* ($K_M^* > K_R^*$) to have ASIC hardware cost C_M below this cutoff, that is $C_M \leq \bar{C}_M$. This same condition is also necessary for the existence of an interior optimum with positive FCL capacity. Section 9 examines where current market parameters fall relative to this cutoff. We will also briefly discuss the case $C_M > \bar{C}_M$ later.

FCL capacity K_M^*. Differentiating (10) with respect to K_M, we get the marginal net benefit of FCL capacity:

$$(R-c) \cdot F(K_F^* - K_M^*;\, K_R^*) - C_M. \tag{11}$$

FOC then gives

$$F(K_F^* - K_M^*;\, K_R^*) = \frac{C_M}{R-c}. \tag{12}$$

Given the parameter restrictions in Assumption 1, we have the RHS of (12) in $(0,1)$. Hence the equation $F(\cdot;\, K_R^*) = \frac{C_M}{R-c}$ has a solution. Furthermore, a comparison to the FOC for K_F (Equation (13) derived below) will confirm that the K_M satisfying (11) is nonnegative if and only if $C_M \leq \bar{C}_M = \frac{(R-c)(\beta+p-c-C_F)}{(\beta+p-c)}$.[5]

Fossil capacity K_F^*. Differentiating (10) with respect to K_F and substituting the FOC for K_M (Equation (11)) yields the marginal benefit of fossil capacity:

$$\Big(R-\beta-p\Big)F(K_F;K_R)+\beta+p-c-C_F-C_M \quad \overset{\text{FOC}}{\Longrightarrow} \quad F(K_F^*;K_R^*) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}. \tag{13}$$

[5]Specifically, it requires $\frac{C_M}{R-c} \leq \frac{\beta+p-c-C_F-C_M}{\beta+p-R}$, which is equivalent to $C_M \leq \frac{(R-c)(\beta+p-c-C_F)}{(\beta+p-c)}$.

Note the presence of the ASIC cost C_M in the numerator: the more expensive it is to install ASIC capacity (high C_M), the lower the optimal $F(K_F^*)$ (lower fossil reliability), and vice versa. When $C_M \to 0$, the formula coincides with the no-mining benchmark (7) except that R replaces c in the denominator, reflecting that mining revenue offsets fossil costs at the margin. The equilibrium blackout probability with mining is therefore:

$$1 - F(K_F^*; K_R^*) = \frac{c + C_F + C_M - R}{\beta + p - R}. \tag{14}$$

A digression on the case with $R \geq c + C_F + C_M$. Equation (13) is derived under the *interior regime* $R < c + C_F + C_M$, in which the right-hand side lies in $(0, 1)$ and an interior optimum $K_F^* \in (0, \bar{D})$ exists. Now we can confirm our earlier intuition that when $R \geq c + C_F + C_M$, the marginal benefit of fossil capacity is always nonnegative (and strictly positive when $R > c + C_F + C_M$) even at $K_F = \bar{D}$. Therefore, in case $R = c + C_F + C_M$, the utility builds at least $\bar{D}$ fossil capacity, which is just enough to cover the maximum possible consumer demand, and is indifferent about building more to serve FCL; In case $R > c + C_F + C_M$, the utility would like to build an infinite amount of fossil capacity to serve FCL – in this case the utility effectively acts as a dedicated computing provider—a qualitatively different market structure that lies outside the present model's scope and is not analyzed further. In both subcases, the utility builds so much fossil capacity that the blackout probability is zero.

In the full-coverage regime the FCL adoption condition $C_M < \bar{C}_M \equiv (R - c)\frac{\beta + p - c - C_F}{\beta + p - c}$ is automatically satisfied—anticipating the formal result in Proposition 1, which shows that $C_M < \bar{C}_M$ is necessary and sufficient for the triple win (lower blackouts, lower carbon, and higher profit). To see this, note that $C_M \leq R - c - C_F$ (the condition $R \geq c + C_F + C_M$ rearranged) and $R - c - C_F < \bar{C}_M$ whenever $\beta + p > R$ (Assumption 1). Hence the full-coverage regime is a *degenerate but favorable* special case in which zero blackouts, lower

carbon, and higher profit are all achieved simultaneously.

For the remainder of the paper we focus on the empirically relevant *interior regime* $R < c + C_F + C_M$; all results carry over to the full-coverage regime with the blackout probability replaced by zero.

Renewable capacity K_R^*. Differentiating (10) with respect to K_R and applying the envelope theorem gives the marginal benefit of renewable capacity:

$$R\int_{K_F^*-K_M^*}^{K_F^*} \frac{\partial F}{\partial K_R}(x; K_R^*)\,\mathrm{d}x + c\int_{-K_R^*}^{K_F^*-K_M^*} \frac{\partial F}{\partial K_R}(x; K_R^*)\,\mathrm{d}x + (\beta+p)\int_{K_F^*}^{\bar{D}} \frac{\partial F}{\partial K_R}(x; K_R^*)\,\mathrm{d}x - C_R. \tag{15}$$

The key insight is that the weight in the FCL dispatch region $(K_F^* - K_M^*, K_F^*)$ is R, whereas in the no-FCL case it is c. Since $R > c$, computing revenue *amplifies* the shadow value of renewable investment, pushing toward a higher K_R^* than the no-FCL optimum; Section 4.3 and Proposition 4 make this comparison precise under specific distributional assumptions.

4.2 Blackout comparison

A blackout occurs when $\tilde{x}$ exceeds fossil capacity K_F. Comparing (8) and (14), straightforward algebra shows:

$$\underbrace{1 - F(K_F^*; K_R^*)}_{\text{with FCL}} = \frac{c + C_F + C_M - R}{\beta + p - R} \lessgtr \frac{C_F}{\beta + p - c} = \underbrace{1 - F(K_F^\circ; K_R^\circ)}_{\text{without FCL}} \quad \Longleftrightarrow \quad C_M \gtrless \bar{C}_M. \tag{16}$$

Therefore, we have that when $C_M < \bar{C}_M$, the equilibrium blackout probability is strictly lower with the FCL program than without. The intuition is that the mining revenue $R > c$ earned from spare fossil capacity raises the per-MW profitability of fossil investment, inducing the utility to build more fossil backup. The net effect (when the ASIC cost C_M is not too high) is a higher equilibrium $F(K_F^*; K_R^*)$, meaning lower blackout probability, rather than a

substitution away from fossil. On the other hand, when $C_M \geq \bar{C}_M$, the utility prefers not to adopt the FCL program, which is equivalent to the case when FCL is banned. In this case, the blackout probability is the same as the no-FCL benchmark. In sum, blackout probability is weakly lower with the FCL program, and strictly lower when $C_M < \bar{C}_M$.

4.3 Carbon emissions

We now compare expected carbon emissions with and without the FCL program. First note that the expected carbon emission without FCL equals c times $K_F^\circ - \int_0^{K_F^\circ} F(x; K_R^\circ)dx$, and that with FCL equals c times $K_F^* - \int_{-K_R^*}^{K_F^*-K_M^*} F(x; K_R^*)dx$. Hence the reduction in carbon emission from the introduction of FCL (i.e., expected carbon emission without FCL minus expected carbon emission with FCL) is proportional to

$$\underbrace{\Big[\int_0^{K_F^\circ}(1-F(x;K_R^\circ))dx - \int_0^{K_F^*}(1-F(x;K_R^*))dx\Big]}_{\text{(I): reduction in carbon emission from conventional use}} - \underbrace{\Big[\int_{K_F^*-K_M^*}^{K_F^*} F(x;K_R^*)dx - \int_{-K_M^*}^{0} F(x;K_R^*)dx\Big]}_{\text{(II): increase in carbon emission by FCL}} \quad (17)$$

From (17), we see that there are two opposing effects of the FCL program on carbon emissions:

Utilization effect (carbon-increasing). The mining program raises fossil utilization by dispatching computing load whenever there is spare fossil capacity, which tends to increase carbon emissions. This is formally captured by Term (II), which is always positive and reduces the reduction in carbon emission from the introduction of FCL.

Capacity effect (carbon-ambiguous). At the joint optimum, the mining program changes both K_F and K_R. More fossil capacity (K_F^* vs. K_F°) tends to raise carbon emission, while more renewable capacity (K_R^* vs. K_R°) shifts the residual demand distribution leftward, tending to lower fossil utilization. The carbon outcome depends on which effect dominates, as Term (I) illustrates.

Net effect. In general, the sign of (17) depends on the relative magnitudes of both the utilization effect and capacity effect, which in turns depends on the distribution of $F(\cdot,\cdot)$ and exogenous parameters. We can nevertheless demonstrate that the introduction of FCL *lowers* carbon emission when $F(\cdot,\cdot)$ represents a uniform distribution and the capital cost of renewables C_R is sufficiently low.

Under the uniform distribution $F(x;K_R)=\frac{x+K_R}{\bar{D}+K_R}$ on $[-K_R,\bar{D}]$, inverting Equation (13) gives:

$$K_F^*=\frac{(\beta+p-c-C_F-C_M)(\bar{D}+K_R^*)}{\beta+p-R}-K_R^*, \tag{18}$$

$$K_M^*=K_F^*-F^{-1}\Big(\frac{C_M}{R-c};K_R^*\Big)=K_F^*-\frac{C_M(\bar{D}+K_R^*)}{R-c}+K_R^*, \tag{19}$$

with K_R^* determined from (15). In comparison, $K_F^\circ=\frac{(\beta+p-c-C_F)(\bar{D}+K_R^\circ)}{\beta+p-c}-K_R^\circ$ without mining, with K_R° determined from (9).

Further simplification gives

$$K_F^\circ = \bar{D}-\frac{C_F}{\beta+p-c}(\bar{D}+K_R^\circ), \tag{20}$$

$$K_R^\circ = \Big(\sqrt{\frac{c}{2C_R-\frac{C_F^2}{\beta+p-c}}}-1\Big)\bar{D}, \tag{21}$$

$$K_F^*-K_M^* = \bar{D}-\frac{R-C_M-c}{R-c}(\bar{D}+K_R^*), \tag{22}$$

$$K_F^* = \bar{D}+\frac{R-c-C_F-C_M}{\beta+p-R}(\bar{D}+K_R^*), \tag{23}$$

and K_R^* is infinite/arbitrary/zero depending whether the following expression is positive/zero/negative:

$$(R-c)\Big(\frac{R-C_M-c}{R-c}\Big)^2+c+(\beta+p-R)\Big(\frac{R-c-C_F-C_M}{\beta+p-R}\Big)^2-2C_R. \tag{24}$$

The knife-edge structure arises because, under the uniform distribution, K_F^* and K_M^* are

proportional to $\bar{D} + K_R^*$ (the linear expressions above), so the marginal return to renewable investment is constant along the joint optimum locus: the utility either builds unbounded renewable capacity (when this constant return exceeds C_R) or invests nothing in renewables (when it falls short).

Therefore, carbon emission without FCL is proportional to

$$\frac{1}{2}\left[\frac{1}{\sqrt{\frac{c}{2C_R - \frac{C_F^2}{\beta+p-c}}}} - (\frac{C_F}{\beta+p-c})^2\sqrt{\frac{c}{2C_R - \frac{C_F^2}{\beta+p-c}}}\right]\bar{D}, \tag{25}$$

and carbon emission with FCL is proportional to

$$\bar{D} + \frac{R-c-C_F-C_M}{\beta+p-R}(\bar{D}+K_R^*) - \frac{1}{2}(\frac{C_M}{R-c})^2(\bar{D}+K_R^*), \tag{26}$$

which is lower than (25) if C_R is adequately small (for example, when (24) is positive, in which case $K_R^* = \infty$: unlimited renewable investment is optimal, residual demand is always in deep surplus, fossil output converges to zero, and so does carbon).[6]

4.4 Conditions for FCL adoption preference

We summarize our analysis so far in the following proposition.

Proposition 1 (Mining preference condition and triple win)**.** *The utility strictly prefers*

[6]More generally, a similar proof logic applies when $F(x; K_R) = \frac{(x+K_R)^\alpha}{(\bar{D}+K_R)^\alpha}$ on $[-K_R, \bar{D}]$ for some positive α (i.e., a power distribution), which nests the uniform distribution as a special case when $\alpha = 1$. The key is that the FOCs for K_F and K_R can be solved in closed form, and the carbon comparison reduces to a parameter restriction that is satisfied when C_R is sufficiently low. Furthermore, such conditions are more likely to hold when α is small (right-skewed $F(\cdot;\cdot)$), which corresponds to the empirically relevant case of rare stress events: the distribution puts most of its mass on deep-surplus states, where the FCL runs on renewable power at zero carbon cost. In contrast, when α is large (left-skewed $F(\cdot;\cdot)$), probability mass concentrates on high residual-demand states, fossil dispatch is frequent, and the condition is harder to satisfy.

allowing the FCL *program if and only if*

$$C_M < \bar{C}_M = \frac{(R-c)(\beta+p-c-C_F)}{\beta+p-c}, \tag{27}$$

where $\bar{C}_M$ *is the adoption threshold introduced in Section 4.1. When* $C_M < \bar{C}_M$*, allowing the* FCL *program simultaneously (i) increases utility profit, (ii)* lowers *the equilibrium blackout probability, and (iii) may reduce carbon emissions (for example when* $\tilde{x}$ *follows a uniform distribution and the fixed cost of renewables* C_R *is sufficiently low) —a triple win. Parts (i) and (ii) hold for any residual-demand distribution satisfying Assumption 1. When* $C_M > \bar{C}_M$*, no* FCL *capacity is optimal within the large-FCL regime maintained in this section, and the outcome coincides with banning* FCL*; Appendix G shows that allowing small-FCL programs widens the adoption threshold to* $\bar{C}_M^b = \bar{C}_M + c\,F(0; K_R^\circ) > \bar{C}_M$ *(see (62)).*

The threshold $\bar{C}_M$ has an instructive factorization. Define $\phi^\circ \equiv F(K_F^\circ; K_R^\circ) = 1 - \frac{C_F}{\beta+p-c}$ as the no-FCL probability of *no* blackout. Then

$$\bar{C}_M = (R-c) \cdot \phi^\circ.$$

The per-MW value of FCL investment equals the per-MWh profit margin $(R-c)$ times the fraction of time the grid is not blacked out (i.e., the probability that spare fossil capacity is available for computing). Intuitively, computing hardware is worth building if its amortized cost C_M is below its expected revenue per MW of capacity.

Comparative statics of $\bar{C}_M$. The viability threshold $\bar{C}_M$ is:

- *Increasing in* R: higher computing revenue (e.g., higher BTC price) widens the viable range for ASIC capital cost;
- *Decreasing in* c: cheaper fossil variable cost raises the effective per-MWh computing margin $R-c$;

- *Increasing in β and p*: higher blackout penalty or retail price raises the no-FCL reliability (ϕ° rises), so the grid has more spare capacity hours available for computing;
- *Decreasing in C_F*: cheaper fossil capital raises no-FCL reliability ϕ° (the utility builds more fossil cheaply), again increasing spare capacity hours.

Remark 3. A necessary condition for $\bar{C}_M > 0$ is $\beta + p > c + C_F$, which is exactly the condition that it is worthwhile to build any fossil backup at all without the FCL program. If the grid would not build fossil capacity absent mining, the demand response mechanism has nothing to work with.

5 Timing Variations: Pre-Installed Capacity

The analysis in Section 4 assumes that the utility can simultaneously adjust fossil, renewables, and FCL capacity. In practice, however, capacity decisions are often made at different points in time. Many grids have legacy fossil fleets or existing renewable plants (e.g., a hydro dam) that cannot be freely adjusted in the short or medium run. This section accommodates this possibility and analyzes three cases of pre-existing (sunk) capacity—pre-installed fossil only (Case A), pre-installed renewables only (Case B), and both capacities pre-installed (Case C). We thus demonstrate how the triple-win results of Section 4 change as the capacity substitution channel is progressively constrained.

5.1 Case A: Pre-installed fossil capacity

Suppose $K_F^{(0)}$ MW of fossil capacity is already installed (a legacy coal or gas fleet). The utility takes $K_F^{(0)}$ as given and optimizes only over K_R and K_M.

Mining and carbon. With $K_F^{(0)}$ fixed, the Demand Response Program raises fossil utilization. Specifically, for states $\tilde{x} \in (K_F^{(0)} - K_M,\ K_F^{(0)})$ the FCL absorbs the residual capacity

gap and drives the fossil plant to run at its full installed capacity $K_F^{(0)}$, whereas without the program the plant would run at partial utilization $\tilde{x} < K_F^{(0)}$ in those same states. For the *same* K_R, this higher utilization rate increases carbon emissions.

However, the joint optimization of K_R now internalizes the higher marginal value of renewable investment (since renewables generate revenue via computing as well as displacing fossil), leading to a higher optimal renewable level than in the no-FCL benchmark. The net carbon effect depends on whether the additional renewable investment reduces fossil use sufficiently to offset the higher utilization rate:

$$\Delta\text{Carbon} < 0 \quad \Longleftrightarrow \quad (R - c) \cdot \left| \frac{\partial\, \mathbb{E}[\text{fossil output}]}{\partial K_R} \right|_{K_R = K_R^{(0)}} > C_R.$$

(The derivative $\partial \mathbb{E}[\text{fossil}]/\partial K_R < 0$, so the absolute value gives the marginal carbon-reduction benefit of renewable investment.) When renewable capital cost C_R is low (as in current markets), this condition is readily satisfied.

Key insight. Carbon improvement requires that the additional renewable investment induced by computing revenue generates enough surplus hours to offset the higher fossil utilization. Grids that are already heavily renewable (large $K_R^{(0)}$) have many surplus hours where FCLs run on clean power at zero carbon cost—making the net effect unambiguously positive in such cases.

5.2 Case B: Pre-installed renewable capacity

Here $K_R^{(0)}$ MW of renewables are fixed (e.g., from prior policy mandates), and the utility optimizes only over K_F and K_M. The FOCs are identical to those of Section 4 (equations

(13) and (11)), evaluated at $K_R = K_R^{(0)}$:

$$F(K_F^*; K_R^{(0)}) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \qquad (R - c) \cdot F(K_F^* - K_M^*; K_R^{(0)}) = C_M. \tag{28}$$

Adoption threshold with fixed renewables. Because K_F remains freely chosen, the no-FCL benchmark satisfies $F(K_F^\circ(K_R^{(0)}); K_R^{(0)}) = \phi^\circ$ for *any* $K_R^{(0)}$ (equation (7)): re-optimization of fossil capacity neutralizes the effect of renewable penetration on the reliability margin itself. What does vary with $K_R^{(0)}$ is the probability of renewable surplus, $F(0; K_R^{(0)})$, which is strictly increasing in $K_R^{(0)}$ (more renewables shift the residual-demand distribution leftward). Since the first marginal unit of FCL capacity necessarily lies in the small-FCL regime ($K_M \to 0 < K_R^{(0)}$), its value includes the surplus-hour margin (Appendix G, equation (61)), and the adoption threshold becomes

$$\bar{C}_M(K_R^{(0)}) \;\equiv\; (R - c)\,\phi^\circ \;+\; c\,F(0;\, K_R^{(0)}) \;=\; \bar{C}_M + c\,F(0;\, K_R^{(0)}) \;>\; \bar{C}_M,$$

which is strictly increasing in $K_R^{(0)}$.

Effect of renewable over-installation. If $K_R^{(0)} > K_R^*$ (more renewables than the jointly optimal level), FCLs spend proportionally more hours in the renewable-surplus region ($\tilde{x} \le 0$), earning R per MWh at zero fossil cost, and the threshold $\bar{C}_M(K_R^{(0)})$ rises accordingly, making the Demand Response Program attractive even at higher hardware capital cost. This has direct policy relevance: grids with large solar/wind buildout and significant curtailment (ERCOT-Texas, Germany, California) have more surplus hours, widening the viable range for FCL investment.

Effect of renewable under-installation. Conversely, if $K_R^{(0)}$ is small, surplus hours are rare, $F(0; K_R^{(0)})$ is small, and the threshold $\bar{C}_M(K_R^{(0)})$ approaches the baseline $\bar{C}_M$ from above.

Mining is still viable if $C_M < \bar{C}_M(K_R^{(0)})$, but the viable range narrows toward the baseline threshold.

5.3 Case C: Both capacities pre-installed

In the short run, both $K_F^{(0)}$ and $K_R^{(0)}$ are fixed and only FCL capacity is chosen. The interior FOC for $K_M^* > 0$ is:

$$(R - c) \cdot F(K_F^{(0)} - K_M^*;\, K_R^{(0)}) = C_M, \tag{29}$$

yielding $K_M^* = K_F^{(0)} - F^{-1}\Big(\frac{C_M}{R-c};\, K_R^{(0)}\Big)$. The FCL program is adopted ($K_M^* > 0$) if and only if $(R - c) \cdot F(K_F^{(0)};\, K_R^{(0)}) > C_M$; otherwise $K_M^* = 0$.

Short-run carbon impact. Since neither capacity adjusts, the only carbon effect is through changed fossil utilization, which is unambiguously positive (the Demand Response Program increases carbon):

$$\Delta\text{Carbon} = c \cdot \mathbb{E}\Big[\min\{(\tilde{x} + K_M^*)^+,\, K_F^{(0)}\} - \min\{\tilde{x}^+,\, K_F^{(0)}\}\Big] > 0, \tag{30}$$

where the expectation aggregates the extra fossil output across all states with $\tilde{x} < K_F^{(0)}$, including renewable-surplus states ($\tilde{x} \leq 0$) in which the dispatched FCL draws fossil power beyond the available surplus.

Remark 4. Equation (30) highlights a crucial distinction: the paper's positive carbon result critically relies on the *capacity substitution channel*. With fully fixed capacity, the Demand Response Program is carbon-neutral only if FCL dispatch never draws on fossil power, requiring a renewable-only design that caps dispatch at the realized surplus, $d \leq (-\tilde{x})^+$ (merely curtailing computing when $\tilde{x} > 0$ is not enough, since a large FCL draws fossil power even in surplus states with $-\tilde{x} < K_M$). Critics who point to high fossil utilization rates at mining facilities near gas plants are typically observing Case C environments where no capacity

adjustment has occurred.

5.4 Summary table: Timing variations

Table 1 summarizes the carbon and profit outcomes across the three cases, and the key condition for carbon improvement.

Table 1: Timing Variations: Carbon and Profit Effects by Capacity Flexibility

Case	**Carbon effect**	**Profit effect**	**Condition for net carbon reduction**
Fixed $K_F^{(0)}$	Ambiguous	Positive	$(R-c)\left\|\partial_K \mathbb{E}[\text{fossil}]\right\|_{K_R^{(0)}} > C_R$
Fixed $K_R^{(0)}$	May decrease when $C_M < \bar{C}_M(K_R^{(0)})$	Positive iff $C_M < \bar{C}_M(K_R^{(0)})$	$K_R^{(0)}$ large $\Rightarrow$ easier; $K_R^{(0)}$ small $\Rightarrow$ harder
Fixed $K_F^{(0)}$, $K_R^{(0)}$	Increase (unambiguous)	Positive	Not achievable without renewable-only Demand Response Program constraint
Baseline (§4): All free	May decrease when $C_M < \bar{C}_M$ (§4.3)	Positive iff $C_M < \bar{C}_M$	$C_M < \bar{C}_M = (R-c)\phi^\circ$; also lowers blackout prob

This table summarizes the direction of carbon and profit changes when an FCL demand response program is introduced, across four capacity-flexibility scenarios. Case A: fossil capacity $K_F^{(0)}$ is pre-installed; renewable K_R and FCL capacity K_M are freely optimized. Case B: renewable capacity $K_R^{(0)}$ is pre-installed; K_F and K_M are freely optimized. Case C: both $K_F^{(0)}$ and $K_R^{(0)}$ are pre-installed; only FCL capacity K_M is chosen. Baseline: all three capacities are jointly optimized (Section 4). The condition for net carbon reduction in Case A requires that mining-induced renewable investment outweighs the higher fossil utilization. The threshold $\bar{C}_M(K_R^{(0)}) \equiv \bar{C}_M + c\,F(0; K_R^{(0)})$ adds the surplus-hour margin to the baseline threshold; it rises with $K_R^{(0)}$, so grids with more renewable capacity face a wider viability range.

The progression from Case C to the baseline captures the key insight: carbon improvement requires the capacity substitution channel.

Remark 5 (Alignment with Knittel, Senga and Wang (2025))**.** This short-run/long-run distinction finds direct computational support in Knittel, Senga and Wang (2025), who use the GenX capacity expansion model to study data center temporal flexibility across three U.S.

regions. In Texas—where projected 2030 renewables supply 54% of generation (39% wind, 15% solar)—full flexibility reduces CO_2 emissions by up to 40%, consistent with our long-run triple-win regime. In the Mid-Atlantic—where renewables supply only 22% of generation—the same flexibility *raises* system-wide emissions by up to 3%, because insufficient renewable capacity means shifted load is absorbed by baseload coal operating more uniformly (higher utilization, lower ramping). This is precisely our Case C logic: when endogenous capacity substitution is absent or limited by the existing generation mix, the demand response program dispatches on spare fossil power and increases carbon. Knittel, Senga and Wang (2025) also confirm that flexibility *always* lowers total system costs regardless of region—consistent with our result that the utility's profit rises whenever it adopts the program—and that cost savings are primarily constrained by the *share* of flexible workload rather than the shifting horizon, a finding that maps to the role of K_M in our model.

The more freely capacity can be adjusted, the more powerful the mechanism.

6 Endogenous Electricity Prices

The baseline result in Section 4 specifies an exogenous electricity price p to regular (non-FCL) customers. In reality, however, electricity prices are typically endogenously determined (at least in the United States) either through (1) *rate-of-return* regulation (in regulated markets such as most southern states) or (2) competitive bidding (in deregulated markets such as California, Texas, Illinois, and many northeastern states). This section studies how our main insight survives different market structures.

6.1 Rate-of-Return Regulation

In traditional regulated electricity markets, electricity prices to customers are typically set through rate cases following *rate-of-return* (ROR) regulation. That is, the regulator sets

electricity prices so as to allow the utility to recover its costs plus a fair return on invested capital. This section replaces the baseline's fixed p with the rate-of-return mechanism, following traditions in regulatory economics (see, e.g., Averch and Johnson (1962) and Joskow (2007)). We show that the triple-win result strengthens under this regime.

Setup: the ROR pricing rule Let $\bar{r} \geq \rho$ denote the allowed rate of return on the *rate base* $\mathcal{B} \equiv C_F K_F + C_R K_R + C_M K_M$ (the amortized value of all three capacity types), and let ρ be the utility's cost of capital (for simplicity and without loss of generality, we may assume $\rho = 0$). The regulator sets the consumer price p so that total revenues cover operating costs plus the allowed return (including both the return *on* capital and the return *of* capital):[7]

$$p \cdot \mathbb{E}[D_{\text{served}}] + R \cdot \mathbb{E}[d(\tilde{x})] \;=\; (1 + \bar{r})\,\mathcal{B} + c \cdot \mathbb{E}[\text{fossil output}]. \tag{31}$$

Solving for the endogenous price:

$$p(K_F, K_R, K_M) \;=\; \frac{(1 + \bar{r})\,\mathcal{B} \;+\; c \cdot \mathbb{E}[\text{fossil}] \;-\; R \cdot \mathbb{E}[d]}{\mathbb{E}[D_{\text{served}}]}. \tag{32}$$

Two features of (32) are worth noting. First, a larger rate base (more capital) raises p: this is the classical AJ over-investment incentive. Second, higher FCL revenue $R \cdot \mathbb{E}[d]$ *lowers* p: mining revenue (or electricity revenue earned from miners) is passed through to consumers as a regulated price reduction.

[7] Placing FCL revenue $R \cdot \mathbb{E}[d]$ on the left-hand side reflects the regulatory convention that non-tariff revenues reduce the consumer revenue requirement dollar-for-dollar. This is consistent with FERC's treatment of demand-response and wholesale market revenues in rate cases (FERC Order 2222, 2020), under which revenues earned by a utility's flexible load program are credited against the cost of service and passed through to retail customers via lower tariffs.

Joint optimization: AJ first-order conditions Plug (32) into (10), we get

$$\Pi^{AJ}(K_R, K_F, K_M) = \bar{r}\,\mathcal{B} - \beta \int_{K_F}^{\bar{D}} \Big(1 - F(x; K_R)\Big)\,\mathrm{d}x. \tag{33}$$

The first term represents the required return paid back to the utility, while the second term represents the blackout penalty. The utility thus has strong incentives to increase K_F and K_R to lower such penalty. Furthermore, in the case of $\bar{r} > 0$ (that is, when the required return calculated for rate cases is higher than the utility's actual cost of capital), the utility has an even stronger incentive to build out capacity, as well established by the Averch-Johnson effect (Averch and Johnson, 1962). Both forces lead to high capital investment, for which the introduction of FCL leads to higher profit, lower blackout probability (zero more precisely), and lower carbon emissions.[8]

Two forces that strengthen the baseline triple win The price pass-through in (32) implies that FCL adoption delivers two distinct benefits under ROR regulation—not just the capacity benefit of the baseline: $\frac{\partial p}{\partial \mathbb{E}[d]} = \frac{-R}{\mathbb{E}[D_{\text{served}}]} < 0$.

1. **Capacity benefit** (same as baseline): FCL capital K_M is included in the rate base $\mathcal{B}$, extending the AJ over-investment distortion to mining hardware and inducing the utility to over-invest in all three capacity types.
2. **Price benefit** (new under ROR): every MWh of FCL output reduces the regulated consumer price by $R/\mathbb{E}[D_{\text{served}}]$. At calibrated values ($R \approx$ \$109/MWh, computing $\approx 10\%$ of grid output), this implies a retail price reduction of roughly \$11/MWh—material in markets where regulated prices average \$50–120/MWh.

Together, the two benefits strengthen the welfare case for allowing regulated utilities to participate in FCL programs, and suggest that the socially optimal FCL investment exceeds

[8] As written, Π^{AJ} is strictly increasing in every capacity stock, so the formal problem has no interior optimum; in practice the rate base is bounded by prudence ("used and useful") reviews, which we do not model. We therefore interpret the AJ first-order effects directionally rather than as an interior characterization.

the utility's private optimum.

AJ triple-win threshold. FCL adoption ($K_M^{\text{AJ}} > 0$) is always optimal whenever $\bar{r} > 0$ and FCL hardware is included in the rate base. This follows directly from the AJ profit formula: $\partial \Pi^{\text{AJ}} / \partial K_M = \bar{r} C_M > 0$ regardless of K_M, so the utility strictly prefers any positive FCL capacity to none. The cost threshold $\bar{C}_M$ from the unregulated baseline disappears entirely—the AJ rate-base return guarantees FCL adoption without any cost-benefit condition on C_M.

Rate-base classification of FCL hardware A key open regulatory question is whether FCL hardware (ASICs, GPU clusters) qualifies as rate base. Rate base traditionally covers assets used to serve regulated consumers; it is unclear whether computing hardware meets this standard.

The answer has first-order consequences:

- **Included in rate base:** the AJ profit formula $\Pi^{\text{AJ}} = \bar{r}\mathcal{B} - \beta \int_{K_F}^{\bar{D}} (1 - F)\, \mathrm{d}x$ applies; since $\partial \Pi^{\text{AJ}} / \partial K_M = \bar{r} C_M > 0$ always, FCL adoption is unconditionally optimal and the cost threshold disappears.
- **Excluded from rate base:** the AJ distortion applies only to K_R and K_F; the FCL hardware FOC reverts to the baseline condition (11) with threshold $\bar{C}_M$.[9]

When excluded, the utility faces a two-tier optimization: regulated capacity choices (K_R, K_F) governed by AJ FOCs and an unregulated hardware choice K_M governed by the baseline FOC. The stakes are first-order: inclusion removes the cost threshold entirely, while exclusion reinstates the baseline threshold $\bar{C}_M$.

Takeaway relative to the baseline (Section 4). Rate-of-return regulation strengthens every dimension of the triple win. The AJ distortion amplifies over-investment in K_F,

[9] Rate-base exclusion arises naturally when the utility sells electricity to third-party mining operators at the regulated tariff p rather than operating computing infrastructure directly. In that arrangement, hardware is owned by miners (not the utility), so $K_M \notin \mathcal{B}$, and the utility's FCL operating margin changes from $R - c$ to $p - c$. See Appendix E for a full development of this alternative.

K_R, and (when included in the rate base) K_M relative to the Section 4 optimum: since $\partial \Pi^{\mathrm{AJ}} / \partial K_j > 0$ for every capacity type, the utility expands all capacity beyond the unregulated optimum, driving blackout probability toward zero and eliminating the cost-threshold condition for FCL adoption. A new price benefit with no counterpart in Section 4 also emerges: FCL revenue is passed through to consumers as a lower regulated price, at a rate of $R/\mathbb{E}[D_{\mathrm{served}}]$ per MWh of FCL output.

6.2 Deregulated Competitive Electricity Markets

Following a sequence of reforms in the 1990s, many electricity markets (e.g., ERCOT in Texas, PJM in the Mid-Atlantic, CAISO in California) have become deregulated, with competitive generation and a system operator that clears spot markets. This section develops the appropriate competitive framework.

Spot price mechanism and FCL participation In a competitive deregulated market, the system-wide electricity price equals the short-run marginal cost of the marginal generator:

$$P(\tilde{x}) = \begin{cases} 0 & \text{if } \tilde{x} \leq 0 \quad \text{(renewable surplus)}, \\ c & \text{if } \tilde{x} \in (0, K_F) \quad \text{(fossil marginal)}, \\ P_{\mathrm{cap}} & \text{if } \tilde{x} > K_F \quad \text{(scarcity)}. \end{cases} \tag{34}$$

The price cap P_{cap} represents either a regulatory ceiling or the value of lost load (VOLL). Fossil generators recover capital costs C_F only through scarcity rents during shortage events.

Competitive equilibrium without FCL In a competitive market, a generator's operating margin varies across demand states. During renewable surplus ($\tilde{x} \leq 0$), the spot price is zero and the fossil plant is not dispatched, earning nothing. During the fossil-marginal region ($\tilde{x} \in (0, K_F)$), the spot price equals marginal cost c, so the generator earns $P - c = 0$

per MWh dispatched. Only during scarcity ($\tilde{x} > K_F$), when fossil capacity is fully utilized and the price rises to P_{cap}, does a generator earn a positive scarcity rent of $P_{\text{cap}} - c$ per unit of capacity. This is the *missing money* problem standard in competitive power markets (Joskow, 2008): fossil generators cannot recover capital costs from energy revenues alone and must rely entirely on scarcity rents.[10] Free entry therefore drives expected scarcity rent equal to the annualized capacity cost:[11]

$$(P_{\text{cap}} - c) \cdot \Pr(\tilde{x} > K_F) = C_F \quad \Longrightarrow \quad \Pr(\text{blackout}) = \frac{C_F}{P_{\text{cap}} - c}. \tag{35}$$

The equilibrium blackout probability is thus pinned entirely by the cost structure—it equals the ratio of the capacity cost to the scarcity rent per unit—and is independent of the demand distribution. This is structurally identical to the regulated FOC (7) with P_{cap} playing the role of $\beta + p$, confirming that the competitive equilibrium replicates the regulated outcome when $P_{\text{cap}} = \beta + p$, i.e., when the price cap reflects the full social cost of lost load.

Competitive equilibrium with FCL We now study the impact of introducing FCL to a competitive electricity market. A key observation to start with is that the competitive spot-price mechanism implements the baseline dispatch rule (3) without central coordination: FCL operators respond to prices in a way that replicates the regulated outcome. To see this, partition demand states as follows:

[10] Deregulated electricity markets vary in how they compensate generators for fixed costs. Some markets, such as PJM (covering the mid-Atlantic and Midwest United States), operate a *capacity market* alongside the spot energy market (which runs on a day-ahead or hour-ahead basis with real-time adjustment). In capacity markets, generators bid for compensation paid years in advance — traditionally three years, though there is a trend toward shorter intervals given rapidly changing market conditions — simply for committing to have capacity available when needed, providing a revenue stream separate from energy sales. Texas (ERCOT), by contrast, operates an *energy-only* market with no capacity payments; generators must recover all fixed costs through spot-market revenues alone, which requires allowing very high prices P_{cap} during scarcity events. Our analysis captures the energy-only setting, where the scarcity rent P_{cap} is the sole mechanism by which fossil generators recover capital costs. Capacity markets can be viewed as an intermediate arrangement between the energy-only competitive market studied below and rate-of-return regulation analyzed earlier.

[11] In (35), only C_F appears on the right-hand side because this is the zero-profit condition for *fossil generators* specifically (who are the marginal generator at the brink of blackout).

- When $\tilde{x} \leq -K_M$, renewable generation covers all demand including FCL, so the spot price is $P = 0$ and FCL runs at full capacity K_M, earning R per MWh.
- When $\tilde{x} \in (-K_M,\, K_F - K_M)$, fossil is the marginal generator but total capacity is sufficient to serve both consumers and FCL at full K_M, so $P = c$ and FCL earns $R - c > 0$ per MWh.
- When $\tilde{x} \in (K_F - K_M,\, K_F)$, running FCL at full K_M would push total demand above K_F and spike the price to $P_{\text{cap}} > R$, making full dispatch unprofitable; FCL therefore curtails to exactly the spare capacity $K_F - \tilde{x}$, the price settles back to c, and FCL earns $R - c > 0$ on the dispatched portion.
- When $\tilde{x} > K_F$, non-FCL demand alone exhausts fossil capacity, the spot price rises to P_{cap}, FCL shuts off entirely, and blackout occurs.

The equilibrium FCL dispatch, $\min(K_M,\, (K_F - \tilde{x})^+)$ for $\tilde{x} < K_F$ and zero otherwise, is identical to the regulated dispatch rule (3): the competitive price mechanism decentralizes the regulated solution without a central controller or the introduction of demand response programs. Blackouts still occur at $\tilde{x} > K_F^*$, so the fossil zero-profit condition (35) takes the same form with FCL as without, and the equilibrium blackout probability $C_F/(P_{\text{cap}} - c)$ is unchanged. FCL operators enter until average operating profit per unit of capacity equals the annualized capital cost:

$$\mathbb{E}\left[\frac{\min(K_M,\, (K_F - \tilde{x})^+)}{K_M} \cdot \Big(R - P(\tilde{x})\Big)\right] = C_M, \tag{36}$$

where $P(\tilde{x}) = 0$ for $\tilde{x} \leq -K_M$, c for $\tilde{x} \in (-K_M, K_F)$, and P_{cap} for $\tilde{x} > K_F$. Condition (36) is the competitive analogue of the regulated FCL optimality condition (11) and pins the equilibrium scale K_M^*.

The renewable free-entry condition completes the characterization of the equilibrium. FCL demand absorbs renewable surplus up to K_M MWh, keeping fossil dispatched for states

$\tilde{x} \in (-K_M, 0)$ that would otherwise be zero-price surplus. When renewable surplus exceeds FCL capacity ($\tilde{x} \leq -K_M$), fossil remains off and the spot price is still 0. FCL therefore shifts the zero-price region from $\tilde{x} \leq 0$ to $\tilde{x} \leq -K_M$, and a renewable generator earns c per MWh for all states in the extended fossil-dispatch region $\tilde{x} \in (-K_M,\, K_F^*)$:

$$c \cdot \mathbb{E}\Big[S \cdot \mathbf{1}_{-K_M < \tilde{x}^* \leq K_F^*}\Big] + P_{\text{cap}} \cdot \mathbb{E}\Big[S \cdot \mathbf{1}_{\tilde{x}^* > K_F^*}\Big] = C_R, \tag{37}$$

where S denotes the stochastic renewable capacity factor. Without FCL, the corresponding condition is $c \cdot \mathbb{E}[S \cdot \mathbf{1}_{0 < \tilde{x}^\circ \leq K_F^\circ}] + P_{\text{cap}} \cdot \mathbb{E}[S \cdot \mathbf{1}_{\tilde{x}^\circ > K_F^\circ}] = C_R$, in which surplus-hour revenue is zero. At the same (K_F, K_R), FCL raises renewable revenue by $c \cdot \mathbb{E}[S \cdot \mathbf{1}_{-K_M < \tilde{x} \leq 0}] > 0$ earned during formerly zero-price hours $\tilde{x} \in (-K_M, 0)$, making renewable investment strictly more attractive. Free entry therefore drives more renewable capacity until the zero-profit condition is restored: $K_R^* > K_R^\circ$.

Equilibrium blackout and carbon. The three free-entry conditions (35), (36), and (37) jointly pin the competitive equilibrium (K_F^*, K_R^*, K_M^*). The *blackout probability* equals $C_F/(P_{\text{cap}} - c)$ regardless of FCL adoption, since it is pinned entirely by the fossil zero-profit condition (35). *Carbon emissions* face two offsetting forces: FCL dispatch uses spare fossil capacity during states $\tilde{x} \in (-K_M, K_F^*)$ that would otherwise be idle, directly increasing fossil output; against this, the higher renewable revenue in the FCL equilibrium induces $K_R^* > K_R^\circ$, shifting residual demand left and reducing fossil utilization. The net carbon impact parallels the regulated baseline analysis of Section 4.3: whether the renewable-investment channel dominates the direct demand effect depends on the relative magnitudes of $K_R^* - K_R^\circ$ and the FCL-induced increase in fossil dispatch.

The "missing money" problem and FCLs. Since FCL operators shut down during scarcity ($\tilde{x} > K_F^*$), their presence does not affect fossil scarcity rents, and the fossil zero-

profit condition (35) takes the same form with or without FCL: the minimum P_{cap} required for fossil generators to break even is unchanged, so FCL adoption neither worsens nor alleviates the missing money problem for *fossil* generators. Renewable generators, however, face an analogous zero-revenue problem during deep surplus hours ($\tilde{x} \leq -K_M$), when the spot price is 0. FCL demand converts the formerly zero-price states $\tilde{x} \in (-K_M, 0)$ into fossil-marginal hours with spot price c, raising renewable revenue by $c \cdot \mathbb{E}[S \cdot \mathbf{1}_{-K_M < \tilde{x} \leq 0}] > 0$ per unit of capacity. In this sense FCL alleviates the missing money problem for *renewable* generators, and this is precisely the mechanism behind $K_R^* > K_R^\circ$.

6.2.1 Carbon taxes and FCL programs as complements

Despite potential carbon reduction benefits from increased renewable capacity buildup, our analysis above also shows that FCL programs directly increases fossil dispatch during spare-capacity hours. The standard policy instrument for internalizing this externality is a carbon tax on fossil output. Because FCL profitability depends on the spot price during fossil-dispatch hours, a carbon tax interacts directly with the FCL participation decision—raising the effective cost of electricity for FCL operators and potentially pushing computing toward zero-carbon hours. We now examine how this interaction shapes the joint design of carbon pricing and FCL programs.

A carbon tax τ per unit of fossil output raises the effective marginal cost to $c+\tau$, shifting the spot price during dispatch. The carbon tax thus interacts with FCLs in two ways:

1. **Direct effect**: The higher dispatch price $(c + \tau)$ reduces the FCL profit margin to $R - (c + \tau)$ during fossil dispatch hours. If $\tau > R - c$, FCL operators shut down whenever fossil is the marginal generator, and compute *only* on renewable surplus ($\tilde{x} \leq 0$). This endogenously tilts computing toward zero-carbon hours without an explicit renewable mandate.
2. **Indirect effect**: The carbon tax shrinks the fossil scarcity margin from $P_{\text{cap}} - c$ to

$P_{\text{cap}} - (c + \tau)$, reducing fossil revenue during scarcity events. This discourages fossil entry and raises the equilibrium blackout probability (as confirmed by $\phi^\circ(\tau)$ decreasing in τ in (38)). A higher carbon tax therefore partially unwinds the reliability gain that FCL programs provide in the regulated baseline.

Carbon taxes and FCL demand response programs are therefore *complementary instruments*: a sufficiently high τ pushes computing exclusively onto renewable surplus, while tax revenues can fund renewable capacity subsidies. Note, however, that for $\tau \geq \bar{\tau} = R - c$, FCL operates only at price 0 (during renewable surplus) and no longer keeps fossil dispatched during formerly zero-price hours. The renewable missing money channel identified above—whereby FCL demand converts surplus hours into fossil-marginal hours at price c, inducing $K_R^* > K_R^\circ$—is therefore eliminated, and equilibrium renewable investment K_R^* reverts toward K_R°.

The following proposition formalizes this complementarity and characterizes the threshold carbon tax that maximally leverages FCL programs for carbon reduction.

Proposition 2 (Pigou tax and FCL complementarity)**.** *Let $\bar{\tau} = R - c$ denote the* critical carbon tax. *The cases* (i)–(iii) *below characterize the regulated baseline of Section 4 under a per-MWh carbon levy τ on fossil output; the qualitative conclusions carry over to the deregulated setting.*

(i) For $\tau < \bar{\tau}$: FCLs operate during both fossil-dispatch and renewable-surplus hours. The triple-win adoption condition becomes

$$C_M \;<\; \bar{C}_M(\tau) \;\equiv\; (R - c - \tau)\,\phi^\circ(\tau), \qquad \phi^\circ(\tau) \;\equiv\; \frac{\beta + p - c - \tau - C_F}{\beta + p - c - \tau}, \tag{38}$$

where $\phi^\circ(\tau) = F(K_F^\circ(\tau); K_R^\circ(\tau))$ is the no-FCL reliability under carbon tax τ. Since both $(R - c - \tau)$ and $\phi^\circ(\tau)$ decrease in τ, $\bar{C}_M(\tau)$ is strictly decreasing: a higher carbon tax narrows *the viable range of FCL hardware costs.*

(ii) At $\tau = \bar{\tau}$: FCLs earn zero net margin during fossil-dispatch hours and operate only

during renewable-surplus hours. Computing is fully decarbonized without any explicit renewable mandate.

(iii) For $\tau > \bar{\tau}$: FCLs shut down whenever fossil is the marginal generator. The Demand Response Program delivers only the renewable-surplus channel, with no capacity-substitution effect on fossil investment. Carbon emissions from fossil generation fall because computing competes only for zero-cost renewable surplus.

Among taxes $\tau \geq 0$, the threshold $\tau = \bar{\tau}$ has a special property: it simultaneously (a) makes all computing zero-carbon by confining FCL dispatch to renewable-surplus hours, and (b) preserves the full renewable-capacity-investment incentive from FCL revenue (since the utility still earns R per MWh on surplus-hour dispatch). For $\tau > \bar{\tau}$, (b) weakens because FCLs earn revenue only during renewable-surplus hours, eliminating the fossil-dispatch margin that partly funds capital recovery. In this sense, $\bar{\tau}$ is a candidate optimum for minimizing steady-state fossil emissions subject to maintaining the FCL investment channel; a formal welfare-optimal tax in the presence of the full carbon externality σ is derived in Proposition 9 of Appendix F.

Proof. At $\tau < \bar{\tau}$, the effective FCL margin during fossil hours is $R - (c + \tau) > 0$, so the Demand Response Program dispatch rule (3) is unchanged. The FCL adoption condition at the no-FCL equilibrium $(K_F^\circ(\tau), K_R^\circ(\tau))$ becomes $(R - c - \tau)\phi^\circ(\tau) = C_M$, establishing (38). That $\phi^\circ(\tau)$ is decreasing in τ follows from differentiating $\phi^\circ(\tau) = 1 - C_F/(\beta + p - c - \tau)$, which is decreasing since $\beta + p - c - \tau$ falls with τ. At $\tau = \bar{\tau}$, the margin during fossil hours is zero, so FCLs are indifferent between running and shutting down during fossil dispatch, and the utility designs the program to operate only on surplus hours. For $\tau > \bar{\tau}$, FCL operators shut down when the spot price exceeds R, i.e., during all fossil and scarcity hours; computing revenue then comes exclusively from renewable surplus at zero electricity cost. □

Remark 6 (Gap between $\bar{\tau}$ and current carbon prices)**.** At the calibrated baseline ($R =$ \$109/MWh, $c =$ \$45/MWh), the critical carbon tax is $\bar{\tau} = R - c =$ \$64/MWh. Converting to

per-tonne CO_2 at natural gas emissions intensity (≈ 0.43 tonne CO_2/MWh), this corresponds to approximately \$149/tonne CO_2—far above current U.S. carbon prices. The U.S. has no federal carbon price; regional programs (RGGI, California cap-and-trade) currently imply allowance prices of roughly \$10–50/tonne CO_2 ($\approx$ \$4–21/MWh), placing them well below $\bar{\tau}$.

Two implications follow. First, in current U.S. markets, carbon pricing alone will not confine FCL operation to renewable-surplus hours: even with existing cap-and-trade programs, the FCL margin during fossil-dispatch hours remains positive and FCLs will continue to dispatch on spare fossil capacity. Second, *renewable-only dispatch requirements*—contractual provisions capping FCL dispatch at the realized renewable surplus $(-\tilde{x})^+$—achieve the same zero-carbon computing outcome as $\tau = \bar{\tau}$ without requiring federal carbon pricing. In the near term, such contractual provisions are a more practically achievable path to zero-carbon FCL operation than waiting for carbon prices to reach $\bar{\tau}$.

6.3 Comparison with regulated market

Table 2 compares the key equilibrium conditions and FCL effects across the two market structures.

Takeaway relative to the baseline (Section 4). The sharpest contrast with the regulated baseline is on blackout probability: competitive entry pins it at $C_F/(P_{\text{cap}}-c)$ regardless of FCL adoption, so the level of reliability does not change compared to the regulated case in Section 4. Instead, FCLs shift the *composition* of reliability: fossil capacity falls from K_F° to $K_F^* < K_F^\circ$, while additional renewable capacity $K_R^* > K_R^\circ$ maintains the same consumer blackout probability $C_F/(P_{\text{cap}} - c)$. This outcome results from the fact that FCL demand converts formerly zero-price surplus hours into fossil-marginal hours at price c, raising renewable revenue and attracting more renewable entry. The carbon tax result further shows that a Pigouvian levy at $\bar{\tau} = R - c$ decarbonizes computing by confining FCL dispatch to

Table 2: Regulated versus Deregulated Markets: FCL Equilibrium Conditions

Feature	**Regulated (§4–§6)**	**Deregulated (§6.2)**
Blackout probability w/o FCL	$\frac{C_F}{\beta+p-c}$	$\frac{C_F}{P_{\text{cap}}-c}$
Blackout probability w/ FCL	$\frac{c+C_F+C_M-R}{\beta+p-R}$ [lower iff $C_M < \bar{C}_M$]	$\frac{C_F}{P_{\text{cap}}-c}$ [equal to no-FCL level; event shifts from $\{\tilde{x} > K_F^\circ\}$ to $\{\tilde{x} > K_F^*\}$ with $K_F^* < K_F^\circ$, but $K_R^* > K_R^\circ$ shifts residual demand leftward, keeping the probability unchanged]
FCL viability condition	$C_M < \bar{C}_M = (R-c)\phi^\circ$	Entry iff $C_M < R\Pr(\tilde{x} \leq 0) + (R-c)\Pr(0<\tilde{x}<K_F^\circ)$; scale K_M^* from (36)
Carbon channel	Capacity substitution + utilization	Utilization + renewable entry ($K_R^* > K_R^\circ$)
Retail price	Exogenous (p) or ROR-regulated	Spot price $P(\tilde{x})$
FCL revenue	R per MWh from Demand Response Program	$R-c$ per MWh during fossil hours, R on surplus
"Missing money" problem	Not applicable (regulated cost recovery)	Unchanged for fossil; alleviated for renewables (surplus hours repriced at c)
Carbon tax complementarity	Carbon tax can substitute for ROR price floor	Carbon tax tilts FCL toward renewable surplus

This table compares equilibrium conditions and FCL effects across the regulated (Sections 4–6.1) and deregulated competitive (Section 6.2) market structures. In the regulated case, β is the value-of-lost-load penalty and p is the exogenous retail price; in the deregulated case, P_{cap} is the competitive scarcity price cap (or capacity payment). Results are structurally isomorphic: replacing $\beta+p$ with P_{cap} maps all regulated expressions to their deregulated counterparts. $\phi^\circ \equiv F(K_F^\circ; K_R^\circ) = 1 - C_F/(\beta+p-c)$ is the no-FCL equilibrium reliability (probability of no blackout without the FCL program). The blackout rows differ across market structures in one key respect: in the regulated case the utility's capacity optimization drives the *level* of the blackout probability below the no-FCL benchmark (when $C_M < \bar{C}_M$); in the competitive deregulated case the zero-profit condition pins the consumer blackout probability at $\frac{C_F}{P_{\text{cap}}-c}$ regardless of FCL adoption, so reliability does not change in equilibrium. The benefit of FCL in the deregulated case operates instead through the *composition* of reliability (less fossil, more demand response) and through the *renewable* missing money channel: FCL demand converts zero-price surplus hours into fossil-marginal hours at price c, raising renewable revenue and inducing additional renewable investment ($K_R^* > K_R^\circ$).

renewable-surplus hours without any explicit renewable mandate, though this threshold is far above current U.S. carbon prices.

7 Grid Scope: Local versus Global Mining

Throughout Sections 4–6.2, we have treated the per-MWh mining revenue R as exogenous to the grid operator. This is reasonable: electricity utilities serve geographically distinct regions, while Bitcoin mining revenue is determined by the global hash rate—an aggregate of computing loads across thousands of independent grids worldwide. For any single utility, changes in its FCL program have negligible effect on R: a 100 MW program represents roughly 0.6% of the ∼16 GW global mining fleet as of 2025 (and even a 1 GW program only ∼6%).

This section asks what changes as the scope of adoption grows. Section 7.1 considers the polar extreme: a single utility that serves all miners worldwide, so that R is fully endogenous to its choices. The FCL program collapses in that case—expanding capacity merely dilutes the per-unit reward proportionally, leaving total mining revenue unchanged. Section 7.2 analyzes the intermediate case of N regional utilities each adopting independently: each individually remains too small to affect R, so the baseline analysis applies utility by utility, but collectively their programs reduce R for everyone, creating a commons externality that places outcomes between the baseline (full adoption, negligible aggregate effect) and the global monopolist (zero adoption). Throughout this section, we will use Bitcoin mining as an example for FCL and model around its specific features.

7.1 Endogenous mining revenue: the global monopolist

In a hypothetical case in which a single utility serves all miners worldwide, mining revenue becomes fully endogenous to its choices. With competitive mining and a fixed total block

reward $\bar{R}$ per period, the per-unit mining revenue is $R(K_M) = \frac{\bar{R}}{\mathbb{E}[d(\tilde{x};K_M)]}$, so that aggregate mining income $R \cdot \mathbb{E}[d] = \bar{R}$ is constant regardless of program size. Expanding the FCL program therefore earns no additional mining revenue in aggregate: $\frac{\partial\Big(R\,\mathbb{E}[d]\Big)}{\partial K_M} = 0$. Appendix C formalizes this and shows that the global monopolist utility optimally sets $K_M^G = 0$: because expanding K_M proportionally dilutes R, the mining-revenue incentive for FCL adoption collapses entirely. This stands in sharp contrast to the baseline, where R is fixed and the triple-win threshold $\bar{C}_M = (R - c)\phi^\circ$ from Proposition 1 supports positive FCL adoption whenever $C_M < \bar{C}_M$.

Remark 7 (Tullock competition and the dispatch rule)**.** The endogenous-R case can be further rationalized via the Tullock structure of miner competition. With n symmetric oligopolistic miners each maximizing $\frac{R\,d_i}{d_i + \sum_{j \neq i} d_j} - d_i P$, the Nash equilibrium aggregate demand is

$$d = \frac{n-1}{n} \cdot \frac{R}{P}, \tag{39}$$

so total miner expenditure $Pd = \frac{n-1}{n} R$ is independent of the electricity price P. In the competitive limit $n \to \infty$, expenditure converges to R, still independent of P. Aggregate miner demand thus falls smoothly in the electricity price, with total expenditure invariant to P; the bang-bang dispatch rule—run when $R > P$, shut down when $R < P$—emerges as the behavior of price-taking miners ($n \to \infty$) facing a common per-MWh revenue R. The key property used in our analysis—that miners willingly absorb electricity whenever R exceeds its price—is therefore robust to the competitive structure of the mining industry.

Remark 8 (Observability and contractibility of R)**.** In practice, R fluctuates with the Bitcoin price, the block-subsidy schedule, and transaction fees—but all of these inputs are observable in real time via the public Bitcoin protocol (as noted in Remark 1). For the Demand Response Program, R can be updated at the dispatch-contract frequency (e.g., hourly), so the condition $R > c$ can be verified and enforced by both the utility and the miner without

private information. Since almost all mining is conducted at large farms under individual grid contracts, the utility can price-discriminate between mining and non-mining loads and embed a live R-contingent dispatch clause directly in the contract (Kawaguchi, Noda and Komiyama, 2022). This contractibility distinguishes Bitcoin mining from AI training loads, where the analogue of R (the marginal value of a compute-hour) is opaque and firm-specific; Section 8 discusses the implications.

7.2 Multi-grid coordination and the mining commons

Now suppose there are N regional utilities, each a local monopoly in its own market, that independently operate FCL programs. Each utility is small enough relative to the global fleet that it reasonably treats R as exogenous—the baseline assumption remains valid at the individual level. Yet when many grids adopt simultaneously, their aggregate computing load raises global hash rate; because Bitcoin's difficulty adjustment targets a fixed block production rate, this proportionally reduces the per-MWh revenue R for every participant. Each grid's FCL expansion thus imposes a negative externality on all other grids through the shared global R—a "mining commons" problem that no individual utility has incentive to internalize.

With N grids each running programs of size δ_i:

$$R \approx \frac{\text{block reward} \times P_{\text{BTC}}}{(\sum_i \delta_i + m) \cdot t_B},$$

where m is the mining power (MW) of out-of-grid miners and $t_B \approx 1/6$ hour is the average block time, so that $(\sum_i \delta_i + m)\, t_B$ is the network's energy consumption per block (MWh). A social planner would internalize this externality and prescribe lower individual Demand Response Program sizes. With U.S. mining accounting for roughly 38% of global hash rate as of 2024, North American grids acting in concert would have significant market power.

Appendix C formalizes the Nash equilibrium across grid operators and characterizes the coordination failure. In that equilibrium, each grid's adoption is individually rational (since R is approximately exogenous to any single grid) but collectively excessive relative to the social optimum—a classic commons problem. From the perspective of the triple-win threshold, this means $\bar{C}_M = (R - c)\phi^\circ$ is the relevant benchmark for any grid acting alone today; as aggregate FCL adoption grows globally, the effective R will decline and the threshold will narrow, eventually requiring higher-quality renewable capacity or lower hardware costs to sustain the triple win.

Carbon accounting and the miner-relocation concern. A natural objection is that local carbon reductions may be illusory: if a grid's FCL program attracts miners from higher-carbon regions, the grid's own fossil investment falls but global emissions may be unchanged or worse. Three features of the Section 4 analysis address this concern. First, the carbon reduction operates through the *capacity substitution channel*: the utility builds more renewables because FCL revenue raises their shadow value (Proposition 1, Step 3). This investment is a local, durable decision that reduces the grid's carbon intensity regardless of where miners eventually locate. Second, once renewable capacity is installed it is sunk—subsequent miner relocation does not reverse it. Third, the carbon benefit is largest in grids with high existing renewable penetration (Section 5.2), precisely the grids most likely to attract renewable-seeking miners, so the relocation effect reinforces rather than undermines the carbon case. No cross-jurisdictional coordination is required; the mechanism is self-contained within a single grid's investment decision.

The two polar cases—price-taking local grid and a single global grid operating all mining—bracket the range of realistic scenarios, with the multi-grid Nash equilibrium falling in between: individually rational adoption that is collectively excessive. Our baseline analysis covers the price-taking case; the global grid case is analyzed in Appendix C.

8 Bitcoin Mining vs AI Training as FCLs

The model developed above applies to any FCL technology, for which both Bitcoin mining and AI training appear as examples. This section further compares Bitcoin mining and AI training as specific implementations of FCL technology, identifying their structural similarities and the key differences that determine which technology provides more effective grid flexibility.

8.1 Bitcoin mining as the paradigm FCL

Bitcoin mining exhibits the three properties that make an ideal FCL:

Temporal flexibility. Bitcoin mining is a memoryless process: each microsecond of hashing delivers an identical expected reward, with no "memory" of previous computation. Switching ASICs off and back on imposes no "ramp-up" costs. This makes Bitcoin mining the ideally flexible load: computing intensity can be adjusted in real time, so that curtailment incurs no adjustment costs.

Revenue transparency. The per-MWh mining revenue R is observable, real-time, and publicly verifiable:

$$R = \frac{\text{block reward} \times P_{\text{BTC}}}{\text{total network energy per block}}.$$

Both numerator and denominator are published in real time by the Bitcoin network. This transparency is essential for Demand Response Program contract design: the utility and regulator can independently verify the revenue rate, set incentive-compatible pricing, and monitor program compliance.[12]

[12]Even if miners do not immediately sell their mining rewards and instead hold them, there are mature option pricing frameworks to account for it.

Scalability and modularity. Bitcoin mining is modular at any scale: a Demand Response Program can enroll participants ranging from a single ASIC (~3 kW) to a 500 MW mining facility, providing smooth scaling of Demand Response Program capacity across a utility's service territory. This is not purely theoretical: programs in Texas, Alberta, and several other jurisdictions demonstrate workable large-scale deployment (see e.g., empirical evidence from Halaburda and Yermack (2023)).

8.2 AI training as an emerging FCL

We now analyze AI training as a potential FCL, in comparison to Bitcoin mining.

Similarities. Large-language-model (LLM) training runs last days to weeks and support checkpointing: pausing mid-run without losing progress, then resuming— analogous to powering down ASICs and restarting. A 50-hour annual curtailment (Duke Energy's estimated threshold for accommodating large data center loads without new generation) is imperceptible in a three-week training run. Hyperscalers (Google, Microsoft, Amazon) can also reroute training workloads across data centers in different grid regions, providing spatial demand response unavailable to geographically fixed Bitcoin miners.

At scale, AI compute is comparable to Bitcoin mining: global data centers consumed approximately 415 TWh in 2024, projected to exceed 800 TWh by 2030 (IEA). U.S. AI-focused data centers alone are forecast to require 90–100 GW of additional capacity by 2030, comparable to Bitcoin's current 16 GW continuous draw.[13]

The inference latency constraint. The most important structural difference between AI and Bitcoin mining is the *inference* workload. AI inference—serving live queries to products

[13] As Van Nieuwerburgh (2026) observes, the financing of these assets is becoming increasingly "infrastructure-like," which may further incentivize their integration into grid-level demand response programs to optimize the returns on such massive capital expenditures.

like ChatGPT, Claude, and Gemini—is latency-constrained: users expect responses within seconds, making inference non-curtailable. As inference increasingly dominates AI electricity consumption (already the majority in 2024–2025), the fraction of AI demand that is truly curtailable shrinks. Bitcoin mining has no latency constraint whatsoever.

Revenue opacity. Bitcoin mining revenue R is transparent, real-time, and market-determined. AI training revenue is the expected future commercial value of the model being trained—speculative, internal, and unverifiable by a grid operator or regulator. There is no direct observable per-MWh analogue of R for AI training, complicating Demand Response Program contract design.

Power surge behavior. GPU clusters exhibit rapid power fluctuations: as GPUs transition between compute phases, power draw can surge or drop 30–60% within milliseconds (IEA 2024 data center report). This creates frequency stability challenges absent from Bitcoin mining's near-constant ASIC power draw, and imposes additional costs on grid operators.

8.3 Model modification for AI loads

Our model accommodates AI by decomposing data center workload into a curtailable portion (training and batch workloads) of fraction $\theta \in (0, 1)$ and a must-run portion (inference, live services) of fraction $1-\theta$. The must-run portion draws power irrespective of grid conditions; we fold it into baseline consumer demand $\tilde{D}$, where it is served with the same priority as ordinary consumers. The enrolled capacity then follows the baseline dispatch rule

$$d_{\mathrm{AI}}(x) = \min\Big\{\max\{K_F - x,\, 0\},\, K_M^{\mathrm{AI}}\Big\}, \tag{40}$$

where K_M^{AI} is enrolled AI data center capacity (MW). Because only the curtailable workload share θ generates dispatch-contingent compute value—the committed share's value does not depend on when or where it runs (it is rerouted or rescheduled when local power is curtailed)—each dispatched MWh carries an effective dispatch-contingent revenue of θR^{AI}, giving an effective net margin of $\theta R^{\text{AI}} - c$ per MWh of dispatch. The following proposition formalizes the AI triple-win condition.

Proposition 3 (AI triple-win threshold)**.** *Suppose the AI data center's curtailable fraction is $\theta \in (0,1)$, its effective per-MWh training revenue is R^{AI}, and its total hardware capital cost is C_M^{AI} per MW. Assume $\theta R^{\text{AI}} > c$ (AI dispatch is profitable on spare fossil capacity) and $\beta + p > \theta R^{\text{AI}}$ (consumer welfare exceeds effective AI computing revenue per MWh). Under the AI dispatch rule (40), the AI FCL program simultaneously*

(i) increases utility profit and

(ii) lowers the equilibrium blackout probability

if and only if

$$C_M^{\text{AI}} \;<\; \bar{C}_M^{\text{AI}} \;\equiv\; \frac{(\theta R^{\text{AI}} - c)\,(\beta + p - c - C_F)}{\beta + p - c} \;=\; (\theta R^{\text{AI}} - c)\,\phi^{\circ}; \tag{41}$$

under the same condition it (iii) may also reduce carbon emissions, subject to the distributional conditions of Proposition 4 (e.g., the uniform distribution with sufficiently low C_R).

Proof. Folding the must-run load $(1-\theta)K_M^{\text{AI}}$ into $\tilde{D}$ and replacing the per-MWh revenue R with the dispatch-contingent revenue θR^{AI}, the AI dispatch rule (40) coincides with the baseline rule (3) with $K_M = K_M^{\text{AI}}$, and the utility's profit function is identical in structure to (5) with $R \to \theta R^{\text{AI}}$. The Assumption 1 conditions become: $\theta R^{\text{AI}} > c$ (AI dispatch profitable on spare fossil capacity); $\beta + p > \theta R^{\text{AI}}$ (consumer welfare exceeds effective AI computing revenue per MWh—this must be imposed as a separate condition for the AI setting, since θR^{AI} need not be bounded by the Bitcoin R); and $\beta + p > c + C_F$ (unchanged). Applying

Proposition 1 with these substitutions yields the threshold $\bar{C}_M^{\text{AI}} = (\theta R^{\text{AI}} - c)\phi^\circ$, and parts (i)–(iii) follow from the same three-step argument. □

Corollary 1 (AI vs. Bitcoin triple-win comparison)**.** *The AI program has a wider viable hardware cost range ($\bar{C}_M^{\text{AI}} > \bar{C}_M$) if and only if $\theta R^{\text{AI}} > R$. At current calibration ($R \approx$ \$109/MWh), AI training must satisfy $R^{\text{AI}} > \frac{R}{\theta}$: at $\theta = 0.5$, this requires $R^{\text{AI}} >$ \$218/MWh—a threshold plausibly met by frontier model training but not by general-purpose inference workloads. For AI programs with $\theta = 0.5$ and $R^{\text{AI}} \approx$ \$200–400/MWh, $\bar{C}_M^{\text{AI}} \approx$ \$51–145/MWh, bracketing the Bitcoin threshold $\bar{C}_M \approx$ \$60/MWh.*

Grids with 50% inference workloads ($\theta = 0.5$) obtain roughly half the dispatch-contingent value per enrolled MW, and correspondingly weaker reliability and carbon benefits, all else equal.

Comparison. The structural comparison reveals that Bitcoin mining dominates AI training as an FCL on three dimensions (temporal flexibility, revenue transparency, hardware modularity), while AI training potentially dominates on two dimensions (spatial flexibility via rerouting, and scale of available capacity if R^{AI} is high). The latter advantage is heavily conditioned on whether training revenue R^{AI} can be contracted and verified—a practical obstacle that does not apply to Bitcoin.

A grid operator choosing between Bitcoin mining and AI training as Demand Response Program participants should prefer Bitcoin mining when transparency and contract verifiability are paramount (regulated utilities, public interest programs), and may prefer AI if large-scale data centers offer credible interruptible service commitments and their inference fraction $1 - \theta$ is demonstrably low.

Table 3 summarizes the structural comparison across the five key FCL dimensions.

Table 3: Bitcoin Mining vs. AI Training as Flexible Computing Loads

Dimension	**Bitcoin Mining**	**AI Training**	**Advantage**
Temporal flexibility	Memoryless hashing; costless on/off cycling at any time	Checkpointing supports pauses, but latency-sensitive inference cannot be curtailed	BTC
Revenue transparency	Real-time, publicly verifiable via Bitcoin network (R = block reward $\times$ $\frac{P_{\text{BTC}}}{\text{energy}}$)	Internal, unverifiable—no direct observable per-MWh analogue of R	BTC
Hardware modularity	Scalable from single ASIC ($\sim$3 kW) to 500 MW facility; smooth Demand Response Program enrollment	GPU clusters exhibit 30–60% power surges between compute phases; less smooth	BTC
Spatial flexibility	Geographically fixed miners cannot reroute	Hyperscalers can shift training across data centers in different grid regions	AI
Revenue scale	Mining margin $R - c \sim$ \$64/MWh (2026 calibration)	Effective margin $\theta R^{\text{AI}} - c \sim$ \$55–155/MWh at $\theta = 0.5$, $R^{\text{AI}} =$ \$200–400/MWh	Conditional on AI

"Advantage" indicates which technology performs better on each dimension for Demand Response Program participation. BTC dominates on the three dimensions critical for contract design under current regulation. AI's spatial flexibility advantage is unique to hyperscale operators and does not apply to most data center deployments.

Life-cycle hardware emissions: a caveat. The model's capital cost parameter C_M captures the amortized *financial* cost of FCL hardware but not its embodied carbon. Manufacturing Bitcoin ASICs (primarily 5–7nm semiconductor processes) and GPU servers each generates significant upstream CO_2 emissions: estimates range from 0.1–0.3 tonne CO_2-equivalent per device, or roughly 5–20 kg CO_2-eq/MWh over the hardware lifetime at typical utilization rates. This embodied carbon is a one-time, *stock* cost (incurred at manufacturing, not during operation) and is therefore orthogonal to the *flow* carbon comparison in our model, which compares steady-state fossil MWh with and without an FCL program. Nonetheless, three observations are worth noting. First, the one-time hardware carbon cost is amortized over 2–5 years of operation: for a miner operating 8,760 hours/year, even a 20 kg CO_2-eq/MWh lifetime carbon cost corresponds to ∼4–10 kg CO_2-eq/MWh annually—an order of magnitude smaller than the ∼400 kg CO_2-eq/MWh from natural gas combustion the FCL displaces. Second, GPU servers used for AI training have higher embodied carbon per MW than ASICs (due to larger die areas and more complex packaging), modestly increasing the life-cycle carbon cost of AI-based FCL programs relative to Bitcoin mining. Third, and most importantly, a full life-cycle analysis would need to assess not just the hardware but the *counterfactual*: would those semiconductors have been manufactured anyway for other uses? If demand for ASICs is inframarginal to semiconductor production, their carbon attribution is near zero. We leave a formal life-cycle assessment to future work and flag this as a limitation of the current flow-carbon analysis.

9 Quantitative Assessment

This section presents the model's calibration and quantifies the implications for the viability of FCL demand response programs.

9.1 Mining revenue R per MWh

Table 4 presents the mining revenue calibration.

Table 4: Mining Revenue R per MWh: Calibration

Parameter	**2022 estimate**	**Calibrated value (2026)**
BTC price	\$22,000	≈\$95,000
Block reward	6.25 BTC	3.125 BTC
Network hash rate	∼400 EH/s	∼800–976 EH/s
Energy per block (average)	≈2,000 MWh	≈2,700 MWh
R (revenue/MWh)	≈\$70	≈\$100–157

This table presents the key revenue parameter R (mining income per MWh of electricity consumed) under current (2026) market conditions. R is computed as block reward value divided by global energy consumed per block: at \$95,000 × 3.125 BTC per block and global mining power of approximately 16,300 MW, a 10-minute block consumes ∼2,717 MWh, giving $R \approx$ \$109/MWh—the model's calibrated central value.

Calculation. With global mining power approximately 16,300 MW (143 TWh/year), a 10-minute block consumes $\frac{16{,}300 \times 600}{3600} \approx 2{,}717$ MWh. At the stated BTC price of \$95,000: $R = \frac{\$95{,}000 \times 3.125}{2{,}717} \approx \$109$/MWh—the calibrated central value, roughly 55% above the 2022 estimate of \$70/MWh.

9.2 Viability threshold $\bar{C}_M$

Table 5 shows how the viability threshold scales with the updated mining margin $R - c$.

Current Antminer S21+ machines (16.5 J/TH efficiency) cost \$25–35/TH, or approximately \$2,500–3,500/MW of compute. Amortized over a 2–3 year hardware lifecycle, $C_M \approx$ \$1,000–1,750/MW-year—well within the calibrated threshold, confirming strong current viability.

Table 5: Triple-Win Threshold $\bar{C}_M$: 2022 versus 2026 Calibration

Parameter	**2022 estimate**	**2026 calibration**
R (mining revenue/MWh)	\$70	\$109
c (fossil variable cost)	\$45	\$45 (stable)
$R - c$ (mining margin)	\$25/MWh	\$64/MWh
$\bar{C}_M$ (ASIC threshold)	≈\$23/MWh	≈\$60/MWh
Viability range in C_M	narrow	$\approx 2.5\times$ wider

This table presents the triple-win threshold $\bar{C}_M \equiv (R-c)\frac{\beta+p-c-C_F}{\beta+p-c}$ under 2022 and 2026 market conditions. The mining margin $R - c$ of \$64/MWh in the calibration implies a viable range of hardware costs C_M roughly 2.5 times wider than in 2022. Other parameters: $\beta + p =$ \$200/MWh, $C_F =$ \$10/MWh. The value $\beta + p =$ \$200/MWh is far below typical value-of-lost-load estimates (e.g., ∼\$9,000/MWh in ERCOT); we use a conservative value because $\bar{C}_M$ is insensitive to $\beta+p$ once $\beta+p \gg c+C_F$ (as $\phi^\circ \to 1$).

9.3 Renewable and fossil capacity costs

Table 6 presents renewable and fossil capital cost data. Renewable CapEx has fallen roughly 40–50% since 2020, reducing the parameter C_R in the model and shifting the jointly optimal renewable capacity K_R^* sharply upward. This implies more surplus hours, more zero-carbon computing, and a stronger capacity substitution effect.

9.4 Summary: Calibration and implications

Current market conditions strongly support the model's conclusions. First, mining revenue R has risen roughly 55% (from \$70 to \$109/MWh), expanding the viable ASIC capital cost range $\bar{C}_M$ by roughly 2.5-fold. Second, renewable CapEx C_R has fallen roughly 40%, making high-K_R^* scenarios economically realistic and dramatically increasing surplus hours. Third, ASIC efficiency has improved approximately 12% per year (J/TH), reducing effective C_M. Together, the model's calibrated carbon reduction estimates likely understate current potential by a factor of 2–3.

A cross-sectional observation from the calibration exercise is also worth noting. The

Table 6: Renewable and Fossil Capacity Costs: 2020 versus 2025–26

Parameter	**Typical 2020**	**Current (2025–26)**
Solar PV CapEx (utility scale)	\$800–1,200/kW	\$388–600/kW
Onshore wind CapEx	\$1,200–1,600/kW	\$850–1,000/kW
CCGT gas CapEx	\$1,000–1,200/kW	\$800–1,000/kW
Solar LCOE	\$40–60/MWh	\$28–66/MWh
Onshore wind LCOE	\$35–55/MWh	\$25–70/MWh

This table presents representative capital expenditure (CapEx) ranges and levelized cost of energy (LCOE) for utility-scale solar PV, onshore wind, and combined-cycle gas turbines (CCGT), comparing 2020 benchmarks to current (2025–26) values. Renewable CapEx has declined approximately 40–50% since 2020, corresponding to a fall in the model parameter C_R (renewable capital cost per MW). A lower C_R shifts the jointly optimal renewable capacity K_R^* upward, increasing surplus hours and strengthening the capacity-substitution channel. Sources: IRENA, NREL, Lazard LCOE.

quantitative benefit of FCL adoption scales with the *variance of residual demand* $\tilde{x}$: in regions where renewable supply and consumer demand are poorly synchronized, the shock-absorbing role of mining is more valuable and the carbon reduction larger. Conversely, in grids where supply and demand move closely together, the FCL mechanism adds less value. This suggests that FCL programs will be most impactful in grids with volatile, intermittent renewables—precisely the environments where fossil backup costs are highest.

9.5 Sensitivity to Bitcoin price volatility

A key concern about the calibration is the extreme volatility of Bitcoin prices. Since mining revenue per MWh is $R = \frac{P_{\text{BTC}} \cdot b}{E_{\text{block}}}$ (BTC price times block reward divided by energy per block), a halving of the BTC price roughly halves R, compressing the mining margin $R - c$ and narrowing the viable ASIC cost range $\bar{C}_M$. Table 7 quantifies this dependence across the full historical price range, holding the post-2024 block reward at 3.125 BTC and global mining power at 16,300 MW (so energy per block $E_{\text{block}} \approx 2{,}717$ MWh).

Table 7: Sensitivity of Mining Revenue to Bitcoin Price

BTC price	R **(\$/MWh)**	$R - c$	$\frac{\bar{C}_M}{\bar{C}_M^{\text{base}}}$	**FCL viability**
\$20,000	\$23	−\$22	—	Not viable ($R < c$)
\$35,000	\$40	−\$5	—	Not viable ($R < c$)
\$45,000	\$52	+\$7	0.11	Marginally viable
\$60,000	\$69	+\$24	0.38	Viable (narrow range)
\$75,000	\$86	+\$41	0.64	Viable
\$95,000	\$109	+\$64	1.00	Viable (baseline)
\$120,000	\$138	+\$93	1.45	Strongly viable
\$150,000	\$172	+\$127	1.98	Very strongly viable

$c = \$45$/MWh (fossil variable cost). R is computed as $\frac{P_{\text{BTC}} \times 3.125\,\text{BTC}}{2{,}717\,\text{MWh}}$. $\frac{\bar{C}_M}{\bar{C}_M^{\text{base}}}$ is normalized to the baseline ($R = \$109$/MWh, BTC = \$95,000) value. A dashed entry ($---$) indicates $R < c$, so FCL dispatch is never profitable and $\bar{C}_M < 0$.

Break-even Bitcoin price. Setting $R = c = \$45$/MWh and solving for $P_{\text{BTC}}^{\min}$:

$$P_{\text{BTC}}^{\min} = \frac{c \cdot E_{\text{block}}}{b} = \frac{45 \times 2{,}717}{3.125} \approx \$39{,}000.$$

Below this threshold, the FCL dispatch rule never activates—mining on spare fossil capacity is unprofitable—and all model results collapse to the no-FCL benchmark. At 2024–2026 BTC prices (roughly \$50,000–\$100,000), the mining margin is positive throughout, confirming that the calibrated viability range is not a knife-edge result.

Endogenous hash rate as a stabilizer. Table 7 understates the resilience of R to price declines, because hash rate responds endogenously to BTC price: when prices fall, less efficient miners exit, reducing global energy consumption per block and thereby *raising* R relative to the fixed-hash-rate calculation. In bear markets, network difficulty adjusts downward within two weeks, partially offsetting the price decline in R. Empirically, R has historically exhibited a coefficient of variation roughly 40–50% lower than the BTC price

itself over multi-month horizons, making the mining margin $R - c$ considerably more stable than raw price volatility would suggest.

Implications for triple-win robustness. Three conclusions follow from this sensitivity analysis. First, the model's qualitative predictions are robust for any BTC price above roughly \$45,000 (post-halving): the triple-win threshold $\bar{C}_M$ is positive, and current ASIC hardware costs ($C_M \approx$ \$1,000–1,750/MW-year) fall well within the viable range. Second, the quantitative size of $\bar{C}_M$ scales roughly linearly with the mining margin $R - c$, so the welfare gains from FCL adoption grow as BTC prices rise—meaning the program is self-reinforcing in bull markets. Third, bear-market periods (BTC below \$45,000) represent a genuine sustainability concern: FCL operators would halt voluntary curtailment since dispatching on fossil reduces their margins, weakening the demand response channel. A well-designed FCL contract should therefore include *price floors* that guarantee a minimum R payment to the FCL operator, insulating the demand-response commitment from spot BTC price volatility.

9.6 Implications for AI data center loads

For AI training with curtailable fraction $\theta = 0.5$ and training-compute revenue $R^{\text{AI}} =$ \$200/MWh (reflecting the commercial value of frontier model training), the effective margin is $\theta R^{\text{AI}} - c = 0.5 \times 200 - 45 =$ \$55/MWh—roughly comparable to the current Bitcoin mining margin. However, at current inferred R^{AI} values the opacity problem means this revenue cannot be directly contracted, making practical Demand Response Program implementation with AI training loads substantially more challenging than with Bitcoin mining.

10 Conclusion

This paper has developed and analyzed an analytical model of flexible computing loads (FCLs) as a mechanism for reducing carbon emissions through demand response in electricity markets. The core result is that, under economically plausible conditions, integrating FCLs via a demand response program simultaneously increases utility profit, reduces carbon emissions, and enables more renewable capacity investment—a triple win that critically relies on the capacity substitution channel.

In sum, our main analytical results can be summarized as follows.

In the *baseline simultaneous optimization*, the mining preference condition $C_M < \bar{C}_M = (R - c)\phi^\circ$ is both necessary and sufficient for the profit and reliability gains; the carbon reduction additionally requires the distributional conditions characterized in Proposition 4 (for example, it holds under the uniform distribution when renewable capital costs are sufficiently low). The threshold $\bar{C}_M$ is informative: it equals the per-MWh profit margin times the probability of non-blackout, reflecting the expected revenue per MW of computing hardware.

Timing variations reveal that the carbon improvement channel critically depends on the endogeneity of fossil capacity. A **crucial caveat** is that when both fossil and renewable capacities are fixed in the short run (Case C), the demand response program *raises* carbon without exception. FCLs dispatch on spare fossil power, increasing the plant's utilization rate, and no capacity-substitution force offsets this effect (formally, $\Delta\text{Carbon} = c \cdot \mathbb{E}\Big[\min\{(\tilde{x} + K_M^*)^+, K_F^{(0)}\} - \min\{\tilde{x}^+, K_F^{(0)}\}\Big] > 0$; see Section 5.3). This is not a knife-edge result: it holds for every interior $K_M^* > 0$ and every demand distribution. The long-run triple win and the short-run carbon cost are two sides of the same mechanism: the FCL revenue that raises fossil utilization today also finances renewable expansion tomorrow. Evaluating FCL programs over a time horizon short enough to exclude capacity adjustment will

therefore record a carbon cost, not a carbon benefit—a point that should be made explicit in any policy discussion of demand response programs for computing loads. Where short-run carbon neutrality is required (e.g., interim emissions targets), a *renewable-only dispatch* requirement—capping FCL dispatch at the realized renewable surplus $(-\tilde{x})^+$—eliminates the short-run carbon cost, though it narrows the FCL operating window and reduces program profitability.

Rate-of-return regulation introduces a double win: computing revenue reduces the required retail electricity price under ROR pass-through, benefiting consumers directly in addition to subsidizing renewable capacity. The Averch–Johnson over-investment in capital inadvertently benefits consumers through higher reliability, and computing revenue amplifies this effect.

Deregulated competitive markets replicate the regulated outcomes with the market's scarcity price replacing the regulatory penalty. Computing loads alleviate the missing money problem facing *renewable* generators—by converting zero-price surplus hours into fossil-marginal hours at price c, they raise renewable revenue and attract additional renewable investment—and are complementary to carbon taxes, which endogenously tilt computing toward zero-carbon hours.

Comparing Bitcoin mining and AI training, Bitcoin mining dominates on the three key flexibility dimensions: temporal flexibility (memoryless hashing), revenue transparency (publicly verifiable block rewards), and hardware modularity (scalable from kilowatts to gigawatts). AI training potentially offers higher revenue and spatial rerouting flexibility, but is constrained by inference latency and revenue opacity.

Quantitatively, current market conditions (Bitcoin near \$95,000, post-halving rewards, dramatically lower renewable CapEx) imply a viable range for FCL adoption approximately 2.5 times wider today than under 2022 conditions.

Policy implications. The analysis suggests several policy actions. Utility commissions should consider streamlined rate treatment for demand response programs that integrate FCL operators, particularly given the double win under ROR regulation. Regulators should evaluate FCL program timelines over a multi-year horizon to allow the capacity substitution channel to operate. Carbon tax levels above the mining margin $R - c$ would endogenously restrict computing to renewable surplus hours, a simpler and more robust mechanism than prescriptive renewable-only dispatch requirements. Finally, given the difficulty of verifying AI training revenues, blockchain-based computing (Bitcoin mining or similar transparent-revenue loads) is currently the more practical FCL candidate for publicly accountable demand response programs.

Limitations and future research. Several important extensions remain. First, stochastic BTC prices (correlated or uncorrelated with electricity prices) introduce real-options considerations in both the FCL investment decision and the renewable capacity choice. Second, multi-grid competition with endogenous hash-rate pricing creates a strategic interaction not captured in the price-taking model. Third, a continuous-time model with capital adjustment costs and learning curves would capture the dynamic aspects of the renewable transition. Fourth, empirical validation—particularly using ERCOT panel data on Bitcoin miners' curtailment behavior and grid outcomes—would provide direct tests of the model's comparative-static predictions.

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A Notation Summary

Table 8 collects all model variables and parameters. Subscripts R, F, M denote *renewable*, *fossil*, and *mining* throughout. Superscripts * and $^\circ$ mark the FCL (joint) optimum and the

no-FCL benchmark, respectively.

Table 8: Notation summary

Symbol	Description
(A) Capacity choices (MW)	
K_R	Renewable (solar/wind) installed capacity
K_F	Fossil backup installed capacity
K_M	Flexible computing load (FCL/mining) capacity
(B) Capital costs ($ per MW)	
C_R	Amortized capital cost of renewable capacity
C_F	Amortized capital cost of fossil capacity
C_M	Amortized capital cost of FCL capacity
(C) Optimality superscripts	
*	FCL-inclusive (joint) social or private optimum
$^\circ$	No-FCL benchmark optimum
$^{(0)}$	Pre-installed (legacy) capacity level
^{AJ}	Averch–Johnson regulated optimum
^{NE}	Nash equilibrium (multi-grid FCL programs)
^{G}	Global monopoly outcome
(D) Dispatch and demand variables	
$\tilde{x}$	Residual demand $\tilde{x} \equiv \tilde{D} - K_R\tilde{S}$
$d(x)$	FCL dispatch at residual demand x
$\tilde{D}$	Stochastic consumer (non-computing) electricity demand
$\bar{D}$	Upper bound of demand support
$\tilde{S}$	Stochastic renewable capacity factor, $\tilde{S} \in [0, 1]$

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Table 8 (continued)

Symbol	Description
(E) Prices and revenues	
p	Regulated retail electricity price
c	Fossil variable cost (per MWh)
R	Computing revenue per MWh of FCL electricity consumed
R^{AI}	Per-MWh compute value of AI training (Section 8)
θ	Curtailable fraction of AI data-center load (Section 8)
P_{cap}	Scarcity price cap in deregulated markets (Section 6.2)
β	Blackout penalty per unit of unserved demand
$\bar{V}$, VOLL	Consumer gross value per MWh; value of lost load (Appendix F.3)
σ	Social cost of carbon per MWh of fossil output (Appendix F.3)
τ	Carbon tax (per MWh fossil generation)
$\bar{\tau}$	Critical carbon tax threshold ($\bar{\tau} = R - c$)
$\bar{r}$	Allowed rate of return (Averch–Johnson regulation)
(F) Distribution functions and reliability	
$F(x; K_R)$	CDF of residual demand $\tilde{x}$ given renewable capacity K_R
$f(x; K_R)$	PDF of residual demand given K_R
ϕ	Reliability: probability of no blackout, $F(K_F; K_R)$
ϕ^*	Reliability at the FCL optimum
ϕ°	Reliability at the no-FCL benchmark
ϕ^{AJ}	Reliability at the Averch–Johnson optimum
(G) Profit and welfare	
Π	Utility expected profit
Π^*	Utility profit at the FCL optimum

continued on next page

Table 8 (continued)

Symbol	Description
Π°	Utility profit at the no-FCL benchmark
q°	Probability of renewable surplus at the no-FCL optimum, $F(0; K_R^\circ)$
(H) Game-theoretic and aggregate quantities	
N	Number of grids operating FCL programs (Appendix C)
n	Number of (symmetric) miners (Remark 7)
m	Out-of-grid mining power, MW (Appendix C)
$Z \equiv NK_M^{NE}$	Total in-grid FCL at the Nash equilibrium
η	In-grid share of global mining power (Appendix C)
ρ	Utility cost of capital
$\bar{C}_M$	Threshold FCL capital cost for adoption ($\bar{C}_M = (R-c)\phi^\circ$)
$\bar{C}_M^{\mathrm{AJ}}$	FCL adoption threshold under Averch–Johnson regulation

B Proofs

B.1 Proof of Proposition 1

Step 1: Profit comparison. Since (K_R^*, K_F^*, K_M^*) is the jointly optimal solution, by revealed preference $\Pi^* \geq \Pi$ evaluated at any feasible point, including $(K_R^\circ, K_F^\circ, K_M = 0)$ (the no-FCL optimum). Therefore $\Pi^* \geq \Pi^\circ$. The inequality is strict if and only if $K_M^* > 0$. To determine when $K_M^* > 0$, evaluate the marginal profit of the *first unit* of FCL capacity at the *no-FCL optimum* (K_R°, K_F°):

$$\left.\frac{\partial \Pi}{\partial K_M}\right|_{(K_R^\circ, K_F^\circ, 0)} = (R-c)\cdot F(K_F^\circ;\, K_R^\circ) - C_M = (R-c)\cdot\phi^\circ - C_M = \bar{C}_M - C_M,$$

using $\phi^\circ = F(K_F^\circ; K_R^\circ) = \frac{\beta+p-c-C_F}{\beta+p-c}$ from the no-FCL FOC (7). This marginal profit is positive iff $C_M < \bar{C}_M$, confirming that the utility strictly gains from introducing the FCL program. Hence $\Pi^* > \Pi^\circ$ iff $C_M < \bar{C}_M$.

Step 2: Blackout probability comparison. The equilibrium blackout probabilities are:

$$1 - F(K_F^\circ; K_R^\circ) = \frac{C_F}{\beta + p - c}, \qquad 1 - F(K_F^*; K_R^*) = \frac{c + C_F + C_M - R}{\beta + p - R}.$$

The difference $(1 - F^*) - (1 - F^\circ)$ equals

$$\frac{c + C_F + C_M - R}{\beta + p - R} - \frac{C_F}{\beta + p - c} = \frac{C_M(\beta + p - c) - (R - c)(\beta + p - c - C_F)}{(\beta + p - R)(\beta + p - c)}.$$

The denominator is positive (both $\beta + p > R$ and $\beta + p > c$). The numerator is negative iff $C_M < (R - c)\frac{\beta+p-c-C_F}{\beta+p-c} = \bar{C}_M$. Hence $1 - F(K_F^*; K_R^*) < 1 - F(K_F^\circ; K_R^\circ)$ iff $C_M < \bar{C}_M$. $\square$ (blackout probability is lower with FCL)

Step 3: Carbon comparison.

Derivation of expected fossil output. By Lemma 1, fossil plant output in state x under the FCL program equals $x + K_M^*$ for $x \in [-K_R^*, K_F^* - K_M^*]$ and K_F^* for $x > K_F^* - K_M^*$. Integrating by parts:

$$\begin{aligned}\mathbb{E}[\text{fossil with FCL}] &= \int_{-K_R^*}^{K_F^*-K_M^*} (x + K_M^*)\, f(x; K_R^*)\, \mathrm{d}x + K_F^* \Big[1 - F(K_F^* - K_M^*; K_R^*)\Big] \\ &= K_F^* - \int_{-K_R^*}^{K_F^*-K_M^*} F(x;\, K_R^*)\, \mathrm{d}x. \end{aligned} \tag{42}$$

Without the FCL program, fossil output in state x is $\min\{x^+, K_F^\circ\}$, giving by the same integration-by-parts identity:

$$\mathbb{E}[\text{fossil without FCL}] = K_F^\circ - \int_0^{K_F^\circ} F(x;\, K_R^\circ)\, \mathrm{d}x. \tag{43}$$

Utilization effect is positive (distribution-free). Fix any capacities (K_R, K_F, K_M) with $K_M > 0$. Comparing fossil output state by state, the FCL program *weakly raises* output in every state and strictly so for all $x < K_F$: in intermediate states $x \in (K_F - K_M, K_F)$ the plant runs at K_F instead of $x < K_F$; in surplus states $x \leq K_F - K_M$ the plant runs at $x + K_M > \max\{x, 0\}$ instead of x^+. Hence $\mathbb{E}[\text{fossil}|(K_R, K_F, K_M)] > \mathbb{E}[\text{fossil}|(K_R, K_F, 0)]$ for any F satisfying Assumption 1. This utilization effect is *entirely distribution-free.*

Capacity substitution effect and net comparison (uniform distribution). The net carbon change compares (42) at (K_R^*, K_F^*, K_M^*) with (43) at (K_R°, K_F°). Because $R > c$, the FOC for renewable capacity (15) assigns a higher shadow value (R vs. c) to each MW of renewable investment in the FCL dispatch region, pushing toward $K_R^* > K_R^\circ$. This additional renewable capacity is the capacity-substitution channel. The quantitative comparison — showing that the capacity-substitution channel dominates when C_R is sufficiently small — is established in Section 4.3 via (25)–(26). □

Remark 9 (Scope of distributional assumptions in Proposition 1)**.** Steps 1 and 2 of the proof (profit and blackout comparisons) are entirely distribution-free: they rely only on the equilibrium CDF values $F(K_F^*; K_R^*)$ and $F(K_F^\circ; K_R^\circ)$, which are determined by the FOCs independently of the distributional shape. Step 3 (carbon) decomposes into two parts. The *utilization effect*—the carbon-increasing force from higher fossil utilization at fixed capacities—is also distribution-free, as shown above. The *capacity substitution effect*—the carbon-decreasing force from the endogenous increase in renewable capacity $K_R^* > K_R^\circ$—uses the uniform distribution to evaluate the relevant integrals in closed form (Section 4.3). The triple-win threshold $\bar{C}_M = (R-c)\phi^\circ$ is itself distribution-free (it depends only on the equilibrium reliability ϕ°); what the uniform distribution provides is a tractable closed-form setting in which the net carbon comparison can be signed when C_R is sufficiently low. Proposition 4 below characterizes exactly which distributional properties drive the carbon result.

Proposition 4 (Carbon condition: general characterization)**.** *For any residual-demand CDF F satisfying Assumption 1, define*

$$\mu^*_{low} \equiv \frac{\int_{-K^*_R}^{K^*_F-K^*_M} x\, f(x;K^*_R)\,\mathrm{d}x}{F(K^*_F-K^*_M;\,K^*_R)} = \frac{(R-c)}{C_M}\int_{-K^*_R}^{K^*_F-K^*_M} x\, f(x;K^*_R)\,\mathrm{d}x,$$

$$\mu^\circ_{mid} \equiv \frac{\int_0^{K^\circ_F} x\, f(x;K^\circ_R)\,\mathrm{d}x}{F(K^\circ_F;K^\circ_R)-F(0;K^\circ_R)} = \frac{1}{\phi^\circ-q^\circ}\int_0^{K^\circ_F} x\, f(x;K^\circ_R)\,\mathrm{d}x,$$

where $q^\circ \equiv F(0;K^\circ_R)$ is the probability of renewable surplus at the no-FCL optimum. Then expected fossil output satisfies

$$\mathbb{E}[\text{fossil with FCL}] = K^*_F - \tfrac{C_M}{R-c}\Big(K^*_F-K^*_M-\mu^*_{low}\Big), \tag{44}$$

$$\mathbb{E}[\text{fossil without FCL}] = K^\circ_F(1-\phi^\circ) + (\phi^\circ-q^\circ)\,\mu^\circ_{mid}. \tag{45}$$

Carbon falls with the FCL program if and only if

$$\frac{C_M}{R-c}\Big(K^*_F-K^*_M-\mu^*_{low}\Big) \;>\; K^*_F - K^\circ_F(1-\phi^\circ) - (\phi^\circ-q^\circ)\,\mu^\circ_{mid}. \tag{C}$$

Under the uniform distribution $F(x;K_R) = \frac{x+K_R}{D+K_R}$, the conditional means equal interval midpoints:

$$\mu^*_{low} = \tfrac{K^*_F-K^*_M-K^*_R}{2}, \qquad \mu^\circ_{mid} = \tfrac{K^\circ_F}{2},$$

and (C) can be evaluated in closed form; Section 4.3 shows that it holds when the renewable capital cost C_R is sufficiently low.

Proof. Apply integration by parts to each fossil integral. For (44), use $\int_a^b (1-F)\,dx = [x(1-F)]_a^b + \int_a^b xf\,dx$ with $a=-K^*_M$, $b=K^*_F-K^*_M$, $F(a;K^*_R)=0$ (large-FCL regime: $-K^*_M$ lies below the support), and $F(b;K^*_R) = C_M/(R-c)$. For (45), apply the same identity with $a=0$, $b=K^\circ_F$, and $F(K^\circ_F;K^\circ_R)=\phi^\circ$. (C) is obtained by subtracting and

rearranging. □

Corollary 2 (Sufficient distributional conditions for the triple win)**.** *The carbon part of the triple win (Proposition 1 (ii)) holds for all $C_M < \bar{C}_M$ under any distribution satisfying both:*

(a) $F(\cdot; K_R^*)$ is concave *on $(-K_R^*,\ K_F^* - K_M^*)$, equivalently $f(\cdot; K_R^*)$ is non-increasing there; and*

(b) $F(\cdot; K_R^\circ)$ is convex *on $(0,\ K_F^\circ)$, equivalently $f(\cdot; K_R^\circ)$ is non-decreasing there.*

Under (a)*: $\mu_{low}^* \leq (K_F^* - K_M^* - K_R^*)/2$, so FCL fossil output is no greater than under the uniform. Under* (b)*: $\mu_{mid}^\circ \geq K_F^\circ/2$, so no-FCL fossil output is no less than under the uniform. Together,* (C) *holds at least as strongly as under the uniform distribution, and the threshold $\bar{C}_M$ is preserved. The uniform distribution satisfies both conditions with equality (linear CDF, constant density). Conditions* (a) *and* (b) *are evaluated at the respective equilibrium capacities; the comparison holds the distribution-free FOC-pinned cutoff probabilities fixed, while a full comparison would also track how the equilibrium capacities themselves vary with F.*

Remark 10 (Economic interpretation of conditions (a) and (b))**.** Condition (a) requires the residual-demand density to be non-increasing within the FCL full-dispatch zone $(-K_R^*, K_F^* - K_M^*)$: deep renewable surplus is more likely than moderate surplus. Economically, the FCL mostly runs on zero-cost renewable power in this zone, limiting the utilization effect on carbon. Condition (b) requires the density to be non-decreasing within the no-FCL fossil zone $(0, K_F^\circ)$: high consumer demand (close to K_F°) is more likely than low demand. Economically, the no-FCL baseline already burns substantial fossil fuel, making it easier for the FCL program to achieve a net carbon reduction.

When $K_F^* - K_M^* < 0$ (which obtains when FCL capacity exceeds fossil capacity): the two zones are disjoint. In this case, conditions (a) and (b) impose restrictions on *different parts of the support* and can be simultaneously satisfied without conflict. Specifically, any unimodal distribution whose mode lies in the positive-demand region satisfies a version of

both conditions when the full-dispatch zone lies entirely in the surplus half of the support. The uniform distribution is the *boundary case*: it satisfies both with equality for all parameter values.

C Global Mining and Nash Equilibrium Extension

This appendix formalizes the multi-grid coordination problem when N grid operators simultaneously run FCL programs, endogenizing the per-MWh mining revenue R.

C.1 Setup

Let grid i operate an FCL program of size δ_i (MW), and measure all mining in MW of power. Total network mining power is $\sum_i \delta_i + m$, where m is the mining power of out-of-grid miners. Since rewards are shared in proportion to hash rate—hence, for given hardware efficiency, in proportion to power—every miner earns a common per-MWh revenue

$$R \;=\; \frac{R_0}{\sum_i \delta_i + m},$$

where R_0 denotes the total mining reward value per hour (block reward value divided by the block time in hours). Each grid's program δ_i thus dilutes R for all participants. In the Nash equilibrium each grid takes other grids' Demand Response Program sizes as given.

C.2 Nash equilibrium and coordination failure

Proposition 5 (Multi-grid Nash equilibrium: existence, uniqueness, and inefficiency)**.** *In the symmetric Nash equilibrium with N identical grids and out-of-grid miners of aggregate*

capacity m, each grid runs an FCL program of size K_M^{NE} satisfying

$$(R^{NE} - c) \cdot F(K_F^{NE} - K_M^{NE};\, K_R^{NE}) = C_M, \qquad R^{NE} = \frac{R_0}{NK_M^{NE} + m}. \tag{46}$$

This symmetric Nash equilibrium exists and is unique. Moreover, $K_M^{SP} < K_M^{NE} < K_M^*$: equilibrium programs are smaller than the price-taking benchmark K_M^* (which ignores the dilution of R), yet larger than the coordinated optimum K_M^{SP} that internalizes the cross-grid revenue externality—adoption is individually rational but collectively excessive.

Proof of Proposition 5. Existence. Define the symmetric best-response function $K_M^*(Z_-)$ as the solution to each grid's private FOC when aggregate out-of-grid-plus-rival FCL is Z_-:

$$\Big(R(Z_- + K_M) - c\Big) \cdot F(K_F^\circ(K_M, Z_-) - K_M;\, K_R^\circ(K_M, Z_-)) = C_M,$$

where $R(Z_- + K_M) = \frac{R_0}{Z_- + K_M}$ and K_F°, K_R° are the jointly optimal capacities at given (K_M, R). By continuity of the best response in Z_- and a fixed-point argument, a symmetric equilibrium $K_M^{NE} = K_M^*(m + (N-1)K_M^{NE})$ exists.

Uniqueness. Let $Z \equiv NK_M^{NE}$ denote total in-grid FCL. We maintain the following condition for the interior Nash equilibrium to exist:

$$\left(\frac{R_0}{m} - c\right) \cdot F(K_F^\circ(R_0/m); K_R^\circ(R_0/m)) > C_M, \tag{Nash-E}$$

i.e., the single-grid FCL program is individually profitable when in-grid mining is negligible ($Z \to 0$, so $R^{NE} \to R_0/m$); this is the Nash analogue of the triple-win condition $C_M < \bar{C}_M$ in the price-taking baseline.

In the symmetric equilibrium, each grid's FOC (46) can be written as

$$L(Z) \;\equiv\; \left(\frac{R_0}{Z + m} - c\right) \cdot \Phi\Big(\frac{Z}{N}\Big) \;=\; C_M, \tag{47}$$

where $\Phi(K_M) \equiv F(K_F^\circ(K_M, R^{NE}(K_M)) - K_M;\, K_R^\circ(K_M, R^{NE}(K_M)))$ is the equilibrium value of the CDF factor. $L(Z)$ is strictly decreasing in Z for two reasons. First, the per-MWh revenue $R^{NE}(Z) \equiv \frac{R_0}{Z+m}$ is strictly decreasing in Z: $\frac{\partial R^{NE}}{\partial Z} = -\frac{R_0}{(Z+m)^2} < 0$, since more aggregate in-grid FCL dilutes the per-MWh reward. Second, $\Phi\Big(\frac{Z}{N}\Big)$ moves with R^{NE} in a direction that reinforces the decrease: from the K_F° FOC, $F(K_F^\circ; K_R^\circ) = \frac{\beta+p-c-C_F}{\beta+p-R^{NE}}$ is increasing in R^{NE}, so it falls as Z rises, pulling Φ downward as well. Hence both factors of $L(Z) = (R^{NE}(Z) - c) \cdot \Phi\Big(\frac{Z}{N}\Big)$ are strictly decreasing in Z, so L itself is strictly decreasing. By condition (Nash-E), $L(0^+) = \Big(\frac{R_0}{m} - c\Big)\Phi(0) > C_M$; since $L(\infty) \to (-c) \cdot \Phi < 0 < C_M$, the intermediate value theorem together with strict monotonicity of L gives exactly one solution Z^{NE}, hence a unique $K_M^{NE} = \frac{Z^{NE}}{N}$.

Inefficiency: $K_M^{SP} < K_M^{NE} < K_M^*$. The price-taking benchmark K_M^* satisfies $\Big(\frac{R_0}{m} - c\Big) \cdot \Phi(K_M^*) = C_M$ (approximately, when $NK_M^* \ll m$; see the single-grid FOC (11)). In the Nash equilibrium the per-MWh revenue is $R^{NE} = \frac{R_0}{NK_M^{NE}+m} < \frac{R_0}{m}$, since aggregate in-grid FCL dilutes the per-MWh mining return. Because L is strictly decreasing, the lower effective revenue means $L(Z) = C_M$ is satisfied at a smaller program: $K_M^{NE} < K_M^*$. For the coordinated optimum, a planner choosing a common program size for all N grids internalizes that expanding any grid's program dilutes the per-MWh revenue earned on *all* grids' inframarginal dispatch. The planner's FOC therefore subtracts the cross-grid externality $\mathcal{E} > 0$ of equation (49) from the private marginal value: $(R^{SP} - c)\,\Phi - \mathcal{E} = C_M$. Since the planner's marginal value lies strictly below the private one at every Z and both are decreasing, the planner stops expanding earlier: $K_M^{SP} < K_M^{NE}$. □

Remark 11. The coordination failure is modest when grids are small relative to the global hash market ($NK_M^{NE} \ll m$, so $\eta \equiv \frac{NK_M^{NE}}{NK_M^{NE}+m} \approx 0$), as is the case currently for most individual jurisdictions. It becomes quantitatively important if the U.S. as a whole acts as a single bloc (effectively $N = 1$, $\eta \approx 0.40$).

C.3 Welfare loss from coordination failure

The Nash equilibrium is collectively inefficient: each grid takes R^{NE} as given from its own perspective, but ignores that its FCL expansion dilutes other grids' mining revenue. This section characterizes the welfare cost of this coordination failure.

Per-unit coordination externality. When grid i expands its FCL by one unit, it reduces the per-MWh revenue of all miners by $|\partial R/\partial \delta_i| = R_0/(NK_M + m)^2$, lowering the mining revenue of every other grid $j \neq i$ by:

$$\varepsilon_{ij} \;=\; -\frac{\partial R}{\partial \delta_i} \cdot \mathbb{E}[d_j] \;=\; \frac{R_0}{(NK_M + m)^2} \cdot \mathbb{E}[d_j] \;=\; \frac{\eta}{N} \cdot R^{NE} \cdot \frac{\mathbb{E}[d_j]}{K_M^{NE}}, \tag{48}$$

where $\eta \equiv \frac{NK_M^{NE}}{NK_M^{NE}+m}$ is the in-grid share of global mining power. Summing across all $N-1$ rival grids, the total revenue externality per unit of FCL expansion by grid i is:

$$\mathcal{E} \;=\; (N-1) \cdot \varepsilon_{ij} \;=\; \frac{(N-1)\,\eta}{N} \cdot R^{NE} \cdot \frac{\mathbb{E}[d^{NE}]}{K_M^{NE}}. \tag{49}$$

Grid i ignores this externality in the Nash equilibrium: its FOC equates the private marginal value of FCL capacity to C_M, whereas the planner's FOC equates it to $C_M + \mathcal{E}$—the hardware cost plus the revenue dilution imposed on other grids. The planner therefore stops expanding earlier: $K_M^{SP} < K_M^{NE}$ (Proposition 5).

Quantitative calibration. Using the calibrated values ($R^{NE} \approx$ \$100/MWh, $\frac{\mathbb{E}[d^{NE}]}{K_M^{NE}} \approx 0.45$, $\eta/N \approx 0.08$ per jurisdiction for $N = 5$ major mining regions, and current U.S. total hash share of $\approx 40\%$):

$$\mathcal{E} \;\approx\; \frac{4}{5} \times 0.08 \times \$100 \times 0.45 \;\approx\; \$2.9 \text{ per MWh of FCL capacity.}$$

This externality is roughly 4% of the per-MWh mining margin $R^{NE} - c \approx \$64/\text{MWh}$, indicating a relatively modest but non-negligible coordination distortion. The aggregate annual revenue-dilution flow across all N grids, at calibrated FCL deployment of $K_M^{NE} \approx 5$ GW per jurisdiction, is

$$N \times \mathcal{E} \times K_M^{NE} \times 8{,}760\,\text{hr/yr} \;\approx\; 5 \times \$2.9 \times 5{,}000 \times 8{,}760 \;\approx\; \$635\text{M per year.}$$

Much of this flow is a transfer among miners (including out-of-grid miners) rather than a deadweight loss; the efficiency cost of over-adoption is second-order, on the order of $\frac{1}{2}\,\mathcal{E}\,(K_M^{NE} - K_M^{SP})$ per grid, where the over-adoption gap satisfies, to first order,

$$K_M^{NE} - K_M^{SP} \;\approx\; \frac{\mathcal{E}}{|L'(Z^{NE})/N|}, \tag{50}$$

with L defined in (47).

An international coordination mechanism that caps program sizes at K_M^{SP} would eliminate the over-adoption externality; given the modest size of $\mathcal{E}$ relative to the mining margin ($\approx$4%), the associated welfare gain is correspondingly modest today, but it grows with the in-grid share η as FCL programs proliferate.

C.4 Single global utility: monopolist case

The sharpest case of endogenous R arises when a single utility ($N = 1$, no out-of-grid miners: $m = 0$) serves the entire global mining fleet. With total block reward $\bar{R}$ per period and competitive mining, the per-unit revenue is

$$R^G = \frac{\bar{R}}{\mathbb{E}[d(\tilde{x}; K_M)]},$$

so aggregate mining income $R^G \mathbb{E}[d] = \bar{R}$ is a constant independent of the program size K_M.

Proposition 6 (Global monopolist)**.** *When a single utility serves all global miners and mining is competitive, the marginal revenue from expanding the FCL program is zero:*

$$\frac{\partial}{\partial K_M}\Big[R^G(K_M)\,\mathbb{E}[d(\tilde{x};K_M)]\Big] = \frac{\partial \bar{R}}{\partial K_M} = 0.$$

The modified first-order condition for K_M becomes

$$-c\,F(K_F - K_M; K_R) = C_M,$$

which has no interior solution when $C_M > 0$ and $c > 0$. The global monopolist therefore sets $K_M^G = 0$: no FCL program is offered.

Proof. In the exogenous-R case, the FOC for K_M is $(R-c)\,F(K_F^* - K_M^*; K_R^*) = C_M$ (equation (11)), where the term $R{\cdot}F(K_F^*-K_M^*; K_R^*)$ is the marginal mining revenue from expanding the FCL program by one unit ($F(K_F^* - K_M^*; K_R^*)$ is the probability of full FCL dispatch, which is the probability the extra unit is actually used). In the global monopolist case, the marginal mining revenue from K_M is $\frac{\partial[R^G(K_M)\,\mathbb{E}[d(\tilde{x};K_M)]]}{\partial K_M} = \frac{\partial \bar{R}}{\partial K_M} = 0$, since total mining income $\bar{R}$ is independent of K_M. Replacing the marginal mining revenue $R\cdot F$ with zero in the exogenous FOC leaves only the cost terms: $-c\,F(K_F^* - K_M^*; K_R^*) = C_M$. Since $c, F, C_M > 0$, this has no solution, and the corner solution $K_M^G = 0$ is the unique optimum. □

Remark 12. With $K_M^G = 0$, the global monopolist's capacity choices (K_R^G, K_F^G) revert to the no-FCL benchmark (Appendix I.1). All three outcomes—blackout probability, carbon emissions, and utility profit—equal their no-FCL levels, and the "triple win" of Proposition 1 is never achieved.

Propositions 5 and 6 together characterize the full range: the FCL program is largest when grids are small relative to global mining (exogenous R, K_M^*), shrinks as any single operator acquires market power ($K_M^{NE} < K_M^*$), and collapses to zero under global monopoly

$(K_M^G = 0)$.

C.5 Partial miner coverage: mixed in-grid and out-of-grid miners

An intermediate case—between the price-taking benchmark and the global monopolist—arises when a grid serves n in-grid miners competing against m out-of-grid miners who face an exogenous electricity price P_o. Let $\mathcal{N}$ and $\mathcal{M}$ denote the in-grid and out-of-grid miner sets, respectively. Each in-grid miner i solves

$$\max_{d_i} \; R \frac{d_i}{d_i + \sum_{j \neq i, j \in \mathcal{N}} d_j + \sum_{k \in \mathcal{M}} d_k} - d_i P,$$

while each out-of-grid miner faces price P_o. Aggregating first-order conditions across all miners yields total network demand

$$\sum_{j \in \mathcal{M} \cup \mathcal{N}} d_j = \frac{(m+n-1)R}{nP + mP_o}, \tag{51}$$

and summing only over in-grid miners gives equilibrium in-grid demand:

$$d \equiv \sum_{j \in \mathcal{N}} d_j = \frac{(m+n-1)R}{nP + mP_o} \cdot n \cdot \frac{mP_o - (m-1)P}{nP + mP_o}. \tag{52}$$

Hence total in-grid miner expenditure is

$$Pd = \frac{(m+n-1)R}{n + \frac{mP_o}{P}} \cdot n \cdot \frac{mP_o - (m-1)P}{nP + mP_o}.$$

When P is close to P_o, this is approximately exogenous (independent of the grid's capacity decisions), recovering the price-taking benchmark. In the limit $\frac{m}{n} \to \infty$ (the focal grid is

negligible relative to the global mining fleet),

$$d \approx \frac{nR}{P_o}\left(1 - \frac{P}{P_o}\right), \qquad Pd \approx nR\,\frac{P}{P_o}\left(1 - \frac{P}{P_o}\right),$$

which is maximized at $P = \frac{P_o}{2}$. Importantly, $d \geq 0$ requires $P \leq P_o$; if the grid sets $P > P_o$ all miners relocate.

Remark 13. Equation (52) highlights a complication absent from the main text: when the focal grid is not negligible (finite $\frac{m}{n}$), total miner revenue Pd is no longer independent of P, so the simplification that mining revenue is exogenous to grid decisions fails. The main-text results remain valid whenever the grid is small relative to the global fleet—the empirically relevant case today—but equation (52) characterizes the correction term for intermediate cases and clarifies the need for multi-grid coordination as FCL programs proliferate.

C.6 Policy implication

In the Nash equilibrium each grid over-adopts relative to the coordinated optimum because it ignores the negative externality its expansion imposes on other grids' revenues. An international coordination mechanism (e.g., agreed limits on FCL program sizes) could improve on the Nash outcome. The price-taking assumption in the main text is accurate for any single grid in the current environment; as FCL programs proliferate, this assumption should be revisited.

D The Case $R \leq c$

The main text assumes $R > c$ (computing revenue per MWh exceeds fossil variable cost), ensuring that FCL operators willingly consume fossil electricity. This appendix analyzes the boundary case $R = c$ and the unlikely case $R < c$.

D.1 Case $R = c$

When $R = c$, the FCL operator is exactly indifferent between running on fossil electricity and not running during fossil-dispatch hours. The FOC for K_M (11) becomes $(R - c) \cdot F = 0 = C_M$, which is satisfied only if $C_M = 0$ (free ASIC hardware). In practice, ASIC hardware always has positive cost, so $R = c$ implies no FCL investment. The demand response program reduces to a pure renewable surplus absorption mechanism (FCLs run only when $\tilde{x} < 0$), which has no effect on fossil capacity decisions but increases renewable investment.

D.2 Case $R < c$

When $R < c$, computing is unprofitable during fossil dispatch hours. No profit-maximizing FCL operator would accept electricity priced at c or above. The FCL program can still be operated if the utility *subsidizes* computing during fossil hours (paying miners $c - R$ per MWh to run), but this changes the economics fundamentally: the Demand Response Program now costs the utility money during fossil hours instead of earning revenue. The revised FOC for fossil capacity becomes:

$$(\beta + p + c - R)\Big[1 - F(K_F; K_R)\Big] = C_F,$$

implying that subsidized FCL programs lead to more fossil capacity built (not less), raising the threshold for the utility's preference condition.

The case $R < c$ therefore represents a qualitatively different regime in which the carbon reduction mechanism is reversed: the utility would need to pay FCL operators to curtail, which is equivalent to a conservation or efficiency program rather than a demand response program. We rule out this case in the main text by Assumption 1.

E Alternative Setup: Utility Serving Third-Party FCL

The baseline model assumes the utility directly operates the FCL program and captures R per MWh of computing output. This appendix develops the alternative in which the utility *sells electricity* to third-party mining operators at the regulated retail tariff p per MWh—the same rate charged to ordinary consumers—and administers FCL curtailment via a Demand Response Program contract, without owning mining hardware.

E.1 Setup

Two strategic agents determine capacity in this alternative:

1. **Utility**: chooses K_R and K_F to maximize profit. Does not own mining hardware; K_M is not a choice variable and does not enter the rate base ($K_M \notin \mathcal{B}$). Charges all electricity customers—including miners—the regulated tariff p per MWh. Operates the FCL curtailment rule (3).
2. **Miners**: invest in computing hardware of capacity K_M at annualized capital cost C_M per MW. Purchase electricity at price p and earn R per MWh of computing output. Accept curtailment under the DRP.

The DRP dispatch rule (3) is unchanged: FCL runs in the spare-capacity window and is curtailed when the grid is stressed ($\tilde{x} > K_F$).

E.2 Utility's Modified Profit Function

Replace R with p in the FCL revenue term of (10) and treat K_M as exogenous:

$$\Pi^p(K_R, K_F;\, K_M) = p\,\mathbb{E}[\tilde{D}] + p\int_{K_F-K_M}^{K_F} F(x; K_R)\,\mathrm{d}x - c\left(K_F - \int_{-K_R}^{K_F-K_M} F(x; K_R)\,\mathrm{d}x\right) - (\beta + p)\int_{K_F}^{\bar{D}} (1 - F(x; K_R))\,\mathrm{d}x - C_F K_F - C_R K_R. \quad (53)$$

Comparing (53) with the no-FCL profit, the incremental utility gain from FCL being on the system is:

$$\Delta\Pi^{p} = (p-c)\int_{K_F-K_M}^{K_F} F(x;K_R)\,\mathrm{d}x \;>\; 0. \tag{54}$$

Since $p > c$, this is strictly positive for any $K_M > 0$: the utility *always* benefits from FCL operators on the system, because miners bring their own hardware and pay p for electricity that would otherwise power idle spare fossil capacity at zero margin. No adoption condition on C_M is required from the utility's perspective.

E.3 Miners' Free-Entry Condition

Miners earn $R-p$ per MWh when dispatched and zero otherwise. Free entry drives expected profit per unit of mining capacity to zero:

$$(R-p)\cdot F(K_F-K_M;\,K_R) = C_M. \tag{55}$$

This is the analogue of the baseline FCL optimality condition (11), with $R-c$ replaced by $R-p$. Miner participation requires $R > p$, which we assume holds. Given the utility's equilibrium K_F and K_R, condition (55) pins the equilibrium mining scale K_M^*.

E.4 Utility's Capacity FOCs and Equilibrium Blackout

Differentiating (53) with respect to K_F and substituting (55) (so that $F(K_F - K_M) = C_M/(R-p)$) yields the equilibrium condition for fossil capacity:

$$\beta\Big(1-F(K_F;\,K_R)\Big) + (p-c)\cdot\frac{R-p-C_M}{R-p} = C_F. \tag{56}$$

This is the analogue of (13): it has the same structure but with $p - c$ replacing $R - c$ in the FCL margin term. Rearranging (56) gives the equilibrium blackout probability:

$$1 - F(K_F^*;\, K_R^*) = \frac{C_F(R-p) - (p-c)(R-p-C_M)}{\beta(R-p)}. \tag{57}$$

A comparison analogous to (16) shows that the blackout probability is strictly lower than in the no-FCL baseline if and only if:

$$C_M \;<\; \bar{C}_M^{p} \;\equiv\; (R-p)\,\phi^{\circ}, \qquad \phi^{\circ} \equiv \frac{\beta + p - c - C_F}{\beta + p - c}. \tag{58}$$

This is the exact analogue of the baseline triple-win threshold $\bar{C}_M = (R-c)\,\phi^{\circ}$, with $R - c$ replaced by $R - p$. Since $R - p < R - c$ (given $p > c$), the viable range of FCL hardware costs is strictly smaller in this alternative: a higher electricity purchase price narrows the miner's margin and therefore the welfare benefit that FCL can deliver.

E.5 Rate-of-Return Regulation

Under ROR regulation, the utility maximizes profit subject to $\Pi^p \leq \bar{r}\,\mathcal{B}$ with $\mathcal{B} = C_F K_F + C_R K_R$ (since $K_M \notin \mathcal{B}$). The binding ROR profit equals $\Pi^{p,\mathrm{AJ}} = \bar{r}(C_F K_F + C_R K_R)$, and the AJ over-investment distortion extends to K_F and K_R only. The utility has no rate-base incentive to expand K_M; equilibrium mining scale is determined entirely by the miner free-entry condition (55) evaluated at the (over-invested) K_F^{AJ}, K_R^{AJ}.

E.6 Summary Comparison

Feature	Baseline (R captured by utility)	Retail-tariff alternative (p charged to miners)
Utility's FCL margin	$R - c$ per MWh	$p - c$ per MWh
Who chooses K_M	Utility	Miners (free entry)
K_M in rate base	Yes (under ROR)	No
Utility always benefits from FCL	Only if $C_M < \bar{C}_M$	Always ($\Delta\Pi^p > 0$)
Miner participation condition	n/a	$C_M < (R-p)F(K_F - K_M)$
Triple-win threshold on C_M	$(R-c)\,\phi^\circ$	$(R-p)\,\phi^\circ$
AJ distortion on K_M	Yes ($\partial\Pi^{\text{AJ}}/\partial K_M = \bar{r}C_M > 0$)	No
AJ distortion on K_F, K_R	Yes	Yes (same logic)
Qualitative directions (blackout, carbon)	As in Propositions 1 and 4	Same, with $R - c \to p - c$

F Alternative Regulatory Objectives

The main text assumes a profit-maximizing regulated utility that internalizes a blackout penalty imposed by the regulator. This appendix considers three alternative regulatory objective functions that are common in practice: (1) minimizing the probability of blackout, (2) minimizing the retail electricity price, and (3) maximizing social welfare.

F.1 Objective 1: Minimizing blackout probability

Suppose the regulator (or the utility under explicit mandate) aims to minimize blackout probability $\Pr(\tilde{x} > K_F; K_R) = 1 - F(K_F; K_R)$ subject to a profit constraint $\Pi \geq \bar{\Pi}$.

Without FCL. Maximizing $F(K_F; K_R)$ subject to the profit constraint yields a corner solution: the utility invests the maximum fossil capacity that breaks even. The resulting blackout probability is lower than under profit maximization for any given K_R, since reliability is the direct objective.

With FCL. Introducing the FCL program changes the profit constraint: computing revenue relaxes it, allowing a higher K_F (more fossil backup) to be built while remaining profitable. The FCL program therefore reduces blackout probability through two channels: (i) direct demand reduction during scarcity events (FCLs shut down when $\tilde{x} > K_F$, exactly analogous to the baseline), and (ii) indirect capacity expansion (higher permitted fossil investment from the relaxed profit constraint).

Proposition 7 (Blackout-minimizing objective)**.** *Under a blackout-minimizing objective with binding profit constraint, allowing FCLs unambiguously reduces blackout probability whenever $C_M < \bar{C}_M$. The threshold $\bar{C}_M$ is unchanged from Proposition 1.*

Proof. The regulator minimizes $1 - F(K_F; K_R)$ subject to the utility earning at least a fixed profit $\bar{\Pi}$. At the constraint boundary, the objective equals $1 - F(K_F; K_R)$ evaluated at the profit-constrained optimum. Introducing FCLs relaxes the profit constraint by adding computing revenue $R \cdot \mathbb{E}[d] > 0$ to the utility's income (whenever $C_M < \bar{C}_M$, so $K_M^* > 0$ and the program is profitable). The relaxed constraint allows the regulator to demand higher fossil capacity K_F (more reliability) at the same profit level. Formally, the Lagrangian for the reliability-minimizing problem includes the profit constraint multiplier; when the constraint binds, the shadow value of computing revenue enters the regulator's choice of K_F identically to the profit-maximization FOC (13). Hence the blackout-minimizing optimum satisfies the same condition $C_M < \bar{C}_M$ for a reliability improvement, as in Proposition 1. □

The result confirms that the triple win (profit, carbon, reliability) survives even when reliability is the primary objective—as long as the profit constraint remains binding.

F.2 Objective 2: Minimizing retail electricity price under ROR regulation

Under ROR regulation, the regulator sets the retail price to allow a fair return. Alternatively, the regulator may aim to minimize the retail price subject to the utility earning at least the cost of capital.

From (32), the retail price is

$$p = \frac{\bar{r}\,\mathcal{B} + c \cdot \mathbb{E}[\text{fossil}] - R \cdot \mathbb{E}[d]}{\mathbb{E}[D_{\text{served}}]}.$$

Minimizing p subject to the rate-base constraint gives the regulator an incentive to maximize $R \cdot \mathbb{E}[d]$—the total computing revenue the utility earns. This creates a direct regulatory preference for larger FCL programs, since each additional MWh of computing revenue is directly passed through to consumers.

Proposition 8 (Price-minimizing objective under ROR)**.** *Under a price-minimizing ROR regulatory objective, the regulator prefers the largest FCL program consistent with the utility's hardware investment constraint. The socially optimal FCL program size exceeds the utility's private optimum whenever* $\frac{\partial p}{\partial K_M} < 0$*, which holds whenever* $R > 0$ *and* $\mathbb{E}[D_{served}] > 0$*.*

Proof. From the ROR price formula (32),

$$\frac{\partial p}{\partial\, \mathbb{E}[d]} = \frac{-R}{\mathbb{E}[D_{\text{served}}]} < 0,$$

since $R > 0$ and $\mathbb{E}[D_{\text{served}}] > 0$. Thus every additional unit of expected FCL dispatch reduces the ROR price, benefiting consumers. The regulator therefore sets K_M as high as possible subject to the utility's participation constraint (non-negative rate-base return), internalizing the price-reduction benefit to consumers that the profit-maximizing utility ignores. Since the consumer surplus gain $-\frac{\partial p}{\partial \mathbb{E}[d]} \cdot \mathbb{E}[D_{\text{served}}] = R > 0$ is not internalized by the profit-

maximizing utility, the utility's private optimum K_M^{AJ} falls short of the regulator's optimum: $K_M^{\text{social}} > K_M^{\text{AJ}}$. □

Implication for rate-base treatment. Proposition 8 provides a rationale for regulators to *include* FCL hardware in the rate base: doing so both encourages over-investment via the AJ mechanism (Section 6.1) and passes the resulting computing revenue through to consumers via lower prices. The net effect on consumer welfare is unambiguously positive as long as the computing margin $R - c > 0$ is maintained.

F.3 Objective 3: Social welfare maximization

Social welfare definition. Let $\bar{V}$ denote consumers' gross willingness to pay per MWh of electricity (assumed constant, so demand is perfectly inelastic at the regulated retail price $p < \bar{V}$), and let $\sigma > 0$ denote the social cost of carbon per MWh of fossil output (an externality not internalized by the utility unless a carbon tax is in force). Social welfare sums consumer surplus, utility profit, and the two relevant externalities:

$$W \equiv \underbrace{\Pi}_{\text{utility profit}} + \underbrace{(\bar{V} - p)\,\mathbb{E}[\tilde{D}_{\text{served}}]}_{\text{consumer surplus}} - \underbrace{(\text{VOLL} - \beta - p)\,\mathbb{E}[(\tilde{x} - K_F)^+]}_{\text{blackout externality gap}} - \underbrace{\sigma\,\mathbb{E}[\text{fossil}]}_{\text{carbon externality}}, \tag{59}$$

where $\text{VOLL} \equiv \bar{V}$ is the value of lost load, interpreted as the all-in social cost per MWh of unserved demand (consumers' lost value together with the utility's own real blackout losses β), and $\mathbb{E}[\tilde{D}_{\text{served}}] = \mathbb{E}[\tilde{D}] - \mathbb{E}[(\tilde{x} - K_F)^+]$ is expected served demand.

The three wedges between utility profit and social welfare have clear economic interpretations:

1. *Consumer surplus wedge* $(\bar{V} - p)\,\mathbb{E}[\tilde{D}_{\text{served}}] > 0$: consumers receive electricity worth $\bar{V}$ per MWh but pay only p; the utility does not capture this surplus.

2. *Blackout externality gap* $(\text{VOLL} - \beta - p)\,\mathbb{E}[(\tilde{x} - K_F)^+] \geq 0$: for each MWh of un-

served demand the utility forgoes revenue p and bears the penalty β as a real cost paid to no one (a deadweight loss—e.g., equipment damage, liability, and regulatory consequences), so it internalizes $\beta + p$ of the total social cost VOLL; the residual $\text{VOLL} - \beta - p \geq 0$ is a negative externality borne by consumers.

3. *Carbon externality* $\sigma\,\mathbb{E}[\text{fossil}] > 0$: each MWh of fossil output imposes social cost σ through greenhouse gas emissions not priced by the utility absent a carbon tax.

Equivalence between the utility and the social planner.

Proposition 9 (Social planner equivalence)**.** *Suppose (i) the regulator sets the blackout penalty* $\beta = \text{VOLL} - p$ *and (ii) a carbon tax* $\tau = \sigma$ *is levied on each MWh of fossil output. Then the utility's profit-maximization problem* (2) *coincides with the social welfare maximization problem* (59)*: every equilibrium capacity triple* (K_R^*, K_F^*, K_M^*) *and every proposition in Sections 4–6.2 characterizes the socially optimal allocation. In particular, the triple-win condition* $C_M < \bar{C}_M$ *is also the condition under which the FCL program improves social welfare.*

Proof. Under condition (i), $\beta + p = \text{VOLL}$, so the blackout externality gap in (59) is zero. Under condition (ii) with carbon tax $\tau = \sigma$, the utility's effective fossil cost rises from c to $c + \tau = c + \sigma$, matching the social cost. The social welfare objective then reduces to

$$W = \Pi(\tau = \sigma) + (\bar{V} - p)\,\mathbb{E}[\tilde{D}] + \text{const},$$

where $(\bar{V} - p)\,\mathbb{E}[\tilde{D}]$ is a constant (independent of (K_R, K_F, K_M)) under the inelastic demand assumption. Consequently, the capacity FOCs of the social welfare problem are identical to those of the utility profit problem with fossil cost $c + \sigma$, which is the utility problem under condition (ii). All existence and comparative-statics results follow directly. □

Remark 14 (Second-best robustness)**.** In practice, β is typically set below VOLL$-p$ (utility faces a blackout penalty smaller than the true consumer loss) and carbon pricing is absent or partial ($\tau < \sigma$). The baseline equilibrium is therefore a *second-best* allocation: the utility underinvests in reliability and overuses fossil relative to the social optimum.

Nonetheless, the FCL program is welfare-improving in the second-best sense whenever $C_M < \bar{C}_M$. To see why, note from the decomposition (59) that introducing FCLs affects welfare through three channels simultaneously:

1. *Profit channel*: $\Delta\Pi > 0$ whenever $C_M < \bar{C}_M$ (Proposition 1).

2. *Carbon channel*: $\Delta(-\sigma\,\mathbb{E}[\text{fossil}]) > 0$ under the distributional conditions of Proposition 4 (e.g., the uniform distribution with sufficiently low C_R), which reduces the unpriced carbon burden.

3. *Blackout channel*: $\Delta(-(\text{VOLL} - \beta - p)\,\mathbb{E}[(\tilde{x} - K_F)^+]) > 0$ (Proposition 1, Step 2), which reduces the residual consumer outage cost.

The profit and reliability channels operate in the same direction whenever $C_M < \bar{C}_M$, and the carbon channel joins them under the conditions of Proposition 4; under those conditions the firm-level triple win is accompanied by a social welfare improvement even when $\beta <$ VOLL$-p$ and $\tau < \sigma$. The FCL program is, in this sense, a Pareto improvement over the second-best baseline.

G Small-FCL Regime ($K_M^* \leq K_R^*$)

The main text analyzes the *large-FCL* regime in which optimal FCL capacity exceeds renewable capacity ($K_M^* > K_R^*$). In this regime $-K_M^*$ lies strictly below the support: since $-K_M^* < -K_R^*$, for every realized state $\tilde{x} \in [-K_R^*, \bar{D}]$ the net load $\tilde{x} + K_M^* \geq K_M^* - K_R^* > 0$, so fossil always generates a positive amount.

When $K_M^* \leq K_R^*$ (the small-FCL regime), there exist *pure renewable-surplus hours*: states $\tilde{x} \leq -K_M^*$ in which the grid is in such deep surplus that even a fully dispatched FCL of size K_M^* leaves some renewable power unused. During those hours the fossil plant is idle, and the FCL runs at full capacity K_M^* earning R per MWh with zero fossil variable cost.

The utility's profit function in this regime is:

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F(x; K_R)\,\mathrm{d}x - c\int_{-K_M}^{K_F-K_M}\Big(1 - F(x; K_R)\Big)\,\mathrm{d}x \\ - (\beta + p)\int_{K_F}^{\bar{D}}\Big(1 - F(x; K_R)\Big)\,\mathrm{d}x - C_F K_F - C_R K_R - C_M K_M. \quad (60)$$

Note that (60) is identical in form to (5) from Section 4: since $-K_M > -K_R$ in the small-FCL regime, the extended CDF equals the ordinary CDF on the integration domain, and the integration-by-parts simplification that converts (5) into (10) is not available (the lower limit $-K_M$ lies within the support rather than below it). The fossil variable cost integral in (60) has limits $(-K_M,\, K_F - K_M)$. The lower limit $-K_M$ reflects that states $\tilde{x} \leq -K_M$ have no fossil dispatch (pure renewable surplus with FCL at full capacity K_M); the upper limit $K_F - K_M$ is the threshold above which the FCL is in partial dispatch rather than at full capacity K_M. Expected fossil output in this regime is $\int_{-K_M}^{K_F-K_M}(1-F(x;K_R))\,dx$.

Computing capacity K_M. Differentiating (60) with respect to K_M gives:

$$(R - c)\,F(K_F - K_M;\, K_R) + c\,F(-K_M;\, K_R) = C_M. \quad (61)$$

The extra term $cF(-K_M; K_R) > 0$ relative to the large-FCL-regime FOC reflects the marginal pure-renewable-surplus hour: a unit increase in K_M^* is dispatched during an additional $f(-K_M^*)dK_M$ hours when no fossil runs, earning R per MWh rather than the $R - c$ margin of the fossil-dispatch region.

Adoption threshold. Evaluating the marginal value of the first unit of FCL capacity at the no-FCL optimum ($K_M \to 0$, which necessarily lies in the small-FCL regime since $K_M < K_R^\circ$) gives the adoption threshold

$$\bar{C}_M^b \;=\; (R-c)\,\phi^\circ \;+\; c\,F(0;K_R^\circ) \;=\; \bar{C}_M + c\,q^\circ \;>\; \bar{C}_M, \tag{62}$$

where $q^\circ \equiv F(0;K_R^\circ)$ is the probability of renewable surplus at the no-FCL optimum. The ability to run on zero-cost renewable surplus makes the first unit of FCL hardware strictly more valuable than the large-FCL-regime threshold $\bar{C}_M$ suggests; grids with many surplus hours (large q°) support correspondingly higher hardware costs.

Fossil capacity K_F. Differentiating (60) with respect to K_F gives:

$$-\Big(\beta+p-R\Big)F(K_F;\,K_R) - (R-c)\,F(K_F-K_M;\,K_R) + (\beta+p-c) - C_F = 0. \tag{63}$$

This parallels Equation (13) in Section 4, with the additional term $-(R-c)F(K_F-K_M;K_R)$ reflecting that a higher K_F also shifts the boundary between the partial-dispatch and full-capacity regions. Substituting (61) to eliminate $F(K_F-K_M;K_R)$:

$$F(K_F;\,K_R) = \frac{\beta+p-c-C_F-C_M+c\,F(-K_M;\,K_R)}{\beta+p-R}. \tag{64}$$

The extra $cF(-K_M;K_R) > 0$ in the numerator raises $F(K_F^*;K_R^*)$ above the large-FCL-regime value, implying *more* fossil capacity and *lower* blackout probability for the same parameter values.

Renewable capacity K_R. Differentiating (60) with respect to K_R:

$$R\int_{K_F-K_M}^{K_F}\frac{\partial F(x;K_R)}{\partial K_R}\,\mathrm{d}x + c\int_{-K_M}^{K_F-K_M}\frac{\partial F(x;K_R)}{\partial K_R}\,\mathrm{d}x + (\beta+p)\int_{K_F}^{\bar{D}}\frac{\partial F(x;K_R)}{\partial K_R}\,\mathrm{d}x = C_R. \tag{65}$$

The lower limit of the fossil integral is $-K_M$ rather than $-K_R$ as in (15). This is because states $\tilde{x} < -K_M$ are pure-surplus hours with no fossil dispatch, so additional renewable capacity there displaces no fossil generation and contributes only through the mining revenue channel.

G.1 Blackout probability and the triple win

From (64), the equilibrium blackout probability under the small-FCL regime is:

$$1 - F(K_F^*;\, K_R^*) = \frac{c + C_F + C_M - R - c\,F(-K_M^*; K_R^*)}{\beta + p - R}. \tag{66}$$

Since $cF(-K_M^*; K_R^*) > 0$, this is strictly lower than the large-FCL-regime blackout probability $\frac{c+C_F+C_M-R}{\beta+p-R}$ for the same parameters. The small-FCL regime therefore delivers an even larger reliability improvement than the large-FCL regime.

Analytical solutions under uniform demand Under the uniform distribution $F(x; K_R) = \frac{x+K_R}{\bar{D}+K_R}$ on $[-K_R, \bar{D}]$, the three FOCs (61)–(65) reduce to the system:

$$(R - c)\,K_F^* + R\,(K_R^* - K_M^*) = C_M\,(\bar{D} + K_R^*), \tag{67}$$

$$(\beta + p - R)\,K_F^* + c\,K_M^* = (\beta + p - c - C_F - C_M)\,\bar{D} + (R - C_F - C_M)\,K_R^*, \tag{68}$$

$$R\,K_M^*\Big(\bar{D} - K_F^* + \tfrac{K_M^*}{2}\Big) + c\,K_F^*\Big(\bar{D} + K_M^* - \tfrac{K_F^*}{2}\Big) + \tfrac{(\beta+p)(\bar{D}-K_F^*)^2}{2} = C_R\,(\bar{D} + K_R^*)^2. \tag{69}$$

Equations (67)–(68) are linear in (K_F^*, K_M^*) given K_R^*. We thus have:

$$K_F^* = \frac{\Big[R(\beta+p-c-C_F)-(R-c)C_M\Big]\bar{D}+\Big[(R-c)(R-C_M)-RC_F\Big]K_R^*}{c(R-c)+R(\beta+p-R)}, \tag{70}$$

$$K_M^* = \frac{\Big[(R-c)(\beta+p-c-C_F)-(\beta+p-c)C_M\Big]\bar{D}+\Big[(\beta+p-c)(R-C_M)-(R-c)C_F\Big]K_R^*}{c(R-c)+R(\beta+p-R)}. \tag{71}$$

Substituting (70)–(71) into (69), and solving for K_R^* (which is nonnegative) gives

$$K_R^* = \left(\sqrt{\frac{c(R-c)(\beta+p-R)}{2C_R(c(R-c)+R(\beta+p-R))-(\beta+p-c)(R-C_M)^2+2C_F(R-c)(R-C_M)-RC_F^2}}-1\right)\bar{D} \tag{72}$$

Recall that in the case when FCL is banned, we have

$$K_F^\circ = \left(1-\frac{\sqrt{c}}{\sqrt{(\beta-c+p)\,(2C_R(\beta-c+p)-C_F^2)}}C_F\right)\bar{D}, \tag{73}$$

$$K_R^\circ = \left(\frac{\sqrt{c(\beta-c+p)}}{\sqrt{2C_R(\beta-c+p)-C_F^2}}-1\right)\bar{D} \tag{74}$$

G.2 Carbon comparison: mining versus no-mining benchmark

Expected fossil generation under the FCL. Net fossil dispatched in state $\tilde{x}$ equals $(\tilde{x}+K_M^*)$ for $\tilde{x}\in(-K_M^*,\,K_F^*-K_M^*]$, and K_F^* for $\tilde{x}>K_F^*-K_M^*$ (FCL fully curtailed or blackout). Under the uniform distribution with density $1/(\bar{D}+K_R^*)$:

$$E[G_F^*] = \frac{1}{\bar{D}+K_R^*}\int_{-K_M^*}^{K_F^*-K_M^*}(\tilde{x}+K_M^*)\,\mathrm{d}\tilde{x}+\frac{K_F^*\Big(\bar{D}-K_F^*+K_M^*\Big)}{\bar{D}+K_R^*}. \tag{75}$$

Setting $u=\tilde{x}+K_M^*$, the integral equals $\int_0^{K_F^*}u\,\mathrm{d}u=K_F^{*2}/2$, so:

$$E[G_F^*] = \frac{K_F^*\Big(\bar{D}+K_M^*-\frac{K_F^*}{2}\Big)}{\bar{D}+K_R^*}. \tag{76}$$

No-mining benchmark ($K_M = 0$). When mining is banned, the utility optimizes over (K_F^0, K_R^0) only. Setting $K_M = 0$ in the small-FCL profit function annihilates the mining revenue term; the FOCs are:

$$(\beta + p - c)\Big(1 - F^0(K_F^0;\, K_R^0)\Big) = C_F, \tag{77}$$

$$\frac{c\, K_F^0\Big(\bar{D} - \frac{K_F^0}{2}\Big) + \frac{(\beta+p)(\bar{D}-K_F^0)^2}{2}}{(\bar{D} + K_R^0)^2} = C_R. \tag{78}$$

Under the uniform distribution, (77) gives $\bar{D} - K_F^0 = C_F(\bar{D} + K_R^0)/(\beta + p - c)$, hence:

$$K_F^0 = \frac{(\beta + p - c - C_F)\,\bar{D} - C_F\, K_R^0}{\beta + p - c}. \tag{79}$$

Substituting (79) into (78) and using $K_F^0(\bar{D} - K_F^0/2) = \Big[\bar{D}^2 - C_F^2(\bar{D} + K_R^0)^2/(\beta + p - c)^2\Big]/2$, the equation reduces to:

$$\frac{c\,\bar{D}^2}{2\,(\bar{D} + K_R^0)^2} + \frac{C_F^2}{2(\beta + p - c)} = C_R, \tag{80}$$

yielding the closed-form solution:

$$K_R^0 = \bar{D}\left(\sqrt{\frac{c(\beta + p - c)}{2C_R(\beta + p - c) - C_F^2}} - 1\right), \qquad K_F^0 \text{ from (79)}. \tag{81}$$

Expected fossil generation in the no-mining case follows by the same argument as (76):

$$E[G_F^0] = \frac{K_F^0\Big(\bar{D} - \frac{K_F^0}{2}\Big)}{\bar{D} + K_R^0}. \tag{82}$$

Carbon comparison. Defining $\Delta G \equiv E[G_F^0] - E[G_F^*]$ (reduction in expected fossil generation from allowing mining):

$$\Delta G = \frac{K_F^0\left(\bar{D} - \frac{K_F^0}{2}\right)}{\bar{D} + K_R^0} - \frac{K_F^*\left(\bar{D} + K_M^* - \frac{K_F^*}{2}\right)}{\bar{D} + K_R^*}. \tag{83}$$

Mining reduces expected carbon ($\Delta G > 0$) when the FCL-induced renewable overbuild ($K_R^* > K_R^0$) and the co-dispatch term K_M^* together reduce fossil reliance sufficiently to outweigh any increase in fossil nameplate capacity K_F^*.

The small-FCL regime applies when $K_M^* \leq K_R^*$, equivalently (from (67)):

$$(R - c)\, K_F^* \leq C_M\, (\bar{D} + K_R^*).$$

Because K_F^* is increasing in scarcity costs $\beta + p$, the small-FCL regime is relevant when *FCL hardware is cheap and scarcity costs are moderate*—the utility builds a large FCL relative to its renewable fleet. The large-FCL regime (the main-text regime) applies when K_F^* is large, corresponding to high scarcity costs or high ASIC cost. Which regime applies for a given grid depends on its parameters; since the small-FCL regime features a lower blackout probability and a wider adoption threshold, the main-text (large-FCL) results are conservative in these respects.

H Profit Function: Derivation and Case Analysis

This appendix derives the simplified profit function (5) from the primitive utility problem (2) and shows how the expression changes across the four economically relevant parameter regimes.

Convention. Extend $F(x; K_R) := 0$ for $x < -K_R$ and $F(x; K_R) := 1$ for $x > \bar{D}$, so

that all integrals below are well-defined over $\mathbb{R}$. Capital costs $C_R K_R + C_F K_F + C_M K_M$ are non-stochastic and carried through unchanged.

Preliminary: universal sub-expressions

Mining revenue (valid for all parameter values):

$$\mathbb{E}\Big[R \cdot d(\tilde{x})\Big] \;=\; R \int_{K_F - K_M}^{K_F} F(x; K_R)\,\mathrm{d}x. \tag{84}$$

Blackout cost (valid for all parameter values; equals zero when $K_F \geq \bar{D}$):

$$\mathbb{E}\Big[(\beta + p)(\tilde{x} - K_F)^+\Big] \;=\; (\beta + p) \int_{K_F}^{\bar{D}} \Big(1 - F(x; K_R)\Big)\,\mathrm{d}x. \tag{85}$$

Fossil variable cost (unified form, via the survival-function identity $\mathbb{E}[\min\{K_F, Z^+\}] = \int_0^{K_F} (1 - F_Z(z))\,dz$ with $Z = \tilde{x} + K_M$):

$$c\,\mathbb{E}\Big[\min\Big\{K_F,\, \max\{\tilde{x} + d(\tilde{x}), 0\}\Big\}\Big] \;=\; c \int_{-K_M}^{K_F - K_M} \Big(1 - F(x; K_R)\Big)\,\mathrm{d}x. \tag{86}$$

Splitting at $x = -K_R$ makes the two regimes explicit:

$$c \int_{-K_M}^{K_F - K_M} (1 - F)\,\mathrm{d}x = \begin{cases} c \displaystyle\int_{-K_M}^{K_F - K_M} (1 - F(x; K_R))\,\mathrm{d}x & K_M \leq K_R, \\[2ex] c(K_M - K_R) + c \displaystyle\int_{-K_R}^{K_F - K_M} (1 - F(x; K_R))\,\mathrm{d}x & K_M > K_R. \end{cases} \tag{87}$$

The term $c(K_M - K_R)$ in the second line is the expected fossil cost incurred during renewable-surplus hours $x \in (-K_R, 0)$ when the FCL draws on the fossil plant even though consumer residual demand is negative.

Small-FCL regime, interior blackout

Parameter requirements:

$$0 \;\le\; K_M \;\le\; K_R, \qquad K_F \;<\; \bar{D}.$$

During hours of deep renewable surplus ($\tilde{x} < -K_M$) the entire FCL load is met by renewables; the fossil plant is idle.

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F(x;K_R)\,\mathrm{d}x - c\int_{-K_M}^{K_F-K_M}\Big(1-F(x;K_R)\Big)\,\mathrm{d}x \\ -\;(\beta+p)\int_{K_F}^{\bar{D}}\Big(1-F(x;K_R)\Big)\,\mathrm{d}x - C_F K_F - C_R K_R - C_M K_M. \quad (88)$$

First-order conditions:

$$\frac{\partial \Pi}{\partial K_M} = 0: \quad (R-c)\,F(K_F-K_M;K_R) + c\,F(-K_M;K_R) = C_M, \quad (89)$$

$$\frac{\partial \Pi}{\partial K_F} = 0: \quad (\beta+p-R)\,F(K_F;K_R) = \beta + p - c - C_F - C_M + c\,F(-K_M;K_R), \quad (90)$$

$$\frac{\partial \Pi}{\partial K_R} = 0: \quad R\int_{K_F-K_M}^{K_F}\tfrac{\partial F}{\partial K_R}\,\mathrm{d}x + c\int_{-K_M}^{K_F-K_M}\tfrac{\partial F}{\partial K_R}\,\mathrm{d}x + (\beta+p)\int_{K_F}^{\bar{D}}\tfrac{\partial F}{\partial K_R}\,\mathrm{d}x = C_R. \quad (91)$$

When $K_M < K_R$, the term $c\,F(-K_M;K_R) > 0$ appears in both (89) and (90), reflecting the probability of deep-surplus hours in which the marginal FCL unit earns revenue R at zero fossil cost. The c-weighted integral in (91) has lower limit $-K_M$, not $-K_R$.

Small-FCL regime, full coverage

Parameter requirements:

$$0 \;\le\; K_M \;\le\; K_R, \qquad K_F \;\ge\; \bar{D}.$$

No blackout occurs; the $(\beta + p)$ integral in (88) vanishes:

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F(x; K_R)\,\mathrm{d}x - c\int_{-K_M}^{K_F-K_M}\Big(1-F(x; K_R)\Big)\,\mathrm{d}x - C_F K_F - C_R K_R - C_M K_M. \tag{92}$$

FOCs (89)–(91) continue to hold with the $(\beta + p)$ integral set to zero.

Large-FCL regime, interior blackout

Parameter requirements:

$$K_R \;<\; K_M \;\leq\; K_F + K_R, \qquad K_F \;<\; \bar{D}.$$

FCL capacity exceeds renewable capacity; the deep-surplus region $\{x < -K_M\}$ lies outside the support $[-K_R, \bar{D}]$, so $F(-K_M; K_R) = 0$. The upper bound $K_M \leq K_F + K_R$ ensures the full-capacity dispatch region $\{x < K_F - K_M\}$ is non-empty on the support; beyond it, $\partial\Pi/\partial K_M = -C_M < 0$ (see the very-large-FCL regime below).

$$\begin{aligned}\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F(x; K_R)\,\mathrm{d}x - c(K_M - K_R) - c\int_{-K_R}^{K_F-K_M}\Big(1-F(x; K_R)\Big)\,\mathrm{d}x \\ -\;(\beta + p)\int_{K_F}^{\bar{D}}\Big(1 - F(x; K_R)\Big)\,\mathrm{d}x - C_F K_F - C_R K_R - C_M K_M.\end{aligned} \tag{93}$$

Relation to (5). The unified formula (5) (lower limit $-K_M$) and the large-FCL expression (93) (lower limit $-K_R$ plus $-c(K_M - K_R)$) are equivalent: splitting $\int_{-K_M}^{K_F-K_M}(1-F)\,\mathrm{d}x$ at $x = -K_R$ yields $-c(K_M - K_R) - c\int_{-K_R}^{K_F-K_M}(1-F)\,\mathrm{d}x$, confirming the two representations are identical. Consequently, the first-order conditions coincide with those in the main text (11), (13), and (15).

Large-FCL regime, full coverage

Parameter requirements:

$$K_R \;<\; K_M \;\le\; K_F + K_R, \qquad K_F \;\ge\; \bar{D}.$$

$$\Pi(K_R, K_F, K_M) = p\,\mathbb{E}[\tilde{D}] + R\int_{K_F-K_M}^{K_F} F(x; K_R)\,\mathrm{d}x - c(K_M - K_R) - c\int_{-K_R}^{K_F-K_M}\Big(1 - F(x; K_R)\Big)\,\mathrm{d}x - C_F K_F - C_R K \tag{94}$$

Section 4.2 identifies the boundary $K_F = \bar{D}$ ($R = c + C_F + C_M$) as the relevant sub-case within this regime.

Very-large-FCL regime ($K_M > K_F + K_R$) — never optimal

Parameter requirements: $K_M > K_F + K_R$.

When $K_M > K_F + K_R$, the full-capacity dispatch region $\{x < K_F - K_M\}$ lies entirely below the support, so the FCL never reaches full capacity. Both I_R and I_C are independent of K_M in this region, while $-C_M K_M$ is strictly decreasing, giving:

$$\left.\frac{\partial \Pi}{\partial K_M}\right|_{K_M > K_F + K_R} = -C_M < 0.$$

The utility therefore never chooses $K_M > K_F + K_R$; the economically relevant range is $K_M \in [0,\, K_F + K_R]$.

Summary of differences between cases

The following table summarizes the key differences in the profit function and first-order conditions between the small-FCL regime and the large-FCL regime.

	Small-FCL ($K_M \le K_R$)	**Large-FCL** ($K_R < K_M \le K_F + K_$
Fossil cost I_C	$c\int_{-K_M}^{K_F-K_M}(1-F)\,\mathrm{d}x$	$c(K_M-K_R)+c\int_{-K_R}^{K_F-K_M}(1-F)\,\mathrm{d}$
FOC for K_M	$(R-c)F(K_F-K_M)+cF(-K_M)=C_M$	$(R-c)F(K_F-K_M)=C_M$
FOC for K_F	$(\beta{+}p{-}R)F(K_F)=\beta{+}p{-}c{-}C_F{-}C_M+cF(-K_M)$	$(\beta{+}p{-}R)F(K_F)=\beta{+}p{-}c{-}C_F{-}C_$
Lower limit, K_R-FOC	$-K_M$	$-K_R$
$F(-K_M;K_R)$	>0 (on support)	$=0$ (below support)

Table 9: Key differences between the small-FCL and large-FCL profit functions and first-order conditions. The full-coverage sub-cases follow by setting the $(\beta+p)$ integral to zero.

I Comprehensive Summary of Results

This appendix consolidates all major analytical results—optimal capacity allocations, black-out probabilities, and carbon comparisons—across every scenario combination analyzed in the paper. Closed-form expressions assume the uniform distribution $F(x;K_R)=\frac{x+K_R}{\bar{D}+K_R}$ on $[-K_R,\bar{D}]$ unless otherwise noted. Throughout, the triple-win threshold is $\bar{C}_M=(R-c)\frac{\beta+p-c-C_F}{\beta+p-c}$, and Assumption 1 ($\beta+p>R>c$, $\beta+p-c>C_F$) is maintained.

I.1 No-FCL Benchmark

Table 10 records the benchmark optimal allocations and carbon outcomes when no FCL program is offered, corresponding to the no-FCL benchmark in Section 4.

I.2 With FCL Program: Large-FCL Regime ($K_M^* > K_R^*$)

The large-FCL regime is the baseline analyzed in Section 4. Tables 11 and 12 cover the interior and full-coverage regimes respectively.

Table 10: No-FCL Benchmark: Optimal Allocations and Outcomes

Quantity	**General**	**Uniform distribution**
Fossil capacity K_F°	$F(K_F^\circ; K_R^\circ) = \dfrac{\beta + p - c - C_F}{\beta + p - c}$	$K_F^\circ = \dfrac{\beta + p - c - C_F}{\beta + p - c}(\bar{D} + K_R^\circ) - K_R^\circ$
Renewable capacity K_R°	$C = (\beta + p)\int_{K_F^\circ}^{\bar{D}} \frac{\partial F}{\partial K_R}\,\mathrm{d}x + c\int_0^{K_F^\circ} \frac{\partial F}{\partial K_R}\,\mathrm{d}x$	Determined by implicit eq. above
Blackout probability	$1 - F(K_F^\circ; K_R^\circ) = \dfrac{C_F}{\beta + p - c}$	Same (exact)
Expected carbon	$c \cdot \mathbb{E}[\min\{K_F^\circ, \tilde{x}^+\}]$	$c \cdot \int_0^{K_F^\circ}(1 - F(x; K_R^\circ))\,\mathrm{d}x$

This table reports optimal fossil capacity K_F°, renewable capacity K_R°, blackout probability, and expected carbon emissions when no FCL demand response program is offered. Results are given for a general residual-demand distribution $F(x; K_R)$ (with $\frac{\partial F}{\partial K_R} > 0$) and for the uniform distribution $F(x; K_R) = \frac{x + K_R}{\bar{D} + K_R}$ on $[-K_R, \bar{D}]$. The blackout probability $\frac{C_F}{\beta + p - c}$ is distribution-free. Assumption 1 ($\beta + p > c + C_F$, $\beta + p > R > c$) is maintained. This benchmark serves as the comparison point for all tables with an FCL program.

I.3 With FCL Program: Small-FCL Regime ($K_M^* \leq K_R^*$)

Full derivations are in Appendix G. Table 13 summarizes the first-order conditions, blackout probability, and triple-win threshold under the small-FCL regime.

I.4 Timing Variations: Pre-Installed Capacity (Section 5)

Table 14 collects the optimal allocations, blackout probabilities, and carbon outcomes for each of the three pre-commitment cases (A, B, C) alongside the simultaneous baseline.

I.5 Rate-of-Return Regulation (Section 6.1)

Under ROR regulation the regulated price p is replaced by an endogenously determined price that allows the utility to earn a fair return $\bar{r}$ on rate base. The Averch–Johnson (AJ) over-investment effect raises the optimal K_F above the unregulated level. Table 15 summarizes how regulation affects the key first-order conditions, the triple-win threshold, and carbon

Table 11: With FCL Program, Large-FCL Regime, Interior Case: Simultaneous Optimum

Quantity	**General**	**Uniform distribution**
Fossil capacity K_F^*	$F(K_F^*; K_R^*) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}$	$K_F^* = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}(\bar{D} + K_R^*) - K_R^*$
Renewable capacity K_R^*	FOC (15): weight R in FCL region	Determined jointly with K_F^*
FCL capacity K_M^*	$(R-c)F(K_F^* - K_M^*; K_R^*) = C_M$	$K_M^* = K_F^* - \frac{C_M(\bar{D}+K_R^*)}{R-c} + K_R^*$
Blackout probability	$\frac{c+C_F+C_M-R}{\beta+p-R}$	Same (exact)
Comparison to no-FCL	Lower iff $C_M < \bar{C}_M$	Same
Carbon comparison	May fall when $C_M < \bar{C}_M$ (Prop. 4; e.g., uniform F with low C_R)	$\bar{C}_M = \frac{(R-c)(\beta+p-c-C_F)}{\beta+p-c}$
Profit comparison	Rises iff $C_M < \bar{C}_M$	Same
Triple win	$C_M < \bar{C}_M \Leftrightarrow$ profit and reliability improve; carbon under Prop. 4 conditions	Proposition 1

This table reports optimal allocations and welfare comparisons under the large-FCL regime ($K_M^* > K_R^*$, the empirically relevant regime) and the interior regime ($R < c + C_F + C_M$), when K_R, K_F, and K_M are all chosen simultaneously. Closed-form expressions use the uniform distribution $F(x; K_R) = \frac{x+K_R}{\bar{D}+K_R}$ on $[-K_R, \bar{D}]$. The triple-win threshold is $\bar{C}_M \equiv (R-c)\frac{\beta+p-c-C_F}{\beta+p-c}$; the blackout probability $\frac{C_F}{\beta+p-c}$ in the no-FCL benchmark (Table 10) exceeds $\frac{c+C_F+C_M-R}{\beta+p-R}$ iff $C_M < \bar{C}_M$. All comparisons are relative to the no-FCL benchmark.

Table 12: With FCL Program, Large-FCL Regime, Full-Coverage Case: Simultaneous Optimum

Quantity	**Result**
Fossil capacity K_F^*	$K_F^* = \bar{D}$ (corner solution)
Blackout probability	0 (zero blackout)
Triple win	Automatically satisfied: $C_M \leq R - c - C_F < \bar{C}_M$
Carbon	Strictly lower than no-FCL (zero fossil waste hours)
Profit	Strictly higher

This table reports the simultaneous optimum under the large-FCL regime and the corner regime, defined by $R \geq c + C_F + C_M$. In this regime the utility builds fossil capacity equal to the maximum demand $\bar{D}$, eliminating blackouts entirely. The triple win is automatically satisfied whenever the full-coverage regime holds, since the adoption condition $C_M \leq R - c - C_F$ is strictly stronger than the interior-regime threshold $\bar{C}_M$. The interior regime ($R < c + C_F + C_M$, Table 11) is the empirically relevant case for moderate ASIC costs.

Table 13: With FCL Program, Small-FCL Regime ($K_M^* \leq K_R^*$): Simultaneous Optimum

Quantity	**Small-FCL Regime**	**Comparison to Large-FCL Regime**
FOC for K_M	$(R - c)F(K_F - K_M; K_R) + cF(-K_M; K_R) = C_M$	Extra term $cF(-K_M; K_R) > 0$
FOC for K_F	$F(K_F; K_R) = \frac{\beta+p-c-C_F-C_M+cF(-K_M^*;K_R^*)}{\beta+p-R}$	Numerator larger by $cF(-K_M^*; K_R^*)$
Blackout probability	$\frac{c+C_F+C_M-R-cF(-K_M^*;K_R^*)}{\beta+p-R}$	**Strictly lower** than large-FCL regime
Adoption threshold $\bar{C}_M^b$	$\bar{C}_M^b = \bar{C}_M + c\,F(0; K_R^\circ) > \bar{C}_M$ (62)	Wider viable range
Condition for small-FCL regime	$K_F^* \leq \frac{C_M(\bar{D} + K_R^*)}{R - c}$ (uniform)	Small C_M or small K_R^*

This table summarizes the first-order conditions and key comparisons under the small-FCL regime ($K_M^* \leq K_R^*$), in which optimal FCL capacity does not exceed renewable capacity, so there exist deep-surplus states ($x \leq -K_M$) in which the FCL runs entirely on renewable power. In this regime the marginal FCL unit is sometimes dispatched on pure renewable surplus, adding an extra term $cF(-K_M; K_R) > 0$ to the FOC for K_M. This extra term lowers the equilibrium blackout probability below the large-FCL-regime level and widens the viable range of hardware costs to $\bar{C}_M^b > \bar{C}_M$. Full derivations appear in Appendix G. The interior-regime condition ($R < c + C_F + C_M$) is maintained.

Table 14: Timing Variations: Optimal Allocations and Carbon Outcomes by Capacity Flexibility

Case	**Optimal K_F, K_M**	**Blackout**	**Carbon**
A: K_F fixed at $K_F^{(0)}$, free K_R, K_M	K_M from FOC (11) at $K_F^{(0)}$; K_R from modified FOC	Falls (K_R rises while $K_F^{(0)}$ fixed)	Ambiguous: $(R - c)\lvert\partial\mathbb{E}[\text{fossil}]/\partial K_R\rvert \gtrless C_R$
B: K_R fixed at $K_R^{(0)}$, free K_F, K_M	$F(K_F; K_R^{(0)}) = \frac{\beta+p-c-C_F-C_M}{\beta+p-R}$; K_M from (11) at $K_R^{(0)}$	Lower iff $C_M < \bar{C}_M(K_R^{(0)})$	May fall when $C_M < \bar{C}_M(K_R^{(0)})$
C: both $K_F^{(0)}$, $K_R^{(0)}$ fixed, free K_M only	K_M from (11) at $(K_F^{(0)}, K_R^{(0)})$; no capacity response	Unchanged (capacity fixed)	**Always rises** (utilization effect only)
Baseline (§4): all free	Table 11	Lower iff $C_M < \bar{C}_M$	May fall when $C_M < \bar{C}_M$ (Prop. 4)

This table reports optimal FCL capacity K_M, fossil capacity K_F, blackout probability, and carbon outcomes for each of the three pre-commitment scenarios analyzed in Section 5, together with the simultaneous baseline (Section 4). Case A: fossil capacity $K_F^{(0)}$ is pre-installed; renewable K_R and FCL capacity K_M are freely optimized. Case B: renewable capacity $K_R^{(0)}$ is pre-installed; K_F and K_M are freely optimized. Case C: both $K_F^{(0)}$ and $K_R^{(0)}$ are fixed; only K_M is chosen. The threshold $\bar{C}_M(K_R^{(0)}) \equiv \bar{C}_M + c\,F(0; K_R^{(0)})$ adds the surplus-hour margin to the baseline threshold (Section 5.2); it rises with $K_R^{(0)}$ and approaches $\bar{C}_M$ as $K_R^{(0)} \to 0$ (less renewable capacity makes adoption harder).

emissions.

Table 15: Rate-of-Return (Averch–Johnson) Regulation: Key Results

Quantity	**Without FCL**	**With FCL**
AJ profit	$\Pi^{\mathrm{AJ}} = \bar{r}\,\mathcal{B} - \beta\int_{K_F}^{\bar{D}}(1-F)\,\mathrm{d}x$, with $\mathcal{B} = C_F K_F + C_R K_R$	Same, with $\mathcal{B} = C_F K_F + C_R K_R + C_M K_M$ if hardware is in the rate base
Over-investment	$\partial\Pi^{\mathrm{AJ}}/\partial K_j > 0$ for $j \in \{F, R\}$: all rate-base capacity expands; blackout probability driven toward zero	Extends to K_M when in rate base
FCL adoption	n/a	Unconditional if K_M in rate base ($\partial\Pi^{\mathrm{AJ}}/\partial K_M = \bar{r}C_M > 0$); baseline threshold $\bar{C}_M$ if excluded
Price benefit	n/a	$\partial p/\partial\mathbb{E}[d] = -R/\mathbb{E}[D_{\mathrm{served}}] < 0$: FCL revenue passed to consumers as a lower regulated price
Carbon	AJ over-investment raises K_F^{AJ}, increases fossil utilization	Renewable over-investment pushes carbon down; net effect as in Section 4.3

This table summarizes optimal allocations and comparisons under rate-of-return (ROR) regulation following the Averch–Johnson (AJ) framework (Section 6.1). The allowed rate of return $\bar{r}$ satisfies $\bar{r} \geq \rho$ where ρ is the utility's cost of capital; when $\bar{r} > 0$, the utility earns a margin on every rate-base dollar, inducing over-investment in all rate-base-eligible capacity. If FCL hardware qualifies as rate base, FCL adoption is unconditional and the cost threshold disappears; if excluded, the baseline threshold $\bar{C}_M$ applies (Section 6.1). Compare with the unregulated baseline (Table 11).

I.6 Deregulated Competitive Markets (Section 6.2)

In deregulated markets, p is replaced by a spot price $P(\tilde{x})$. A scarcity price P_{cap} (capacity payment) plays the role of $\beta + p$ in the regulated model. All regulated-market results

are isomorphic with $P_{\text{cap}} \leftrightarrow \beta + p$. Table 16 lays out the zero-profit conditions, blackout probabilities, and triple-win threshold under competitive market equilibrium.

Table 16: Deregulated Competitive Market: Equilibrium Conditions with and without FCL

Condition	**Without FCL**	**With FCL**
Fossil zero-profit	$(P_{\text{cap}} - c)\Pr(\tilde{x} > K_F) = C_F$	Same form (FCL curtails during scarcity)
FCL free entry	n/a	Entry iff $C_M < R\,q^\circ + (R - c)(\phi^\circ_{\text{comp}} - q^\circ)$; scale K^*_M pinned by (36)
Blackout probability	$\dfrac{C_F}{P_{\text{cap}} - c}$	$\dfrac{C_F}{P_{\text{cap}} - c}$ (unchanged: pinned by fossil zero-profit; composition shifts, $K^*_F < K^\circ_F$, $K^*_R > K^\circ_R$)
Renewable free entry	Zero revenue in surplus hours ($P = 0$ for $\tilde{x} \leq 0$)	Surplus hours $\tilde{x} \in (-K_M, 0)$ repriced at c, inducing $K^*_R > K^\circ_R$ (37)
"Missing money"	Scarcity rents $P_{\text{cap}} - c$ cover fixed cost C_F	Unchanged for fossil; alleviated for renewables

This table presents equilibrium conditions in a deregulated competitive electricity market (Section 6.2). Fossil generators earn zero profit when scarcity rents $(P_{\text{cap}} - c)\Pr(\tilde{x} > K_F)$ exactly cover fixed cost C_F; FCL operators enter until average computing margin equals hardware cost C_M (equation (36)). Here $\phi^\circ_{\text{comp}} \equiv 1 - C_F/(P_{\text{cap}} - c)$ is the no-FCL competitive reliability and $q^\circ \equiv F(0; K^\circ_R)$ the probability of renewable surplus, both evaluated at the no-FCL equilibrium. P_{cap} is the scarcity price cap (or value of lost load). Unlike the regulated baseline, the consumer blackout probability is pinned at $C_F/(P_{\text{cap}} - c)$ regardless of FCL adoption: the benefit of FCL operates through the composition of capacity (more renewables, less fossil) and the renewable missing-money channel.

I.7 Grid Scope: Local vs. Global Hash Market (Section 7)

Table 17 contrasts the key assumption on mining revenue R and its implication for FCL program design across the single local grid, multi-grid Nash, social planner, and global monopolist scenarios.

Table 17: Grid Scope and FCL Program Design: Local versus Global Hash Market

Scenario	**Assumption on R**	**Key implication**
Single local grid	R exogenous (price-taker)	Sections 4–6.2 apply unchanged
Multi-grid coordination	R endogenous: falls with $\sum_i \delta_i$	Nash equilibrium $K_M^{SP} < K_M^{NE} < K_M^*$ (smaller than the price-taking benchmark, larger than the coordinated optimum)
Social planner	R endogenous, internalized	Coordinated optimum K_M^{SP} below Nash: internalizes cross-grid revenue dilution
Single global grid	One grid controls all mining, R endogenous; $R^G = \frac{\bar{R}}{\mathbb{E}[d]}$	$K_M^G = 0$ (FCL not offered): total mining income $\bar{R}$ is constant, marginal mining revenue vanishes; capacity reverts to no-FCL benchmark (Prop. 6)

This table contrasts the assumption on mining revenue R and its implication for optimal FCL program design across four scenarios analyzed in Section 7. In the single-grid (local) case, the utility serves a small fraction of global miners so R is pinned by global Bitcoin markets (exogenous). In the multi-grid Nash case, N independent grids each choose δ_i, and mining revenue R falls endogenously in aggregate FCL capacity $\sum_i \delta_i$; the Nash equilibrium is collectively excessive relative to the coordinated optimum (a commons problem), though smaller than the price-taking benchmark would suggest. In the global monopolist case, one grid controls all mining: total mining income $\bar{R} = R \cdot \mathbb{E}[d]$ is constant, so the marginal mining-revenue incentive vanishes and $K_M^G = 0$ (Proposition 6).

I.8 Master Comparison Table

Table 18 provides a unified comparison of blackout probabilities and the triple-win condition across all main scenarios, evaluated in the interior regime at the simultaneous optimum unless noted. In each cell, the adoption condition for the triple win is the scenario-specific threshold $\bar{C}_M^{\text{scenario}}$.

J Corrected Quantitative Assessment (Revised Section 9)

Note: This appendix is a complete corrected version of Section 9 for comparison purposes. The only substantive changes are in the hardware-cost calibration (replacing \$2,500–3,500/MW with \$1.5–2.1M/MW), the amortized-C_M calculation, and the viability conclusions that follow from it. All other text is reproduced without change.

J.1 Mining revenue R per MWh

Table 19 presents the mining revenue calibration.

Calculation. With global mining power approximately 16,300 MW (143 TWh/year), a 10-minute block consumes $\frac{16{,}300\times 600}{3600} \approx 2{,}717$ MWh. At the stated BTC price of \$95,000: $R = \frac{\$95{,}000\times 3.125}{2{,}717} \approx \$109$/MWh—the calibrated central value, roughly 55% above the 2022 estimate of \$70/MWh.

J.2 Viability threshold $\bar{C}_M$

Table 20 shows how the viability threshold scales with the updated mining margin $R - c$.

Corrected hardware-cost calibration. Current Antminer S21+ machines (16.5 J/TH efficiency) cost \$25–35/TH. The correct conversion to \$/MW of installed electrical capacity

Table 18: Master Summary: Blackout Probabilities and Triple-Win Threshold across Scenarios

Scenario	**Blackout (no FCL)**	**Blackout (with FCL)**	**Triple-win threshold $\bar{C}_M^{\cdot}$**
Baseline regulated, exog. p	$\frac{C_F}{\beta+p-c}$	$\frac{c+C_F+C_M-R}{\beta+p-R}$	$(R-c)\frac{\beta+p-c-C_F}{\beta+p-c}$
Small-FCL regime	$\frac{C_F}{\beta+p-c}$	$\frac{c+C_F+C_M-R-cF(-K_M^*;K_R^*)}{\beta+p-R}$	$\bar{C}_M^b = \bar{C}_M + c\,q^\circ > \bar{C}_M$
Capacity case A (K_F fixed)	$1-F(K_F^{(0)};K_R^\circ(K_F^{(0)}))$	Falls (K_R rises, $K_F^{(0)}$ fixed)	$(R-c)\left\lvert\partial\mathbb{E}[\text{fossil}]/\partial K_R\right\rvert_{K_R^{(0)}} > C_R$
Capacity case B (K_R fixed)	$1-F(K_F^\circ;K_R^{(0)})$	Lower iff $C_M < \bar{C}_M(K_R^{(0)})$	$\bar{C}_M(K_R^{(0)}) = \bar{C}_M + c\,F(0;K_R^{(0)})$; wider if $K_R^{(0)}$ large
Capacity case C (both fixed)	$1-F(K_F^{(0)};K_R^{(0)})$	Unchanged (carbon **rises**)	Not achievable without renewable-only DRP
ROR regulation	Driven toward zero (AJ over-investment)	Toward zero; adoption unconditional if K_M in rate base	None if K_M in rate base; $\bar{C}_M$ if excluded
Deregulated competitive	$\frac{C_F}{P_{\text{cap}}-c}$	Pinned at $\frac{C_F}{P_{\text{cap}}-c}$ (zero-profit pins level; composition shifts)	Entry iff $C_M < R\,q^\circ + (R-c)(\phi^\circ_{\text{comp}} - q^\circ)$
Nash (multi-grid)	$\frac{C_F}{\beta+p-c}$	Lower iff $C_M < \bar{C}_M(R^{NE})$; collectively excessive vs. coordinated optimum, below price-taking benchmark	$\bar{C}_M^{NE} < \bar{C}_M$ (endogenous $R^{NE} < R$ shrinks threshold)
Global monopolist	$\frac{C_F}{\beta+p-c}$	No FCL program ($K_M^G = 0$); equals no-FCL level	No viable threshold (Prop. 6)

This table provides a unified comparison of blackout probabilities and triple-win thresholds across all main scenarios. All results assume the interior regime ($R < c + C_F + C_M$) unless otherwise noted. "Blackout (no FCL)" and "Blackout (with FCL)" report the equilibrium probability of $\tilde{x} > K_F$ at the respective optima. $\phi^\circ \equiv F(K_F^\circ; K_R^\circ)$ denotes the no-blackout probability in the no-FCL benchmark, $\phi^\circ_{\text{comp}} \equiv 1 - C_F/(P_{\text{cap}} - c)$ its competitive analogue, and $q^\circ \equiv F(0; K_R^\circ)$ the probability of renewable surplus. P_{cap} is the competitive scarcity price; regulated results obtain as the special case $P_{\text{cap}} = \beta + p$.

Table 19: Mining Revenue R per MWh: Calibration (corrected appendix)

Parameter	**2022 estimate**	**Calibrated value (2026)**
BTC price	\$22,000	≈\$95,000
Block reward	6.25 BTC	3.125 BTC
Network hash rate	∼400 EH/s	∼800–976 EH/s
Energy per block (average)	≈2,000 MWh	≈2,700 MWh
R (revenue/MWh)	≈\$70	≈\$100–157

This table presents the key revenue parameter R (mining income per MWh of electricity consumed) under current (2026) market conditions. R is computed as block reward value divided by global energy consumed per block: at \$95,000 × 3.125 BTC per block and global mining power of approximately 16,300 MW, a 10-minute block consumes ∼2,717 MWh, giving $R \approx$ \$109/MWh—the model's calibrated central value.

Table 20: Triple-Win Threshold $\bar{C}_M$: 2022 versus 2026 Calibration (corrected appendix)

Parameter	**2022 estimate**	**2026 calibration**
R (mining revenue/MWh)	\$70	\$109
c (fossil variable cost)	\$45	\$45 (stable)
$R - c$ (mining margin)	\$25/MWh	\$64/MWh
$\bar{C}_M$ (ASIC threshold)	≈\$23/MWh	≈\$60/MWh
Viability range in C_M	narrow	≈ 2.5× wider

This table presents the triple-win threshold $\bar{C}_M \equiv (R-c)\frac{\beta+p-c-C_F}{\beta+p-c}$ under 2022 and 2026 market conditions. The mining margin $R - c$ of \$64/MWh in the calibration implies a viable range of hardware costs C_M roughly 2.5 times wider than in 2022. Other parameters: $\beta + p =$ \$200/MWh, $C_F =$ \$10/MWh. C_M and $\bar{C}_M$ are both in \$/MWh: the per-MW-hour amortized capital cost compared to the per-MW-hour net margin threshold.

uses the fact that at 16.5 W per TH/s, one MW of electrical draw supports

$$\frac{1{,}000{,}000 \text{ W}}{16.5 \text{ W per TH/s}} \;=\; 60{,}606 \text{ TH/s per MW},$$

so that

$$\text{Hardware cost per MW} \;=\; \frac{\$25\text{–}35}{\text{TH}} \;\times\; 60{,}606\,\frac{\text{TH}}{\text{MW}} \;\approx\; \$1.5\text{–}2.1\text{M/MW}.$$

Premium or flagship-generation ASICs (e.g., Antminer S21 XP at $\approx$ \$40/TH) can reach approximately \$2.4M/MW; the range \$1.5–2.5M/MW brackets current market conditions.

The model parameter C_M is the amortized capital cost per MWh of operating time (the K_M FOC equates C_M to a flow payoff in \$/MWh). For a hardware lifecycle of T years:

$$C_M \;=\; \frac{\text{Hardware cost (\$/MW)}}{T \times 8{,}760 \text{ h/year}}.$$

Table 21 reports C_M across hardware costs and lifecycle assumptions and compares each to the calibrated threshold $\bar{C}_M \approx \$60$/MWh.

These numbers imply a *near-threshold* rather than clearly comfortable viability. At current market prices and a 3-year lifecycle—the commonly assumed depreciation horizon—the lower bound (\$1.5M/MW) just clears the threshold ($C_M \approx \$57/\text{MWh} < \bar{C}_M \approx \$60/\text{MWh}$), while the midpoint and upper end exceed it. Viability is therefore sensitive to three parameters that are trending favorably: ASIC prices (declining roughly 15–20% annually as manufacturing scales), ASIC efficiency (improving $\approx$12% per year in J/TH, raising TH per MW), and the mining margin $R - c$ (which at \$95,000 BTC is 2.5$\times$ its 2022 level).

Table 21: Corrected ASIC Hardware Cost Parameter C_M vs. Viability Threshold $\bar{C}_M \approx \$60$/MWh

Hardware cost	C_M **(\$/MWh) by lifecycle**				**Viable at**
(\$/MW)	**2 yr**	**3 yr**	**4 yr**	**5 yr**	**lifecycle**
\$1.5M	\$85.6	\$57.1	\$42.8	\$34.2	≥ 3 yr
\$2.1M	\$119.9	\$79.9	\$59.9	\$47.9	≥ 4 yr
\$2.5M	\$142.7	\$95.1	\$71.4	\$57.1	≥ 5 yr
Threshold: $\bar{C}_M \approx \$60/MWh$					

Entries are C_M = hardware cost/$(T \times 8{,}760)$, rounded to one decimal place. Shaded cells (where $C_M < \bar{C}_M$) indicate parameter combinations consistent with the triple-win. Hardware cost range \$1.5–2.5M/MW corresponds to \$25–40/TH at 16.5 J/TH efficiency. The lifecycle column reports the minimum years required for $C_M < \bar{C}_M \approx \$60$/MWh.

J.3 Renewable and fossil capacity costs

Table 22 presents renewable and fossil capital cost data. Renewable CapEx has fallen roughly 40–50% since 2020, reducing the parameter C_R in the model and shifting the jointly optimal renewable capacity K_R^* sharply upward. This implies more surplus hours, more zero-carbon computing, and a stronger capacity substitution effect.

J.4 Summary: Calibration and implications

Current market conditions are broadly supportive of the model's conclusions, though the hardware-cost comparison is more nuanced than the original section implies. First, mining revenue R has risen roughly 55% (from \$70 to \$109/MWh), expanding the viable ASIC capital cost range $\bar{C}_M$ by roughly 2.5-fold relative to 2022 conditions. Second, renewable CapEx C_R has fallen roughly 40%, making high-K_R^* scenarios economically realistic and dramatically increasing surplus hours. Third, ASIC efficiency has improved approximately 12% per year (J/TH), raising TH/s per MW of electrical draw and directly reducing C_M—a

Table 22: Renewable and Fossil Capacity Costs: 2020 versus 2025–26 (corrected appendix)

Parameter	**Typical 2020**	**Current (2025–26)**
Solar PV CapEx (utility scale)	\$800–1,200/kW	\$388–600/kW
Onshore wind CapEx	\$1,200–1,600/kW	\$850–1,000/kW
CCGT gas CapEx	\$1,000–1,200/kW	\$800–1,000/kW
Solar LCOE	\$40–60/MWh	\$28–66/MWh
Onshore wind LCOE	\$35–55/MWh	\$25–70/MWh

This table presents representative capital expenditure (CapEx) ranges and levelized cost of energy (LCOE) for utility-scale solar PV, onshore wind, and combined-cycle gas turbines (CCGT), comparing 2020 benchmarks to current (2025–26) values. Renewable CapEx has declined approximately 40–50% since 2020, corresponding to a fall in the model parameter C_R (renewable capital cost per MW). A lower C_R shifts the jointly optimal renewable capacity K_R^* upward, increasing surplus hours and strengthening the capacity-substitution channel. Sources: IRENA, NREL, Lazard LCOE.

trend whose importance is amplified in the corrected calibration because viability is near-threshold rather than comfortable. Together, these three trends create a converging window: the triple-win condition $C_M < \bar{C}_M$ is achievable today at the lower end of the hardware-cost and lifecycle range, and is strengthening over time as ASIC prices fall and mining margins remain elevated. The model's carbon-reduction estimates are therefore directionally robust, though the quantitative case rests on continued favorable ASIC-efficiency trends.

A cross-sectional observation from the calibration exercise is also worth noting. The quantitative benefit of FCL adoption scales with the *variance of residual demand* $\tilde{x}$: in regions where renewable supply and consumer demand are poorly synchronized, the shock-absorbing role of mining is more valuable and the carbon reduction larger. Conversely, in grids where supply and demand move closely together, the FCL mechanism adds less value. This suggests that FCL programs will be most impactful in grids with volatile, intermittent renewables—precisely the environments where fossil backup costs are highest.

J.5 Sensitivity to Bitcoin price volatility

A key concern about the calibration is the extreme volatility of Bitcoin prices. Since mining revenue per MWh is $R = \frac{P_{\text{BTC}} \cdot b}{E_{\text{block}}}$ (BTC price times block reward divided by energy per block), a halving of the BTC price roughly halves R, compressing the mining margin $R - c$ and narrowing the viable ASIC cost range $\bar{C}_M$. Table 23 quantifies this dependence across the full historical price range, holding the post-2024 block reward at 3.125 BTC and global mining power at 16,300 MW (so energy per block $E_{\text{block}} \approx 2{,}717$ MWh).

Table 23: Sensitivity of Mining Revenue to Bitcoin Price (corrected appendix)

BTC price	R **(\$/MWh)**	$R - c$	$\frac{\bar{C}_M}{\bar{C}_M^{\text{base}}}$	**FCL viability**
\$20,000	\$23	−\$22	—	Not viable ($R < c$)
\$35,000	\$40	−\$5	—	Not viable ($R < c$)
\$45,000	\$52	+\$7	0.11	Marginally viable
\$60,000	\$69	+\$24	0.38	Viable (narrow range)
\$75,000	\$86	+\$41	0.64	Viable
\$95,000	\$109	+\$64	1.00	Near-threshold (baseline)
\$120,000	\$138	+\$93	1.45	Viable with margin
\$150,000	\$172	+\$127	1.98	Strongly viable

$c = \$45$/MWh (fossil variable cost). R is computed as $\frac{P_{\text{BTC}} \times 3.125\,\text{BTC}}{2{,}717\,\text{MWh}}$. $\frac{\bar{C}_M}{\bar{C}_M^{\text{base}}}$ is normalized to the baseline ($R = \$109$/MWh, BTC = \$95,000) value. A dashed entry ($---$) indicates $R < c$, so FCL dispatch is never profitable and $\bar{C}_M < 0$. "Near-threshold (baseline)" reflects the corrected finding that $C_M \approx \bar{C}_M$ at current hardware costs and typical 3-year lifecycles.

Break-even Bitcoin price. Setting $R = c = \$45$/MWh and solving for $P_{\text{BTC}}^{\min}$:

$$P_{\text{BTC}}^{\min} = \frac{c \cdot E_{\text{block}}}{b} = \frac{45 \times 2{,}717}{3.125} \approx \$39{,}000.$$

Below this threshold, the FCL dispatch rule never activates—mining on spare fossil capacity is unprofitable—and all model results collapse to the no-FCL benchmark. At 2024–2026 BTC

prices (roughly \$50,000–\$100,000), the mining margin is positive throughout, confirming that the calibrated viability range is not a knife-edge result.

Endogenous hash rate as a stabilizer. Table 23 understates the resilience of R to price declines, because hash rate responds endogenously to BTC price: when prices fall, less efficient miners exit, reducing global energy consumption per block and thereby *raising* R relative to the fixed-hash-rate calculation. In bear markets, network difficulty adjusts downward within two weeks, partially offsetting the price decline in R. Empirically, R has historically exhibited a coefficient of variation roughly 40–50% lower than the BTC price itself over multi-month horizons, making the mining margin $R - c$ considerably more stable than raw price volatility would suggest.

Implications for triple-win robustness (corrected). Three conclusions follow from this sensitivity analysis. First, the model's qualitative predictions are robust for any BTC price above roughly \$45,000 (post-halving): the triple-win threshold $\bar{C}_M$ is positive. Whether current ASIC hardware costs fall within the viable range depends on lifecycle and specific hardware price: at \$1.5M/MW and a 3-year lifecycle, $C_M \approx$ \$57/MWh just clears the \$60/MWh threshold; at \$2.1M/MW the same lifecycle gives $C_M \approx$ \$80/MWh, requiring either a longer depreciation horizon (≥ 4 years) or lower hardware prices to satisfy $C_M < \bar{C}_M$. Second, the quantitative size of $\bar{C}_M$ scales roughly linearly with the mining margin $R - c$, so the welfare gains from FCL adoption grow as BTC prices rise—meaning the program is self-reinforcing in bull markets. At BTC above \$120,000 ($\bar{C}_M \approx$ \$87/MWh), even the \$2.5M/MW–3-year scenario is clearly viable. Third, bear-market periods (BTC below \$45,000) represent a genuine sustainability concern: FCL operators would halt voluntary curtailment since dispatching on fossil reduces their margins, weakening the demand response channel. A well-designed FCL contract should therefore include *price floors* that guarantee a minimum R payment to the FCL operator, insulating the demand-response commitment

from spot BTC price volatility.

J.6 Implications for AI data center loads

For AI training with curtailable fraction $\theta = 0.5$ and training-compute revenue $R^{\text{AI}} =$ \$200/MWh (reflecting the commercial value of frontier model training), the effective margin is $\theta R^{\text{AI}} - c = 0.5 \times 200 - 45 =$ \$55/MWh—roughly comparable to the current Bitcoin mining margin and similarly near the threshold under the corrected hardware calibration. However, at current inferred R^{AI} values the opacity problem means this revenue cannot be directly contracted, making practical Demand Response Program implementation with AI training loads substantially more challenging than with Bitcoin mining.